Machine Learning
Study Notes – Syllabus-Mapped Portion

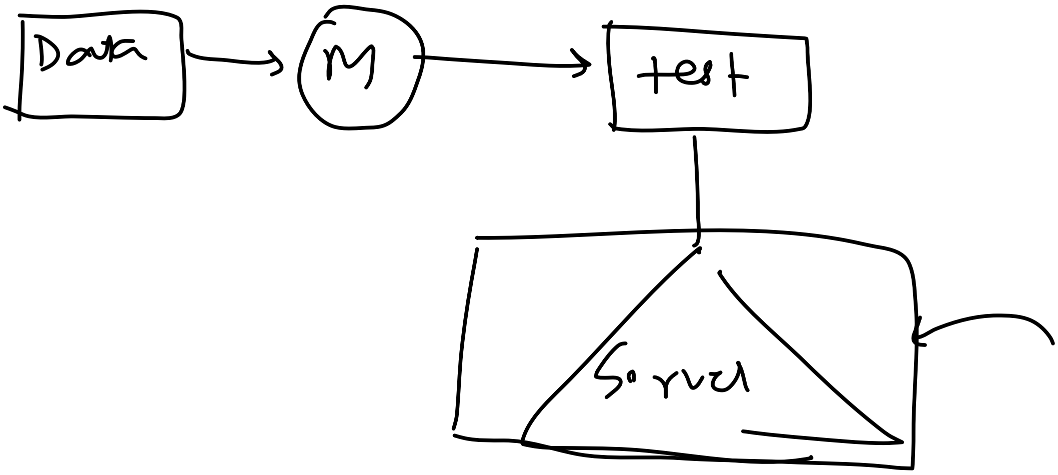
Source: ML Linked Index Notes (354 Topics · 436 Pages)

Extracted pages covering the full course syllabus (5 Units · 28 Hours)

Note: k-NN and Decision Trees are referenced conceptually in this notes set within the supervised learning discussions. Pages are extracted in syllabus order.

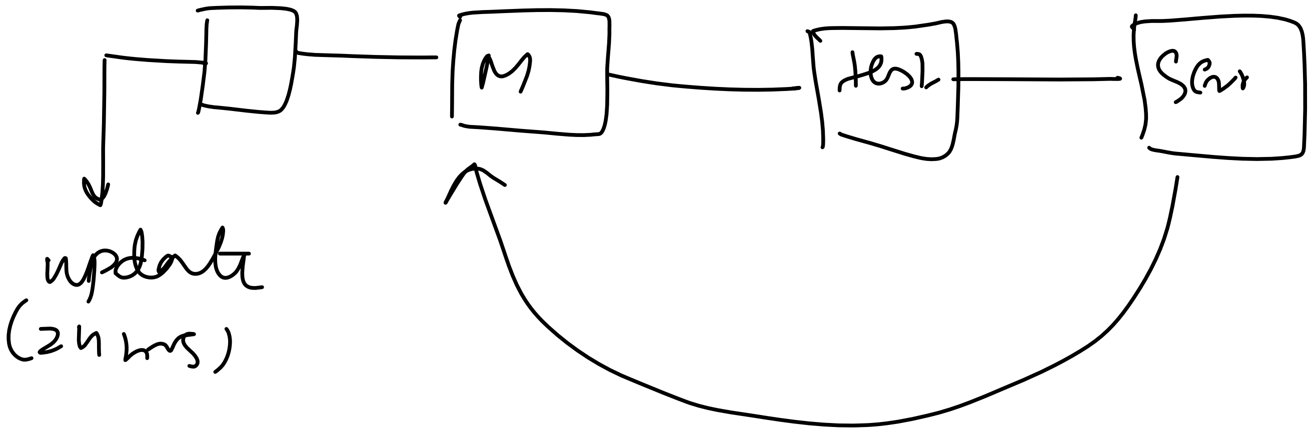
2. Batch/Offline ML

Wednesday, March 17, 2021 5:31 PM



3. The problem with Batch Learning

Wednesday, March 17, 2021 5:47 PM



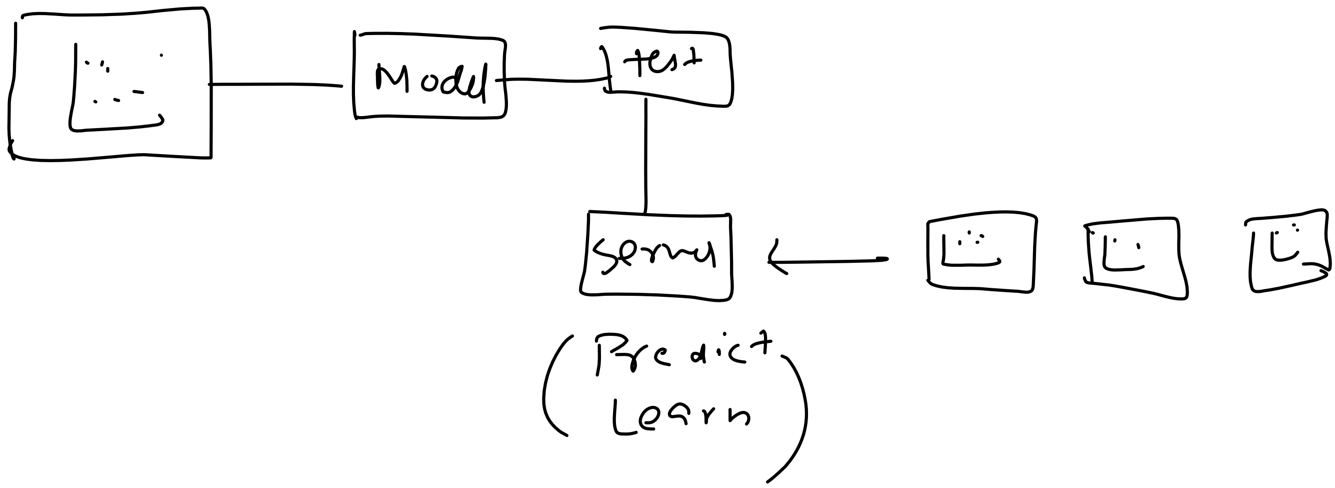
4. Disadvantages of Batch ML

Wednesday, March 17, 2021 5:32 PM

1. Lots of Data
2. Hardware Limitation
3. Availability

1. Online Machine Learning

Thursday, March 18, 2021 4:27 PM



2. When to use?

Thursday, March 18, 2021 4:33 PM

1. Where there is a concept drift
2. Cost Effective
3. Faster solution

3. How to implement?

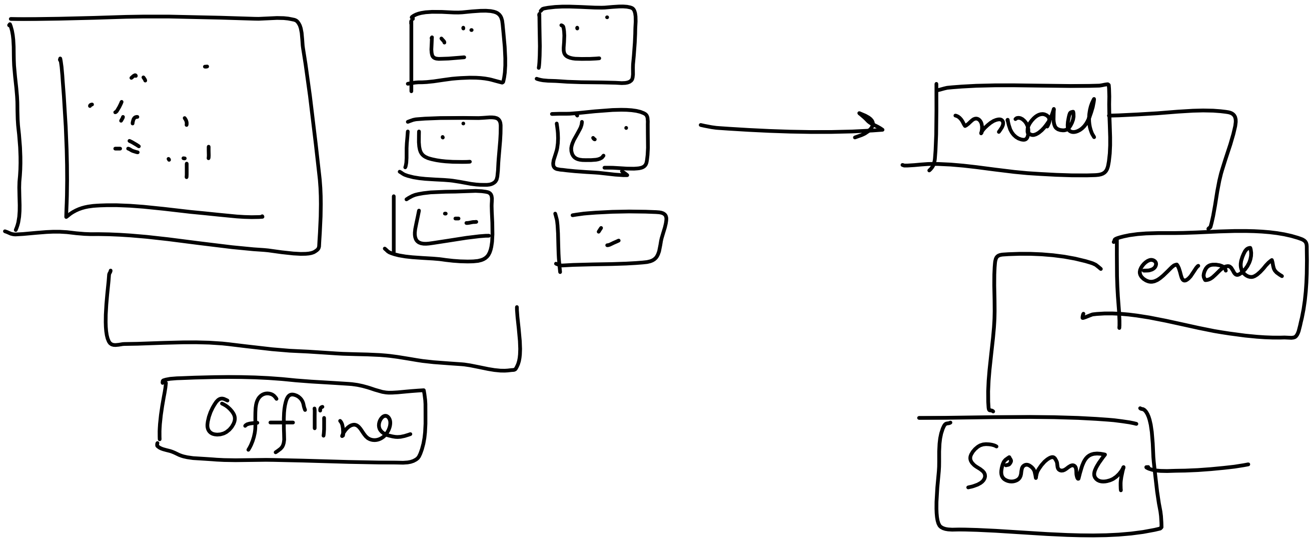
Thursday, March 18, 2021 4:28 PM

4. Learning Rate

Thursday, March 18, 2021 4:28 PM

5. Out of Core Learning

Thursday, March 18, 2021 4:28 PM



6. Disadvantage

Thursday, March 18, 2021 4:29 PM

1. Tricky to use
2. Risky

7. Batch Vs Online Learning

Thursday, March 18, 2021 4:29 PM

Offline Learning	Features	Online Learning
Less complex as model is constant	**Complexity**	Dynamic complexity as the model keeps evolving over time
Fewer computations, single time batch-based training	**Computational Power**	Continuous data ingestions result in consequent model refinement computations
Easier to implement	**Use in Production**	Difficult to implement and manage
Image Classification or anything related to Machine Learning - where data patterns remains constant without sudden concept drifts	**Applications**	Used in finance, economics, heath where new data patterns are constantly emerging
Industry proven tools. E.g. Sci-kit, TensorFlow, Pytorch, Keras, Spark Mlib	**Tools**	Active research/New project tools: E.g. MOA, SAMOA, scikit-multiflow, streamDM

Image courtesy - https://www.iunera.com/kraken/fabric/simple-introduction-to-online-learning-in-machine-learning/

1. Instance Vs Model Based Learning

Friday, March 19, 2021 4:05 PM

Learning

memorizing

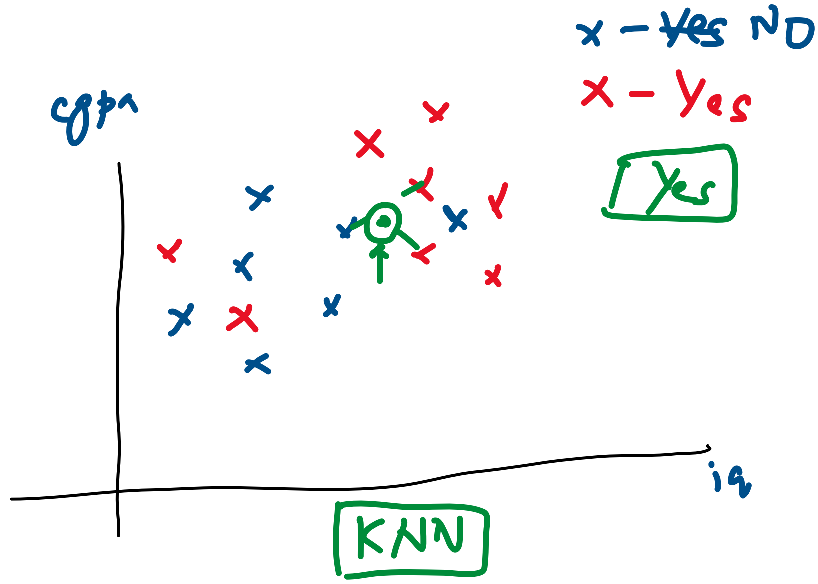
generalizing

2. Instance Based

Friday, March 19, 2021 4:06 PM

iq	cgpa	placement Y/N
80	8	Y
70	7	N

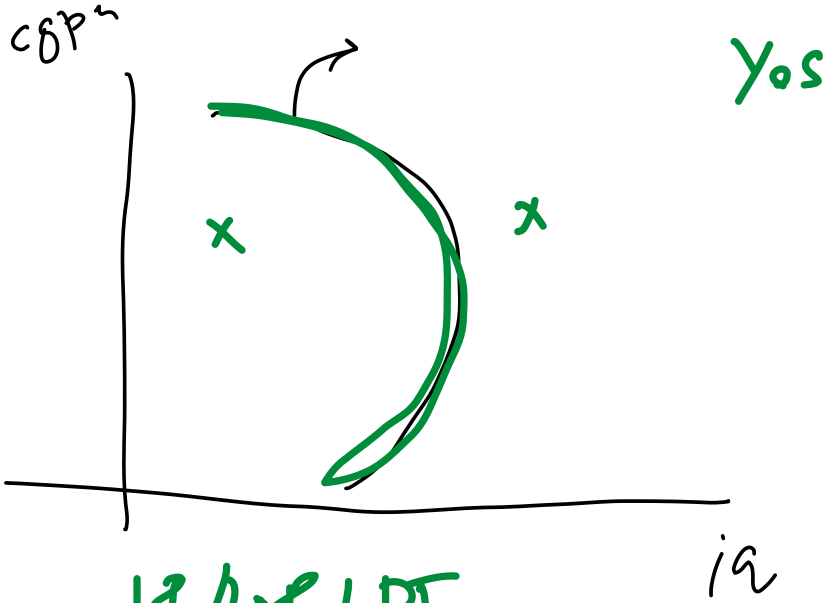
$\boxed{7.5, 103}$



3. Model Based

Friday, March 19, 2021 4:06 PM

iq | cgpa | placement



LR / LoR | DT

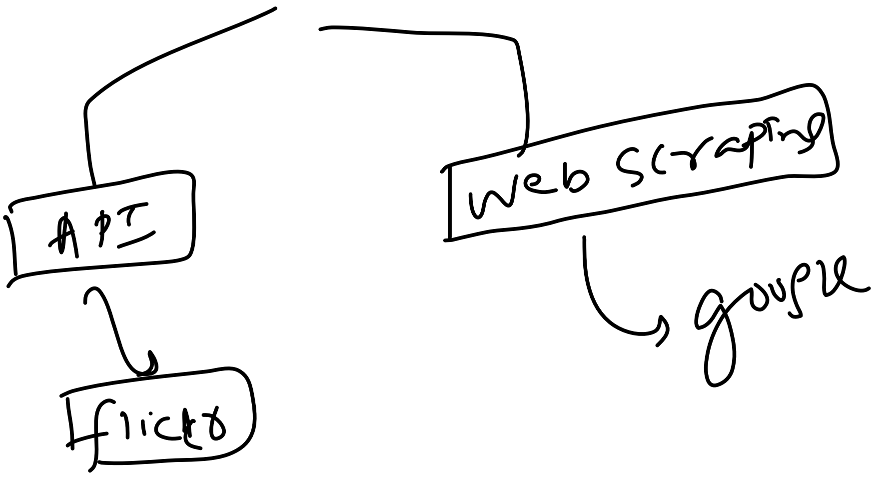
4. Differences

Friday, March 19, 2021 4:06 PM

Usual/Conventional Machine Learning	**Instance Based Learning**
Prepare the data for model training	Prepare the data for model training. No difference here
Train model from training data to estimate model parameters i.e. discover patterns	Do not train model. Pattern discovery postponed until scoring query received
Store the model in suitable form	There is no model to store
Generalize the rules in form of model, even before scoring instance is seen	No generalization before scoring. Only generalize for each scoring instance individually as and when seen
Predict for unseen scoring instance using model	Predict for unseen scoring instance using training data directly
Can throw away input/training data after model training	Input/training data must be kept since each query uses part or full set of training observations
Requires a known model form	May not have explicit model form
Storing models generally requires less storage	Storing training data generally requires more storage

1. Data Collection

Saturday, March 20, 2021 5:59 PM



2. Insufficient Data/Labelled Data

Saturday, March 20, 2021 6:00 PM

NLP

A

(100)

M1

B

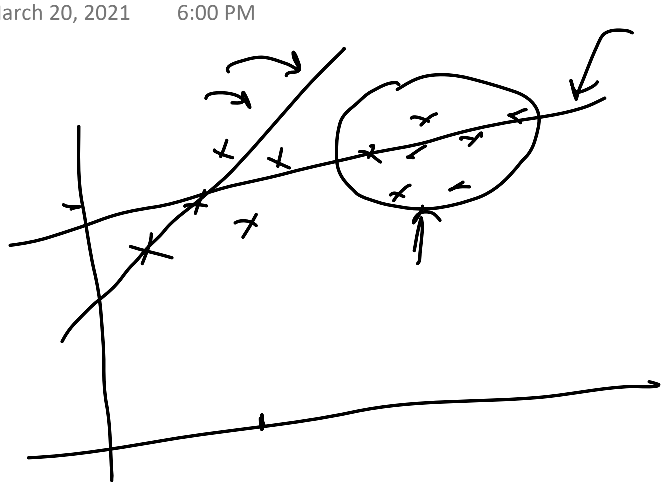
(10^6)

M2

to, t80,

3. Non Representative Data

Saturday, March 20, 2021 6:00 PM



Sampling Noise

Sampling bias

4. Poor Quality Data

Saturday, March 20, 2021 6:00 PM

60%

5. Irrelevant Features

Saturday, March 20, 2021 6:00 PM

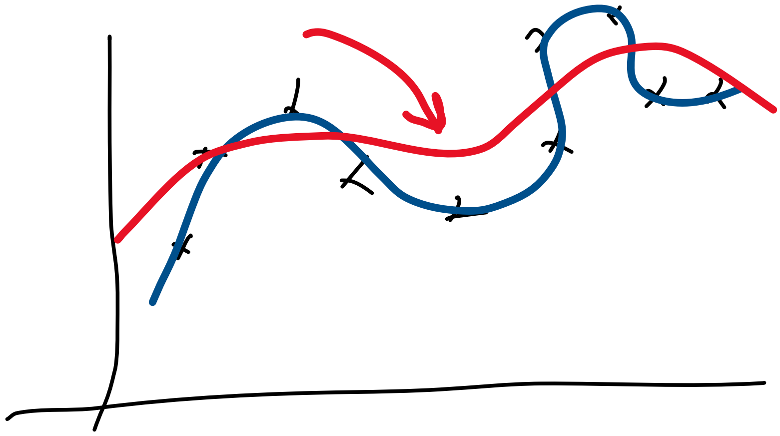
Garbage In
Garbage out

Age | wt | ht || $\boxed{\text{location}}$

bmi ~ feature engineering

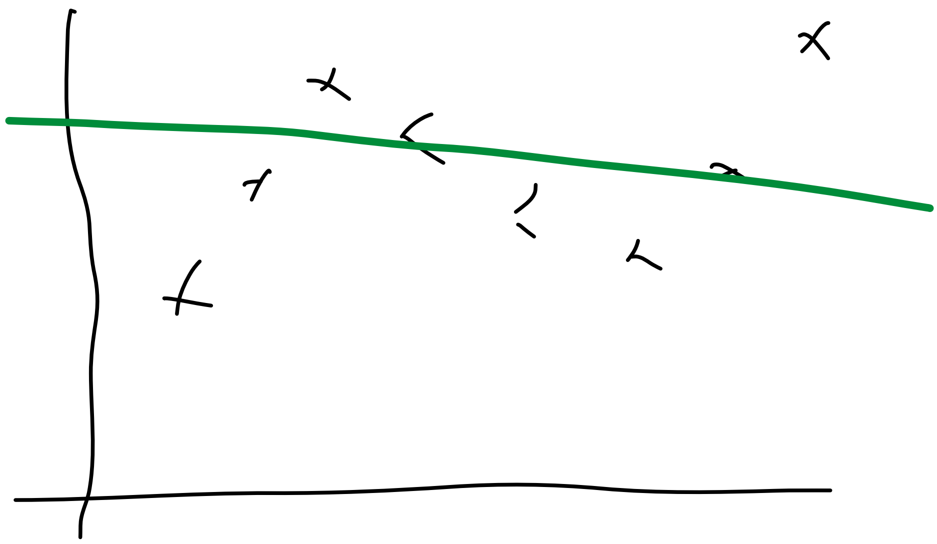
6. Overfitting

Saturday, March 20, 2021 6:01 PM



7. Underfitting

Saturday, March 20, 2021 6:01 PM

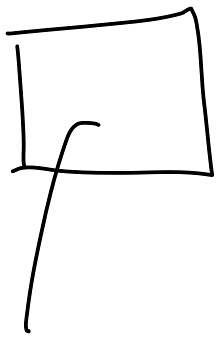


8. Software Integration

Saturday, March 20, 2021 6:01 PM

9. Offline Learning/ Deployment

Saturday, March 20, 2021 6:01 PM



10. Cost Involved

Saturday, March 20, 2021 6:01 PM

Devops

MLops

1. Retail - Amazon/Big Bazaar

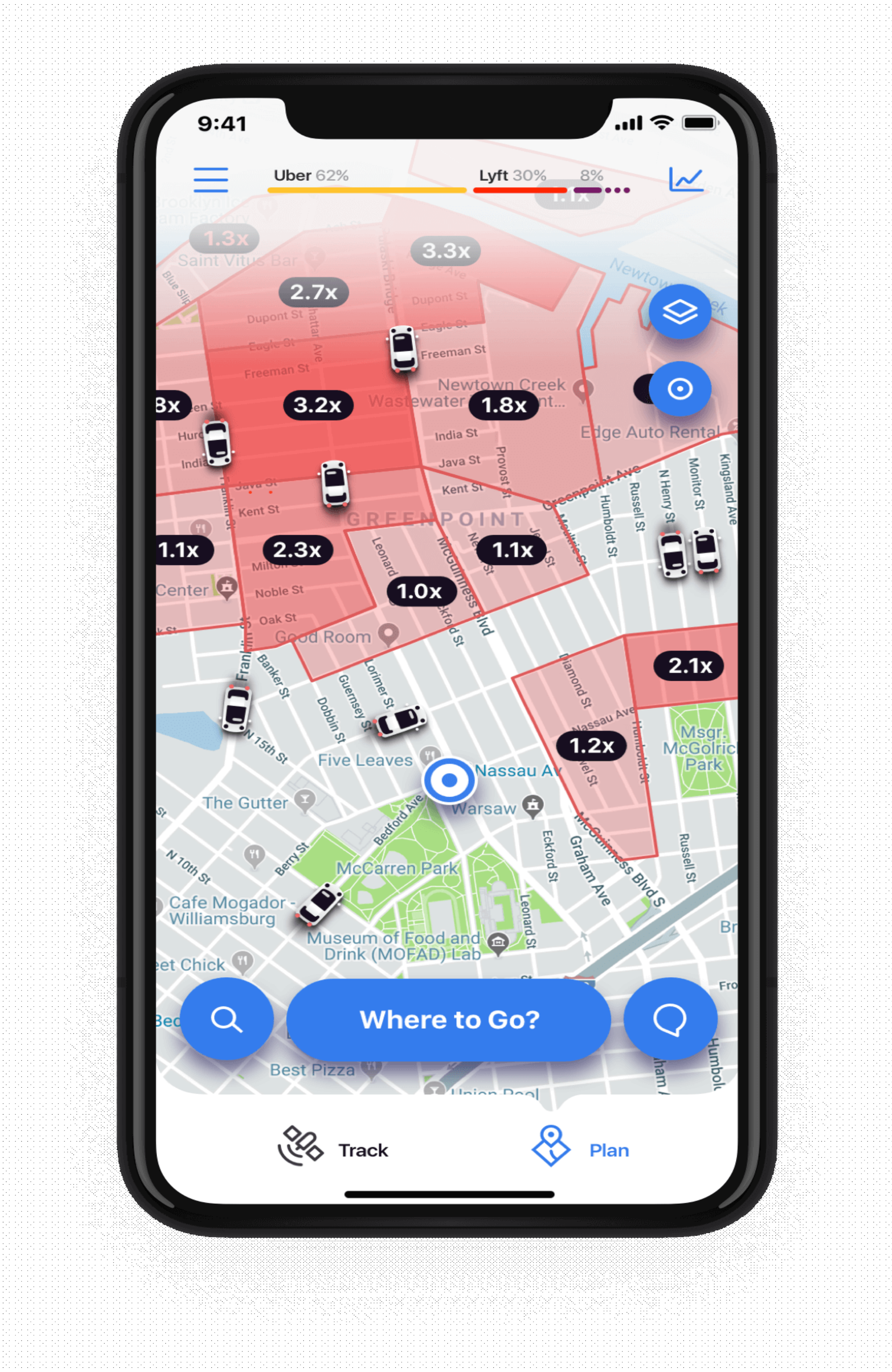
Monday, March 22, 2021 6:07 PM

2. Banking and Finance

Monday, March 22, 2021 6:07 PM

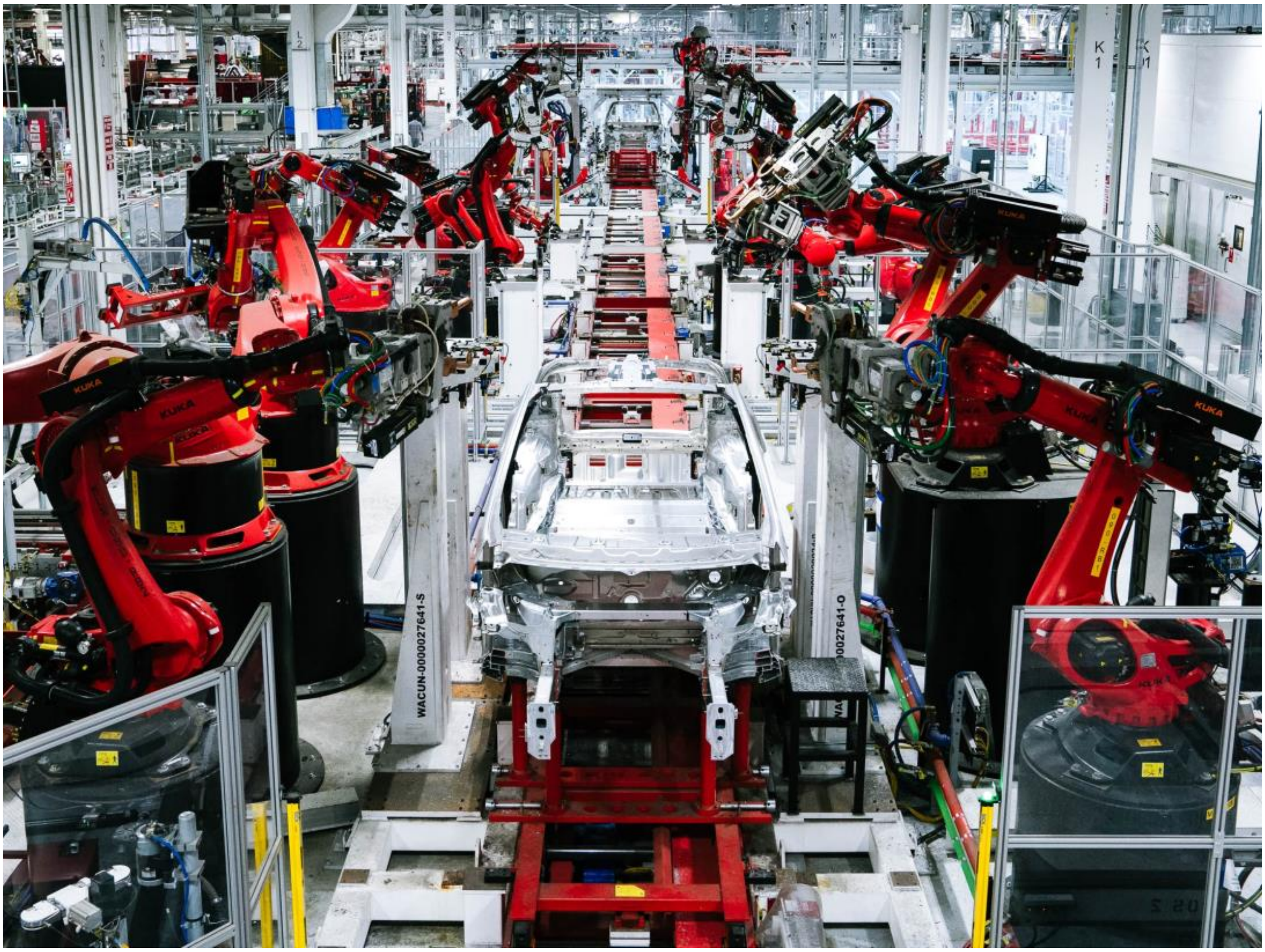
3. Transport - OLA

Monday, March 22, 2021 6:07 PM



4. Manufacturing - Tesla

Monday, March 22, 2021 6:08 PM



5. Consumer Internet - Twitter

Monday, March 22, 2021 6:08 PM

Machine Learning Development Life Cycle(MLDLC/MLDC)

Tuesday, March 23, 2021 12:09 PM

SDLC

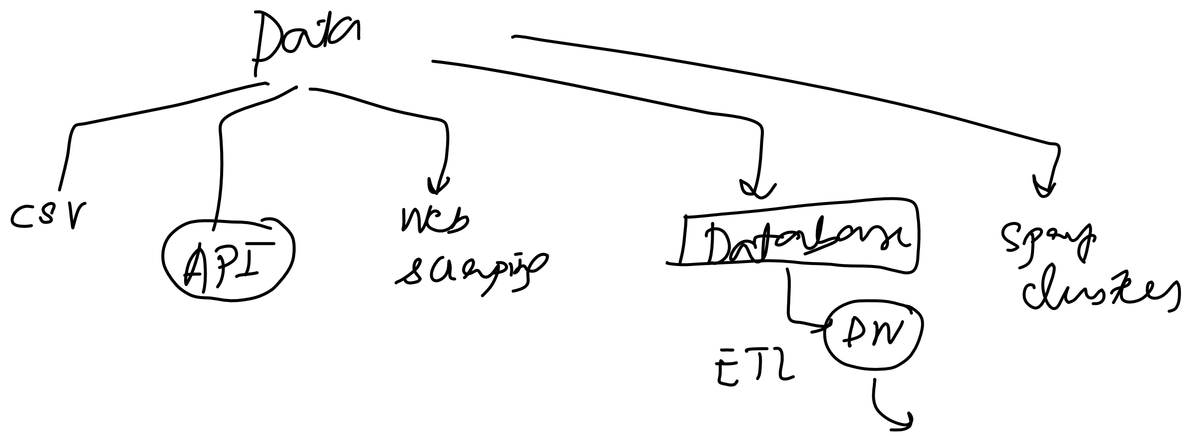
MLDLC

1. Frame the Problem

Tuesday, March 23, 2021 12:10 PM

2. Gathering Data

Tuesday, March 23, 2021 12:11 PM



3. Data Preprocessing

Tuesday, March 23, 2021 12:11 PM

- → Remove duplicates
- → Remove missing val
- → Outliers
- → Scale

4. Exploratory Data Analysis

Tuesday, March 23, 2021 12:11 PM

Vizs

Univariate/Biavariate

Outlier detection

Imbalance →

5. Feature Engineering and Selection

Tuesday, March 23, 2021 12:12 PM

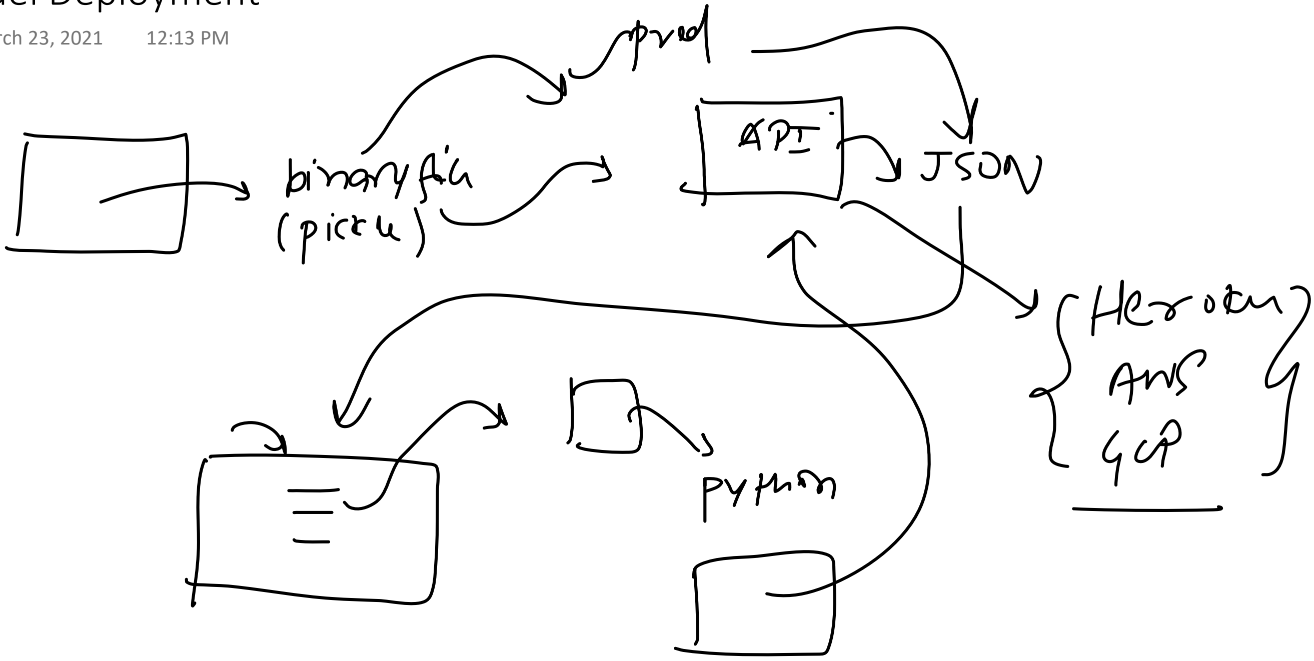
6. Model Training,Evalation and Selection

Tuesday, March 23, 2021 12:12 PM

$\boxed{\text{Ensemble learning}}$

7. Model Deployment

Tuesday, March 23, 2021 12:13 PM



8. Testing

Tuesday, March 23, 2021 12:14 PM

$\boxed{\text{A /B testing}}$

9. Optimize

Tuesday, March 23, 2021 12:15 PM

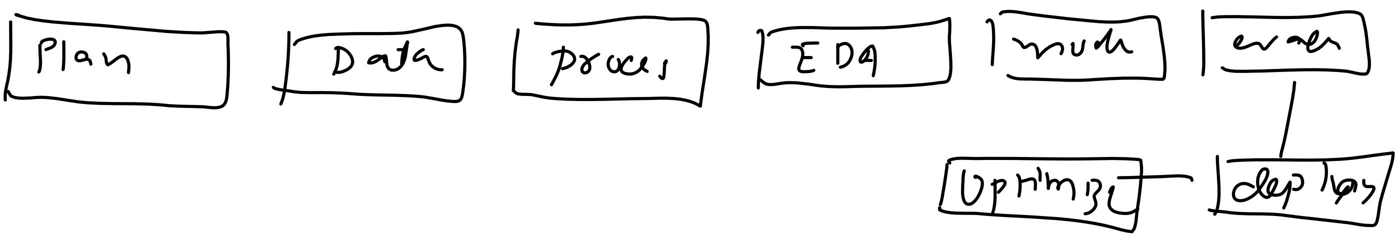
Retrain

Rotting

- Backup
- Data
- load balancing

1. Various Data Based Job Roles

Wednesday, March 24, 2021 1:25 PM

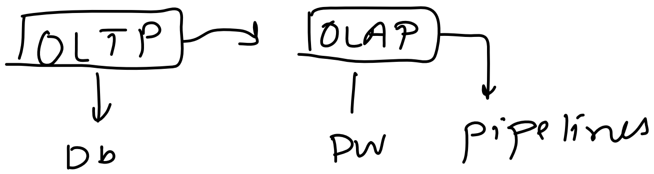


1. Data Engineer

Wednesday, March 24, 2021 1:25 PM

Job Roles

- Scrape Data from the given sources.
- Move/Store the data in optimal servers/warehouses.
- Build data pipelines/APIs for easy access to the data.
- Handle databases/data warehouses.



Skills Required

- Strong grasp of algorithms and data structures
- Programming Languages (Java/R/Python/Scala) and script writing
- Advanced DBMS's
- BIG DATA Tools (Apache Spark, Hadoop, Apache Kafka, Apache Hive)
- Cloud Platforms (Amazon Web Services, Google Cloud Platform)
- Distributed Systems
- Data Pipelines

2. Data Analyst

Wednesday, March 24, 2021 1:26 PM

Responsibilities of a Data Analyst

- *Cleaning and organizing Raw data.*
- *Analyzing data to derive insights.*
- *Creating data visualizations.*
- *Producing and maintaining reports.*
- *Collaborating with teams/colleagues based on the insight gained.*
- *Optimizing data collection procedures*

Skills

- *Statistical Programming*
- *Programming Languages (R/SAS/Python)*
- *Creative and Analytical Thinking*
- *Business Acumen — Medium to High preferred*
- *Strong Communication Skills.*
- *Data Mining, Cleaning, and Munging*
- *Data Visualization*
- *Data Story Telling*
- *SQL*
- *Advanced Microsoft Excel*

1. Business Problem to ML Problem

Monday, March 29, 2021 7:29 AM

Netflix

Churn rate ↓

4%

3.75

3.5

100 → 2%

98

2%

Increase revenue

4% → 3.75%

2. Type of Problem

Monday, March 29, 2021 7:30 AM

Big picture → end product → prediction

1) Supervised

Classification

Regression

3. Current Solution

Monday, March 29, 2021 7:30 AM

Churn rate → $\boxed{5\%}$ → $\boxed{\pm 10\%}$

$\boxed{6\%}$

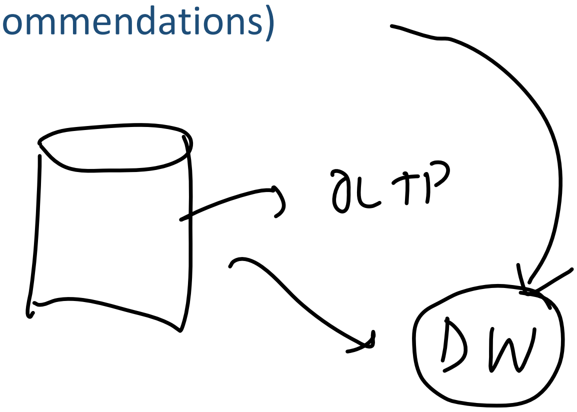
→ $\boxed{\text{factors}}$

4. Getting Data

Monday, March 29, 2021 7:30 AM

1. Watch time
2. Search but did not find
3. Content left in the middle
4. Clicked on recommendations(order of recommendations)

Data engineer

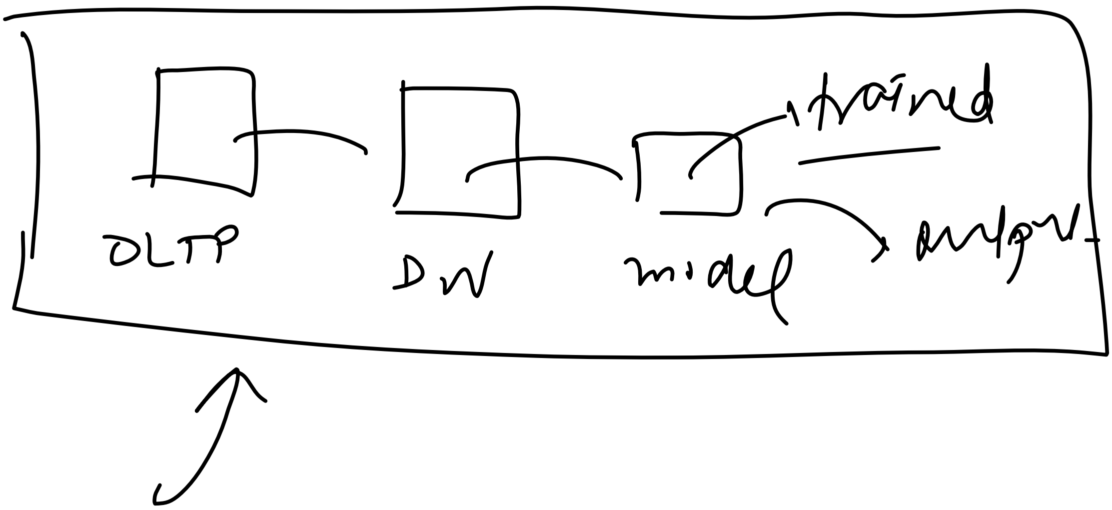


5. Metrics to measure

Monday, March 29, 2021 7:30 AM

6. Online Vs Batch?

Monday, March 29, 2021 7:31 AM

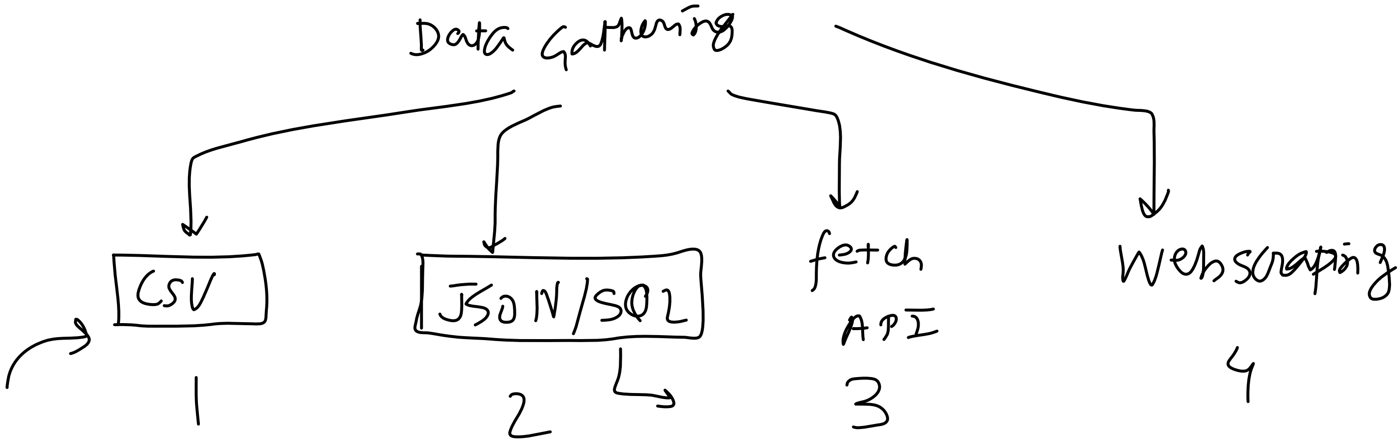


7. Check Assumptions

Monday, March 29, 2021 7:31 AM

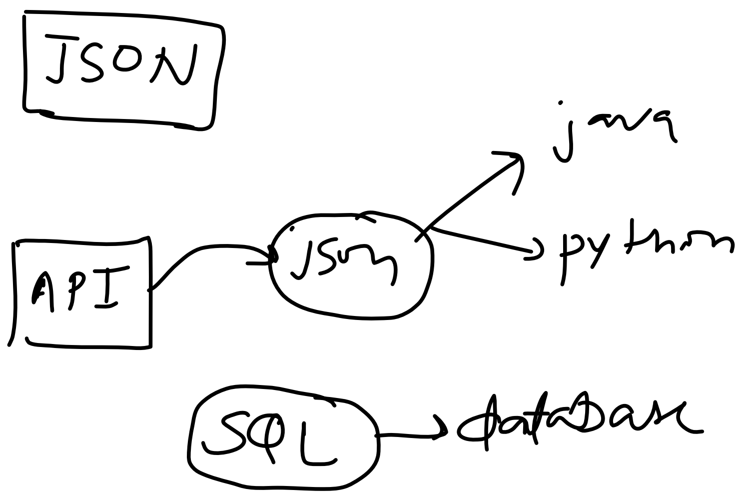
Gathering Data

Tuesday, March 30, 2021 5:35 PM



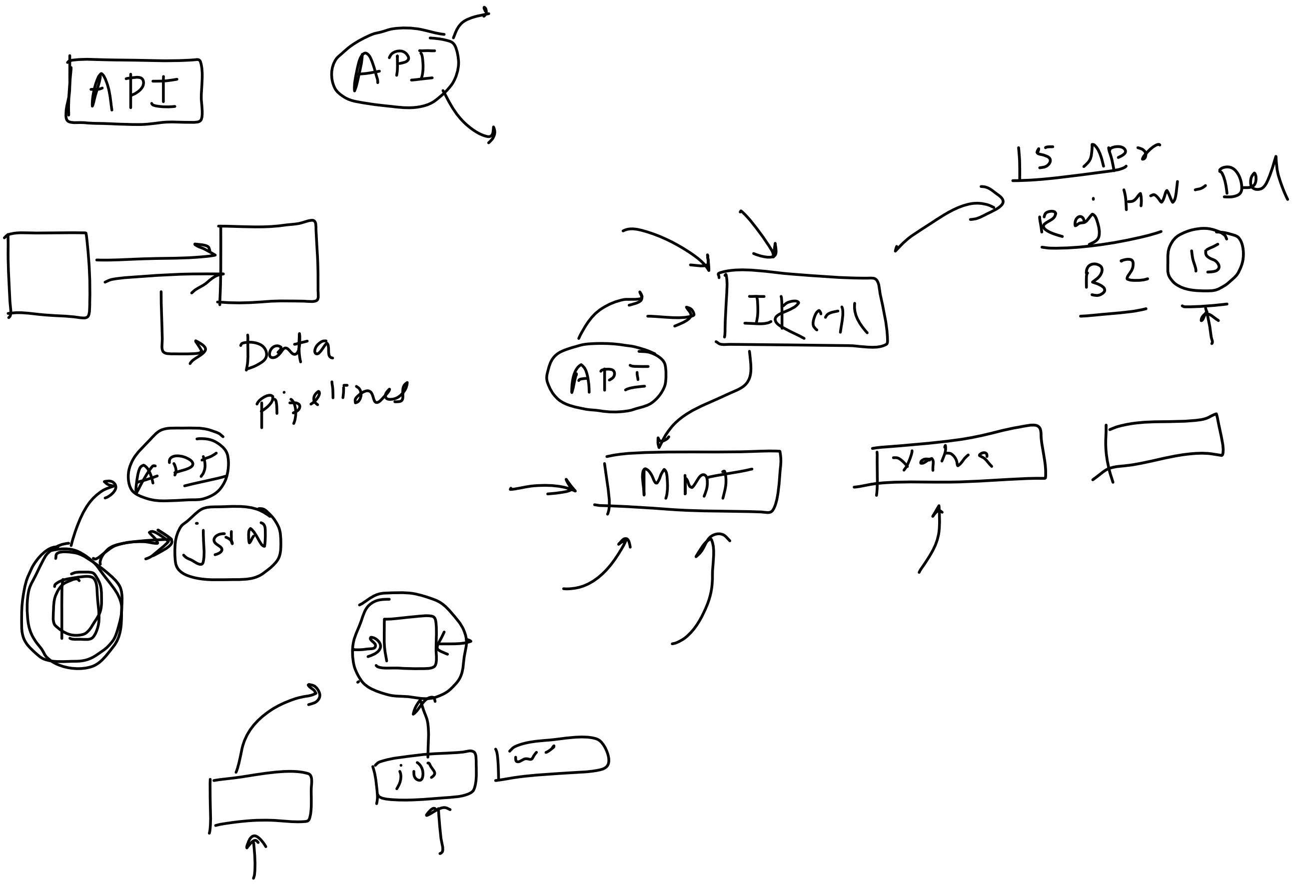
JSON/SQL

Wednesday, March 31, 2021 8:10 AM



What is an API

Thursday, April 1, 2021 6:55 AM



Understanding Your Data

Saturday, April 3, 2021 9:07 AM

[Asking basic questions]

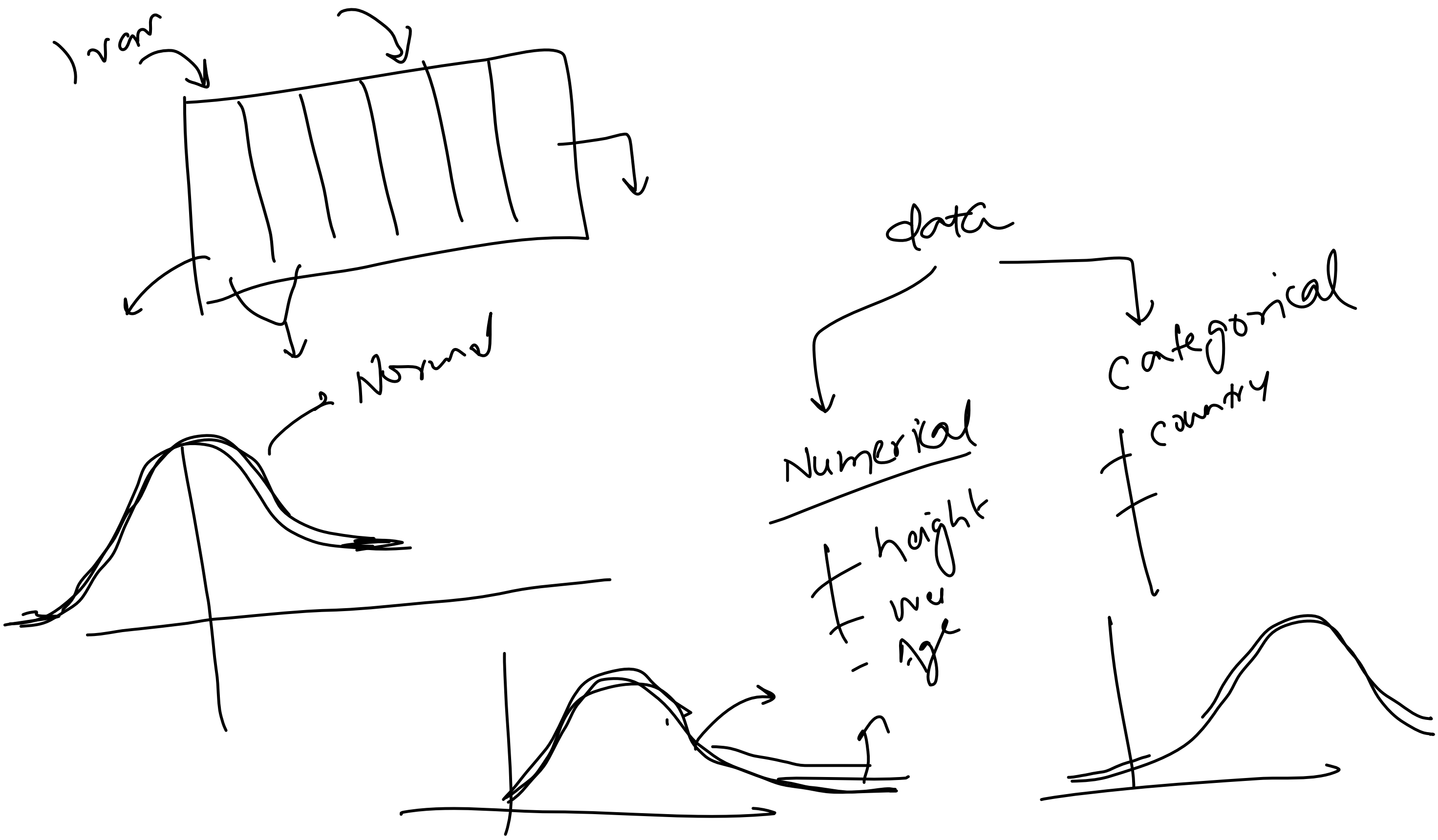
EDA {univariate}

EDA {Multivariate}

[Pandas Profiler]

Univariate Analysis

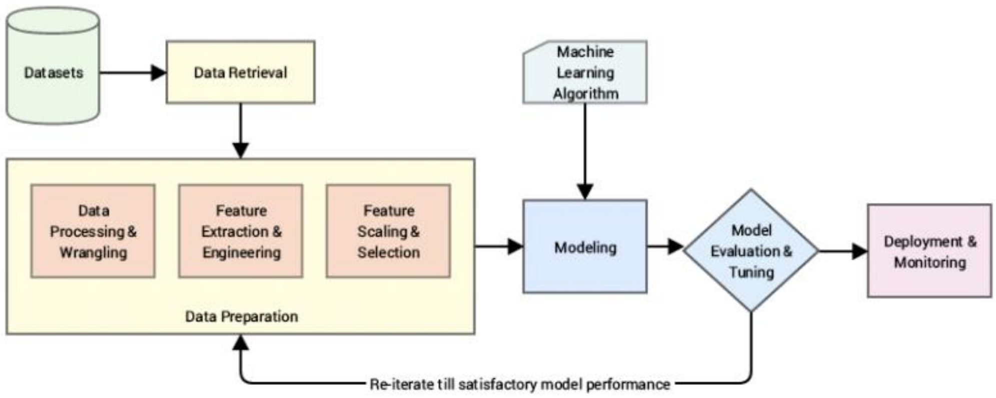
Sunday, April 4, 2021 5:39 PM

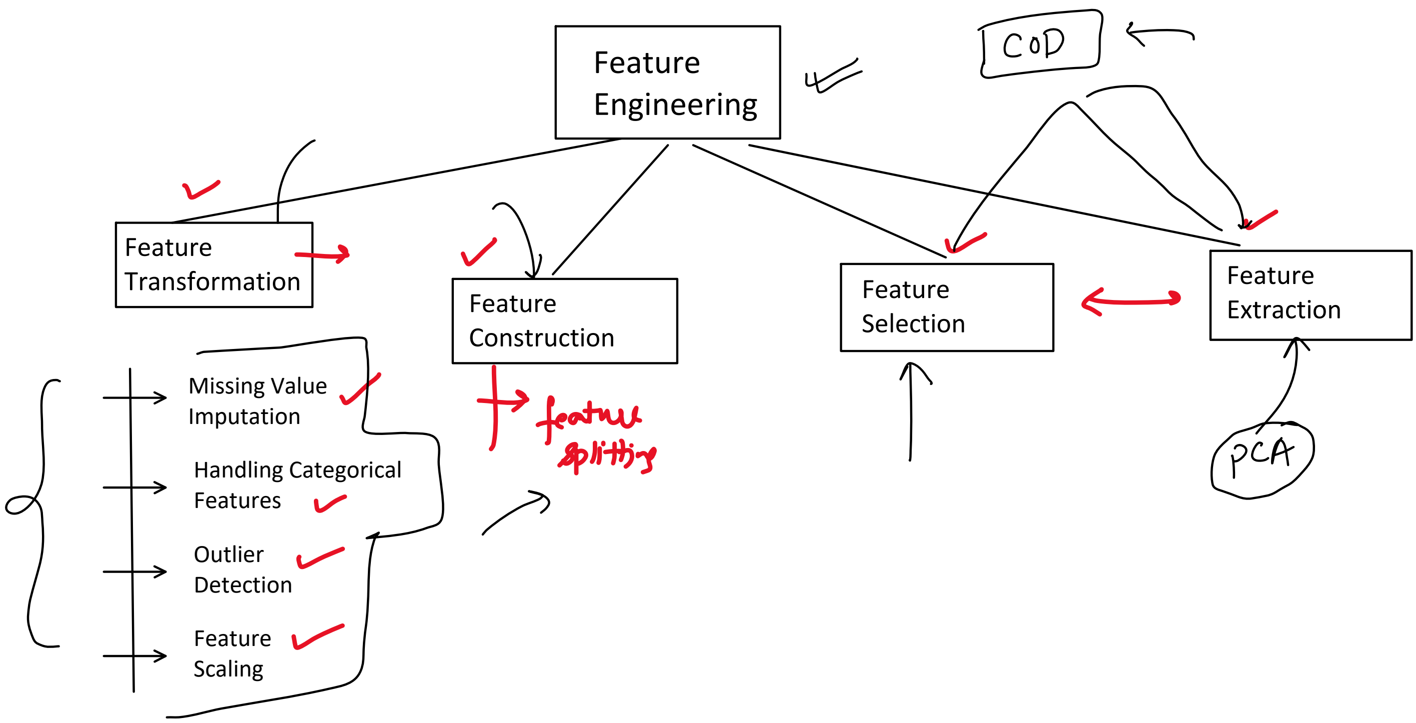


Feature Engineering

Thursday, April 8, 2021 6:02 PM

Feature engineering is the process of using domain knowledge to extract features from raw data. These features can be used to improve the performance of machine learning algorithms.





1.1 Missing Values Imputation

Thursday, April 8, 2021 7:05 PM

Average_Age = 26.0

ID	City	Age	Married ?
1	Lisbon	25	0
2	Berlin	25	1
3	Lisbon	30	1
4	Lisbon	30	1
5	Berlin	18	0
6	Lisbon	NaN	0
7	Berlin	30	1
8	Berlin	NaN	0
9	Berlin	25	1
10	Madrid	25	1

ID	City	Age	Married ?
1	Lisbon	25	0
2	Berlin	25	1
3	Lisbon	30	1
4	Lisbon	30	1
5	Berlin	18	0
6	Lisbon	26	0
7	Berlin	30	1
8	Berlin	26	0
9	Berlin	25	1
10	Madrid	25	1

1.2 Handling Categorical Values

Thursday, April 8, 2021 7:06 PM

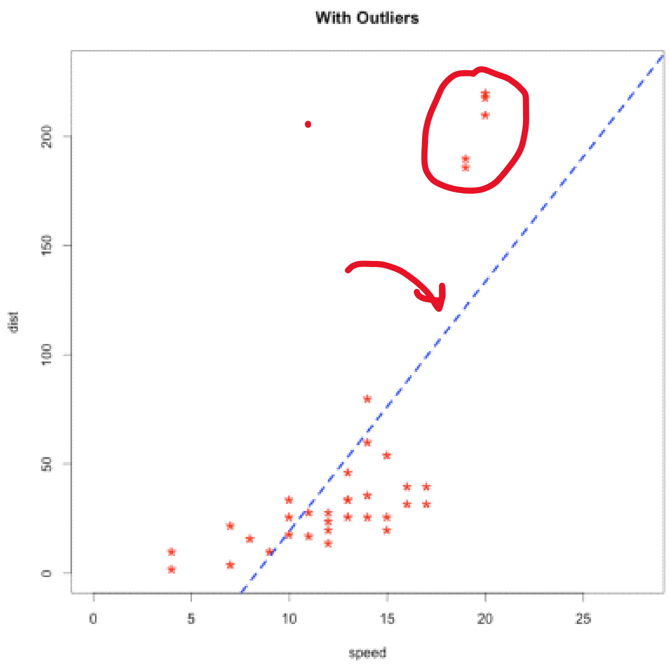
Index	Animal
0	Dog
1	Cat
2	Sheep
3	Horse
4	Lion

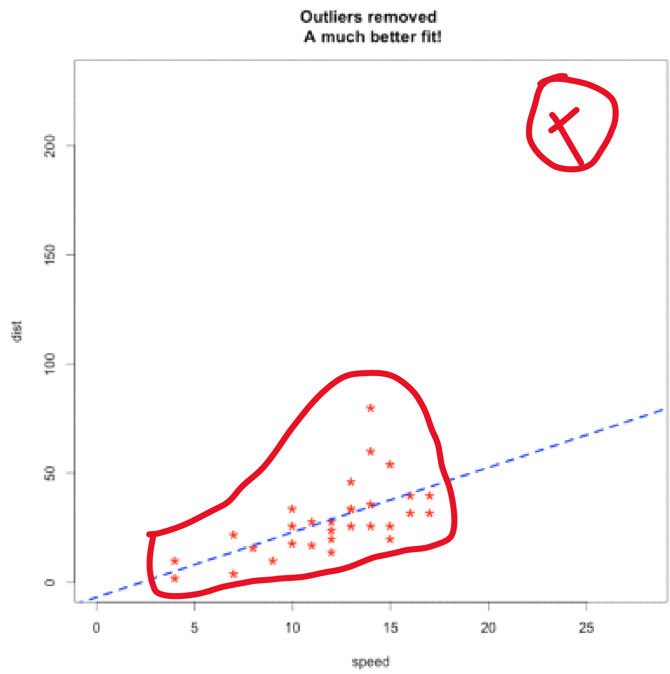
One-Hot code

Index	Dog	Cat	Sheep	Lion	Horse
0	1	0	0	0	0
1	0	1	0	0	0
2	0	0	1	0	0
3	0	0	0	0	1
4	0	0	0	1	0

1.3 Outlier Detection

Thursday, April 8, 2021 7:07 PM



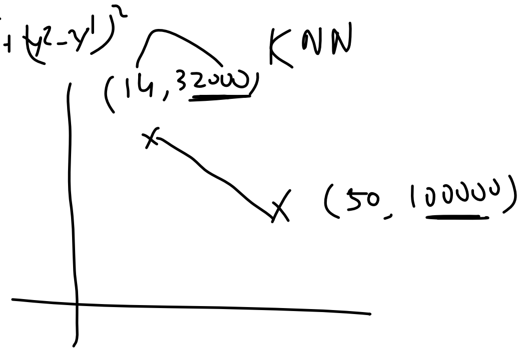


1.4 Feature Scaling

Thursday, April 8, 2021 7:08 PM

$$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

KNN



-1 to 1

	A	B	C	D
1	Country	Age	Salary	Purchased
2	France	44	72000	No
3	Spain	27	48000	Yes
4	Germany	30	54000	No
5	Spain	38	61000	No
6	Germany	40		Yes
7	France	35	58000	Yes
8	Spain		52000	No
9	France	48	79000	Yes
10	Germany	50	83000	No
11	France	37	67000	Yes

2. Feature Construction

Thursday, April 8, 2021 7:09 PM

P. Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, Mrs. John Bradley (Florence Brigg	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2.	7.925		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr. James	male		0	0	330877	8.4583		Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmin	female	27	0	2	347742	11.1333		S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708		C

family

1
4
1

0 - Alone
0-4 - Small
4< large

3. Feature Selection

Thursday, April 8, 2021 7:11 PM

MNIST

```
In [5]: train_df = pd.read_csv(f'{PATH}train.csv', header = 0)
```

```
In [19]: train_df
```

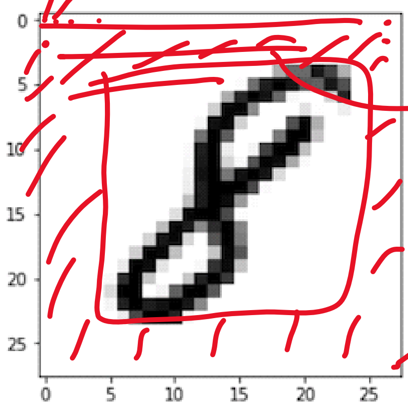
784 features

Out[19]:

	label	pixel0	pixel1	pixel2	pixel3	pixel4	pixel5	pixel6	pixel7	pixel8	...	pixel774
0	1	0	0	0	0	0	0	0	0	0	...	0
1	0	0	0	0	0	0	0	0	0	0	...	0
2	1	0	0	0	0	0	0	0	0	0	...	0
3	4	0	0	0	0	0	0	0	0	0	...	0

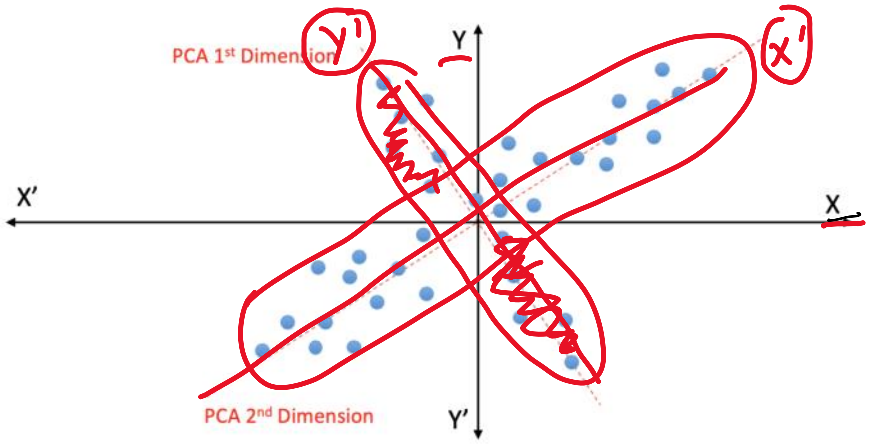
50000 imgs

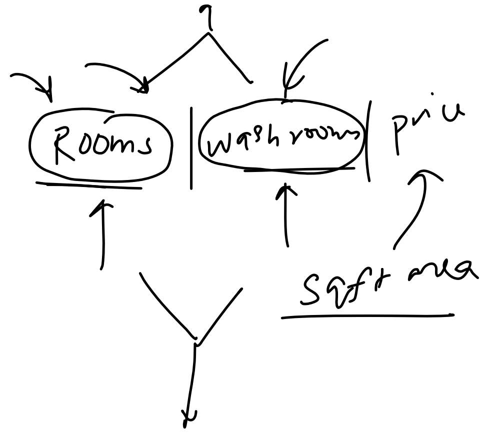
$\boxed{28 \times 28} = 784$



4. Feature Extraction

Thursday, April 8, 2021 7:13 PM

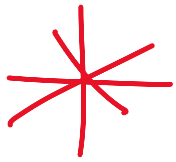




PCA

LDA

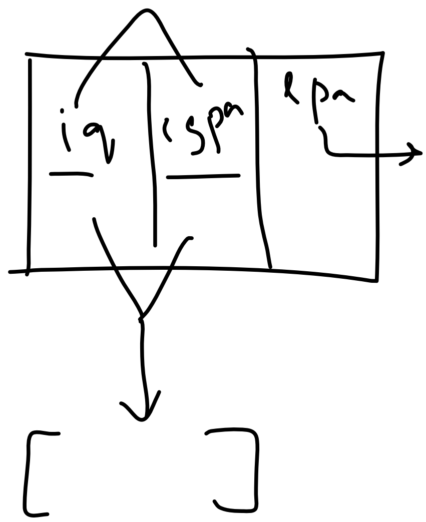
tsne

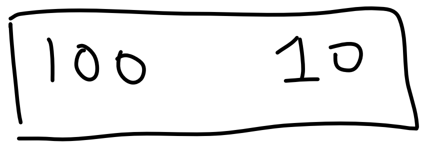


What is Feature Scaling?

Friday, April 9, 2021 4:21 PM

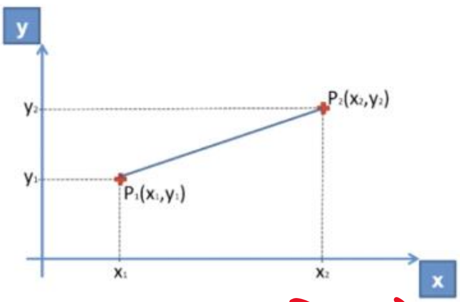
Feature Scaling is a technique to standardize the independent features present in the data in a fixed range





Why do we need Feature Scaling?

Friday, April 9, 2021 4:27 PM



Euclidean Distance between P_1 and $P_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Let x be the no. of Salary and y be the no. of Age

Example: x1&y1 are in row 2, x2&y2 are in row 9

(x2-x1)^2= (83000-48000)^2

$\boxed{=1225000000}$

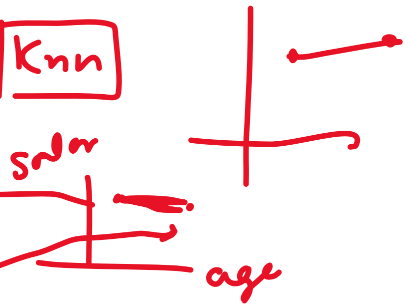
(y2-y1)^2=(50-27)^2

$\boxed{=529}$

Knn

age	salary	purchase
→ 50	83000	1
→ 27	48000	0

$\boxed{\text{Knn}}$



Types of Feature Scaling

Friday, April 9, 2021 4:27 PM

feature scaling

Standardization

Normalization

- Minmax scaler
- Robust scaler

Standardization - Intuition

Friday, April 9, 2021 4:27 PM

Also called as Z-score Normalization

$\frac{15-32}{10}$

X_i Age	salary		age'
27	x_1	2.3	
15	x_2	-1.2	
33	x_1		
63			
90		32	
05			
⋮			

500 values → 500 num

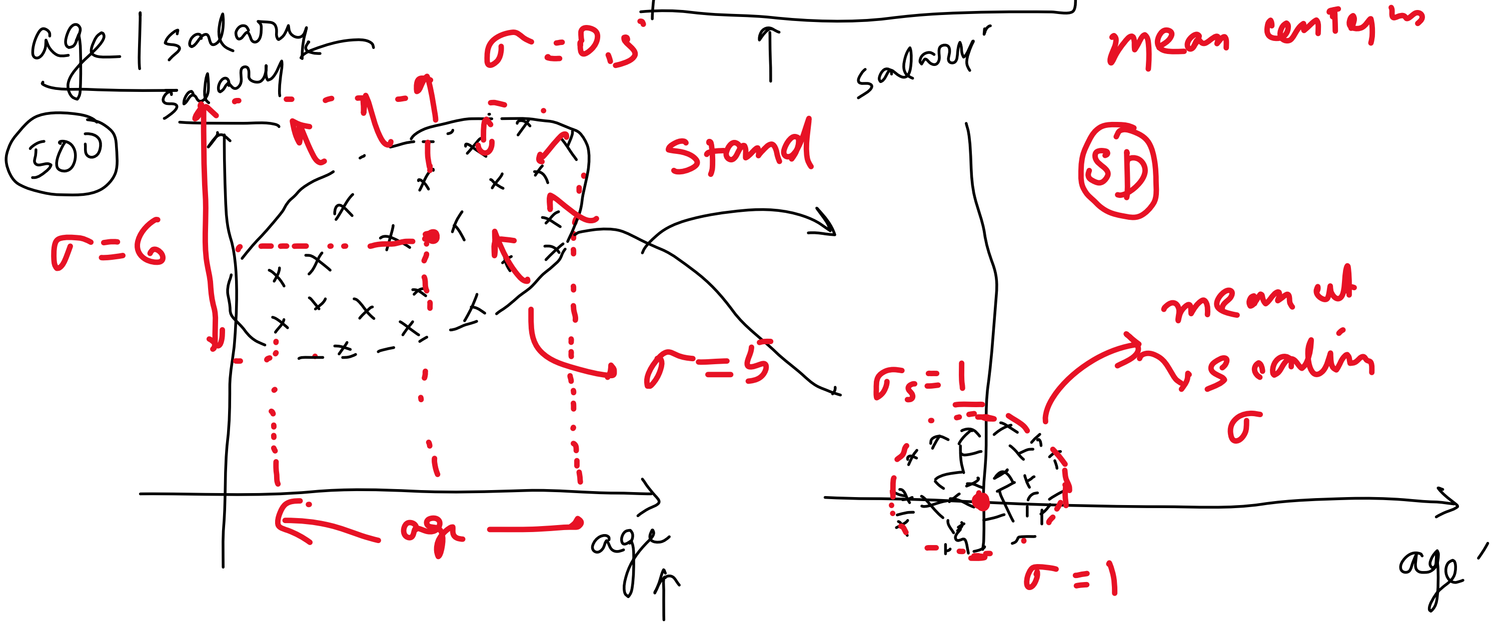
$$x_i' = \frac{X_i - \bar{X}}{\sigma}$$

$\bar{X}$ → mean

σ → SD → 10

$$x_i' = \frac{27-32}{10}$$

$$\mu = 0, \sigma = 1$$

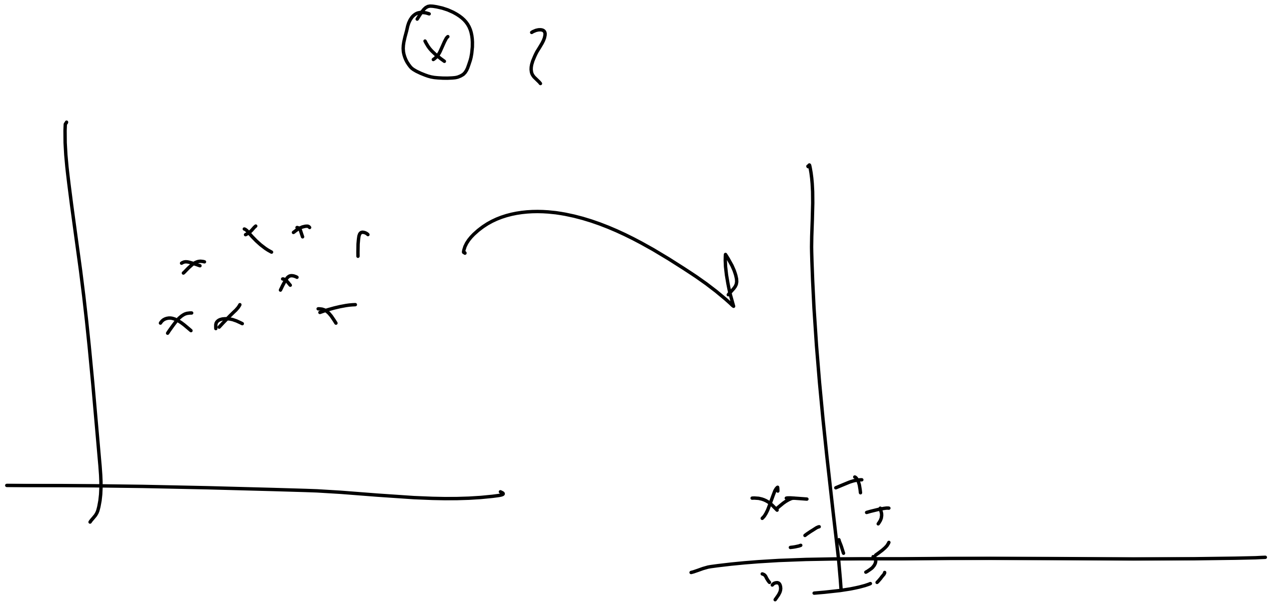


Example

Friday, April 9, 2021 4:28 PM

Impact of Outliers

Friday, April 9, 2021 4:28 PM



When to use Standardization?

Friday, April 9, 2021 4:28 PM

$\boxed{a} > \boxed{b}$ 15 ~~000~~ 7000

Algorithm(s)	Reason of applying feature scaling
1. K-Means	Use the Euclidean distance measure.
2. K-Nearest-Neighbours	Measure the distances between pairs of samples and these distances are influenced by the measurement units
3. Principal Component Analysis (PCA)	Try to get the feature with maximum variance
4. Artificial Neural Network	Apply Gradient Descent
5. Gradient Descent weights	Theta calculation becomes faster after feature scaling and the learning rate in the update equation of Stochastic Gradient Descent is the same for every parameter

Decision Tree / Random Forest

GB, XG boost

Normalization

Saturday, April 10, 2021 1:15 PM

Normalization is a technique often applied as part of data preparation for machine learning. The goal of normalization is to change the values of numeric columns in the dataset to use a common scale, without distorting differences in the ranges of values or losing information

- MinMaxScaling
- Mean normalization
- Max absolute
- Robust scaling

MinMaxScaling - Intuition

Saturday, April 10, 2021 1:16 PM

Weight

$\boxed{130}$ → Normalize

Min max scaling

$X_{min} = 0$ → $[0, 1]$

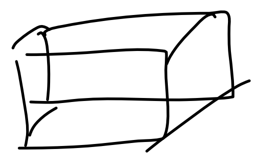
67
81
61
32
54
⋮

$$X_i' = \frac{X_i - \cancel{X_{min}}}{\cancel{X_{max}} - X_{min}} = 1 \quad X_{max}$$

min = 32
max = 130

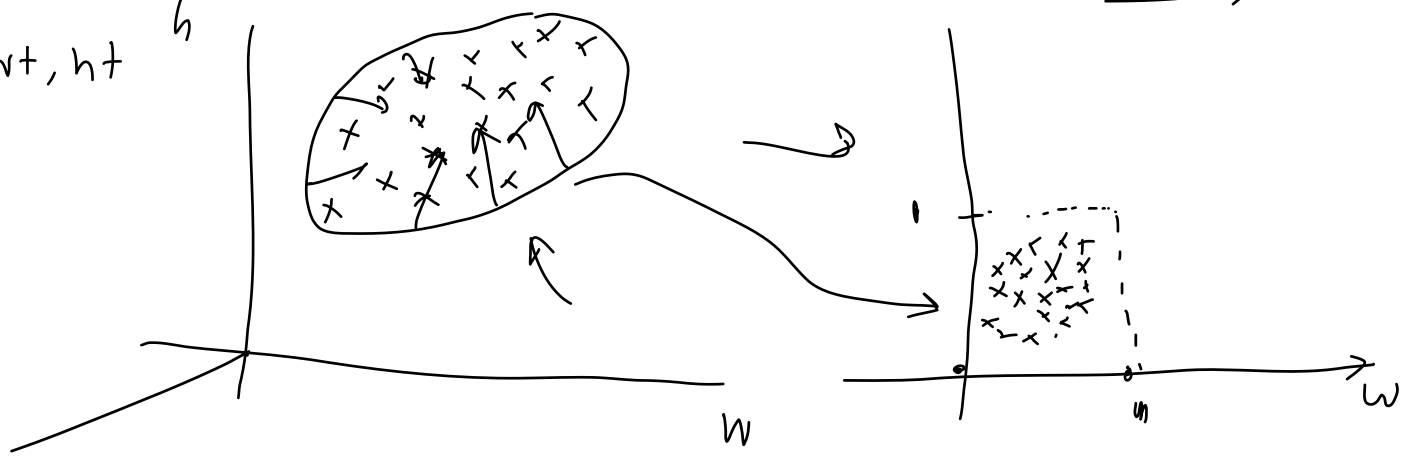
$$\frac{67 - 32}{130 - 32}$$

$$= \frac{130 - 32}{130 - 32} = 1$$



100
wts

wt, ht



Code Example

Saturday, April 10, 2021 1:17 PM

wine_data

Mean Normalization

Saturday, April 10, 2021 1:16 PM

wt

200 $\rightarrow$ normal

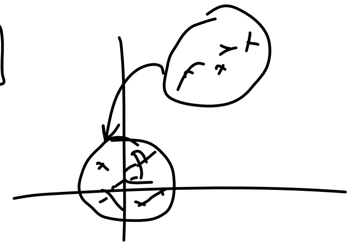
100

val < mean

-ve | $\boxed{+ve}$

mean center

$$x_i' = \frac{\boxed{X_i - X_{mean}}}{X_{max} - X_{min}} \quad [-1 \text{ to } 1]$$



sklearn $\rightarrow$ $\boxed{\text{centered data}}$ $\rightarrow$ Standardization

MaxAbsScaling

Saturday, April 10, 2021 1:17 PM

Wt
200
100
300

$$\boxed{x_i' = \frac{X_i}{|X_{max}|}}$$

MaxAbsScaler

sparse data

0's

Robust Scaling

Saturday, April 10, 2021 1:17 PM

wt

200

300

100

$$x_i' = \frac{x_i - x_{median}}{IQR \{ 75^{th} \text{ per} - 25^{th} \text{ per} \}}$$

<u>Robustscaler</u>

→ Robust to <u>Outliers</u>

Normalization Vs Standardization

Saturday, April 10, 2021 1:17 PM

1) Is feature scaling required?

2) MinMax

image → CNN → $\boxed{0 - 255}$ → minmax

Encoding Categorical Variables

Monday, April 12, 2021 3:53 PM

$\boxed{\text{Cat}}$

Feature Engg

Feanture transform

→ feature sca

→ $\boxed{\text{Encoding categorical}}$

Ordinal encoding

data

Num

Cat

Nominal

$\boxed{\text{Ordinal}}$

Excell
good
bad

$\boxed{\text{State}}$

One hot encoding

Label encoding

→ Ord

Ordinal Encoding

Monday, April 12, 2021 3:54 PM

Nominal OHE

ordinal encoding

Education

Education	
HS	0
UG	1
PG	2
PG	2
UG	1
HS	0
UG	2

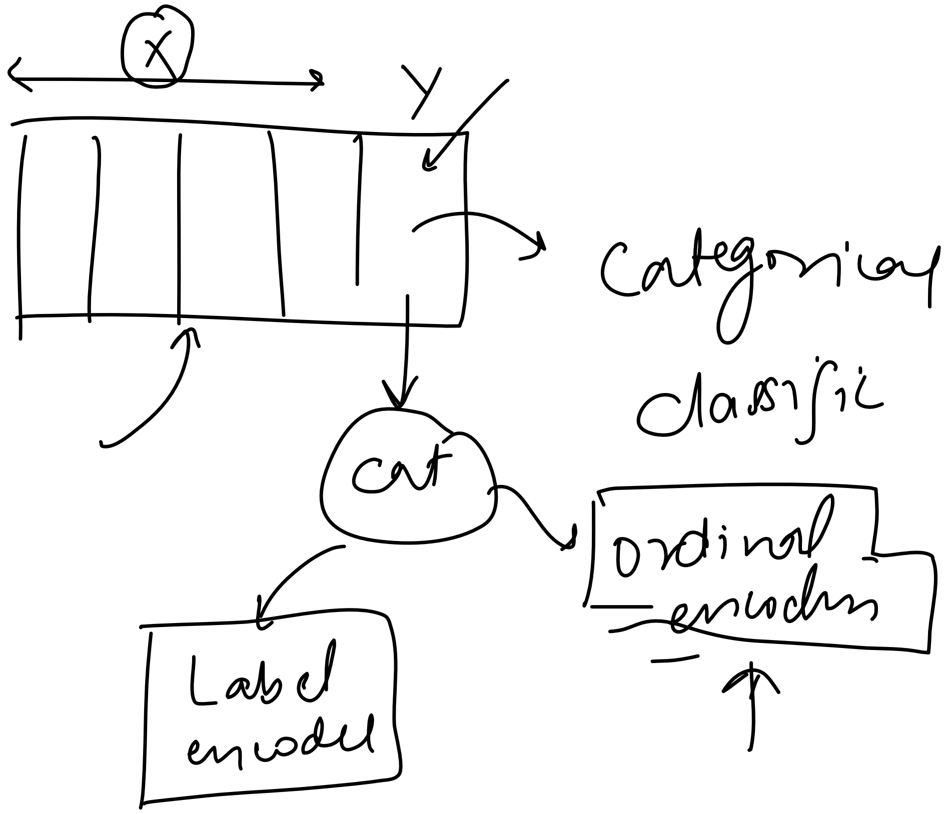
PG > UG > HS

$$\begin{Bmatrix} PG \rightarrow 2 \\ UG - 1 \\ HS \rightarrow 0 \end{Bmatrix}$$

Label Encoding

Monday, April 12, 2021 3:54 PM

Ordinal encoder



Example

Monday, April 12, 2021 3:54 PM

OneHotEncoding

Tuesday, April 13, 2021 12:24 PM

Color	Target
Yellow	0
Yellow	1
Blue	1
Yellow	1
Red	1
Yellow	0
Red	1
Red	0
Yellow	1
Blue	0

➜

Color_Y	Color_B	Color_R	Target
1	0	0	0
1	0	0	1
0	1	0	1
1	0	0	1
0	0	1	1
1	0	0	0
0	0	1	1
0	0	1	0
1	0	0	1
0	1	0	0

One Hot Encoding

[Y, B, R]
0 1 2

[1, 0, 0] → Yellow
[0, 1, 0] → blue
[0, 0, 1] → red

$\boxed{\text{50 diff}}$

Dummy Variable Trap

Tuesday, April 13, 2021 12:26 PM

Color	Target
Yellow	0
Yellow	1
Blue	1
Yellow	1
Red	1
Yellow	0
Red	1
Red	0
Yellow	1
Blue	0

Multic

n cat

(n-1) cols

~~Color_Y~~	Color_B	Color_R	Target
1	0	0	0
1	0	0	1
0	1	0	1
1	0	0	1
0	0	1	1
1	0	0	0
0	0	1	1
0	0	1	0
1	0	0	1
0	1	0	0

One Hot Encoding

n

n cols

(n-1) cols

Multicollinearity →

	B	R
Y -	0	0
B -	1	0
R -	0	1

$\boxed{\Sigma = 1}$ → linear model

Linear Reg

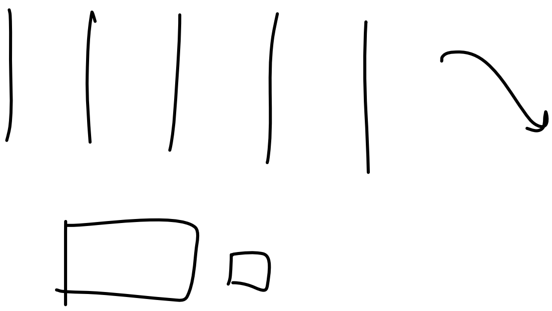
Logistic

OHE using most frequent variables

Tuesday, April 13, 2021 12:31 PM

brand → nominal → 40 brands

cat → OHE



frequent cat

→ others → 10

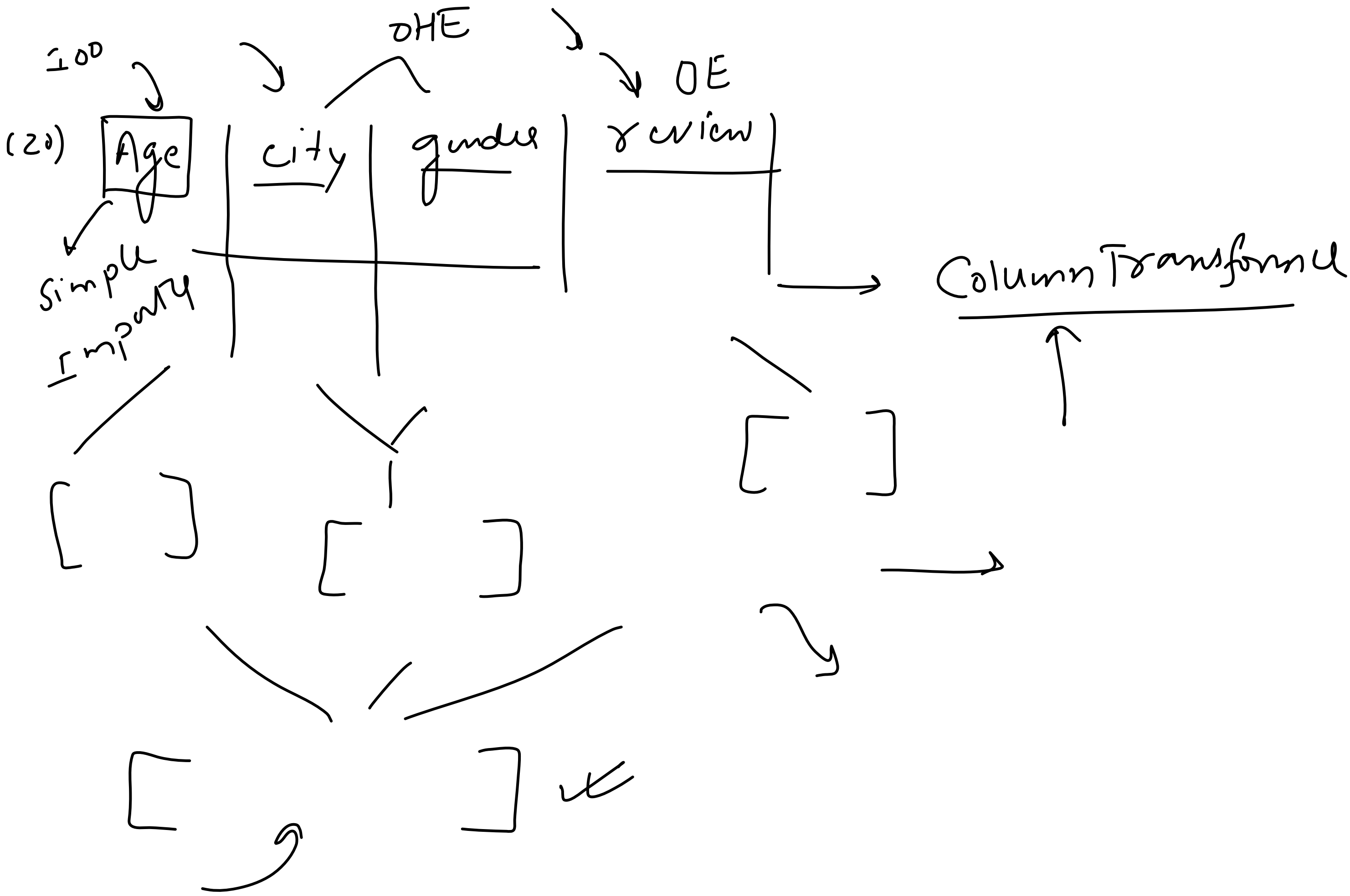
Example

Tuesday, April 13, 2021 12:26 PM

	brand	km_driven	fuel	owner	selling_price
0	Maruti	145500	Diesel	First Owner	450000
1	Skoda	120000	Diesel	Second Owner	370000
2	Honda	140000	Petrol	Third Owner	158000
3	Hyundai	127000	Diesel	First Owner	225000
4	Maruti	120000	Petrol	First Owner	130000

Column Transformer

Wednesday, April 14, 2021 5:16 PM



Internship Project Discussion

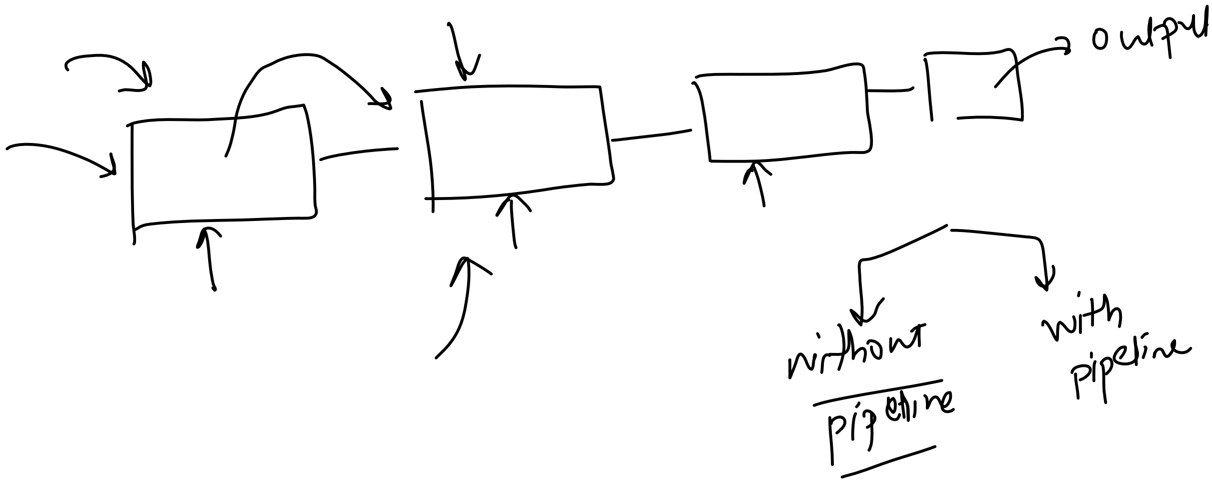
Wednesday, April 14, 2021 6:41 PM

Scikit Learn Pipelines

Thursday, April 15, 2021 5:59 PM

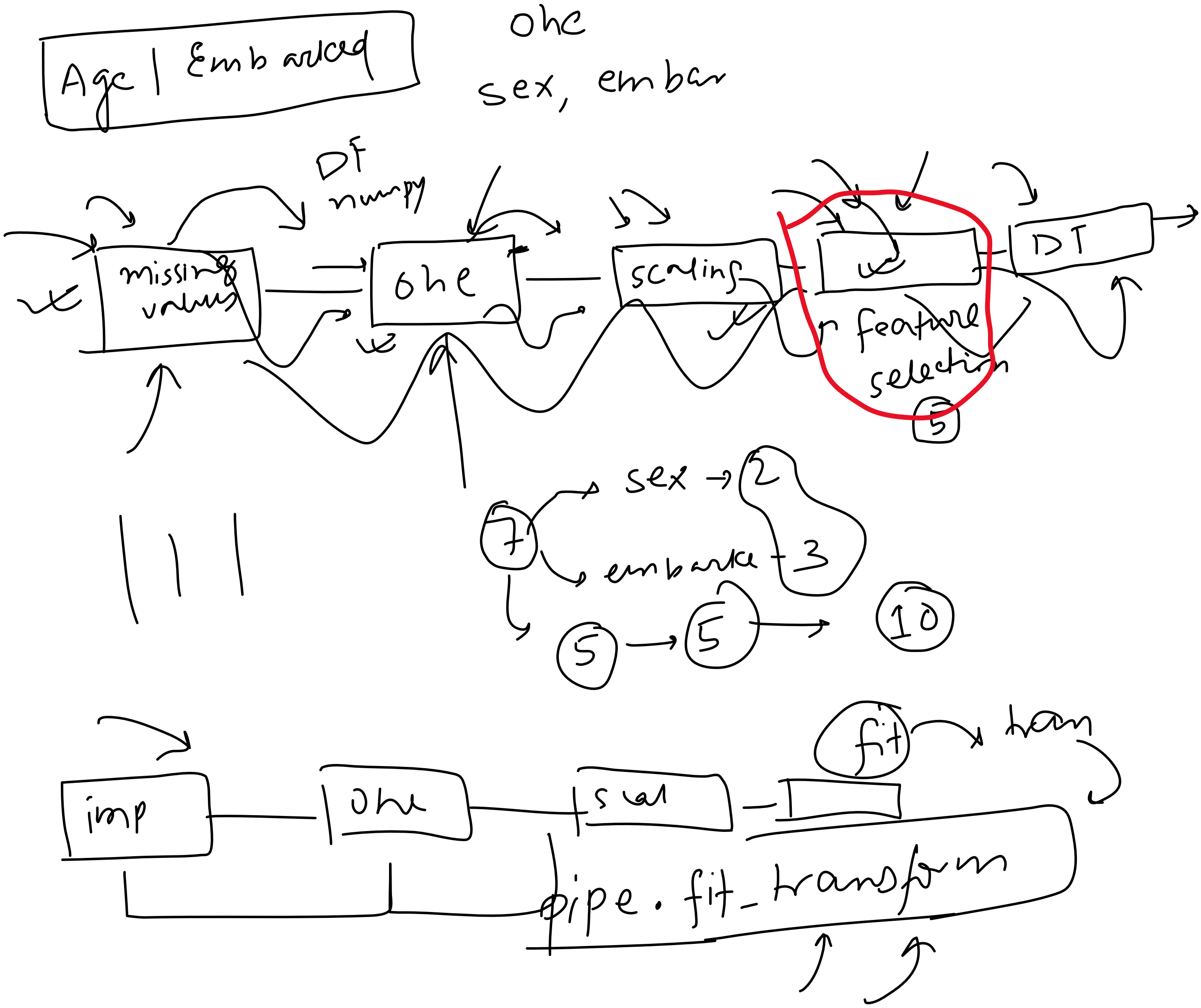
Pipelines chains together multiple steps so that the output of each step is used as input to the next step.

Pipelines makes it easy to apply the same preprocessing to train and test!



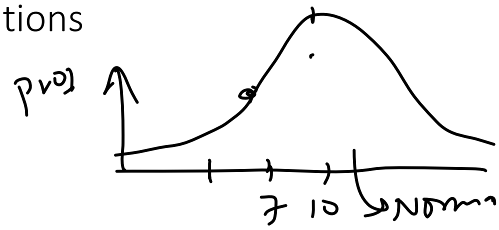
Strategy

Thursday, April 15, 2021 6:58 PM



Mathematical Transformations

Friday, April 16, 2021 5:21 PM

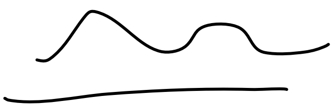


FE

↳ feature transformations

↳ Mathematical trans

Normal distribution



1) log trans
2) Reciprocal
3) power (sq / sqrt)

custom

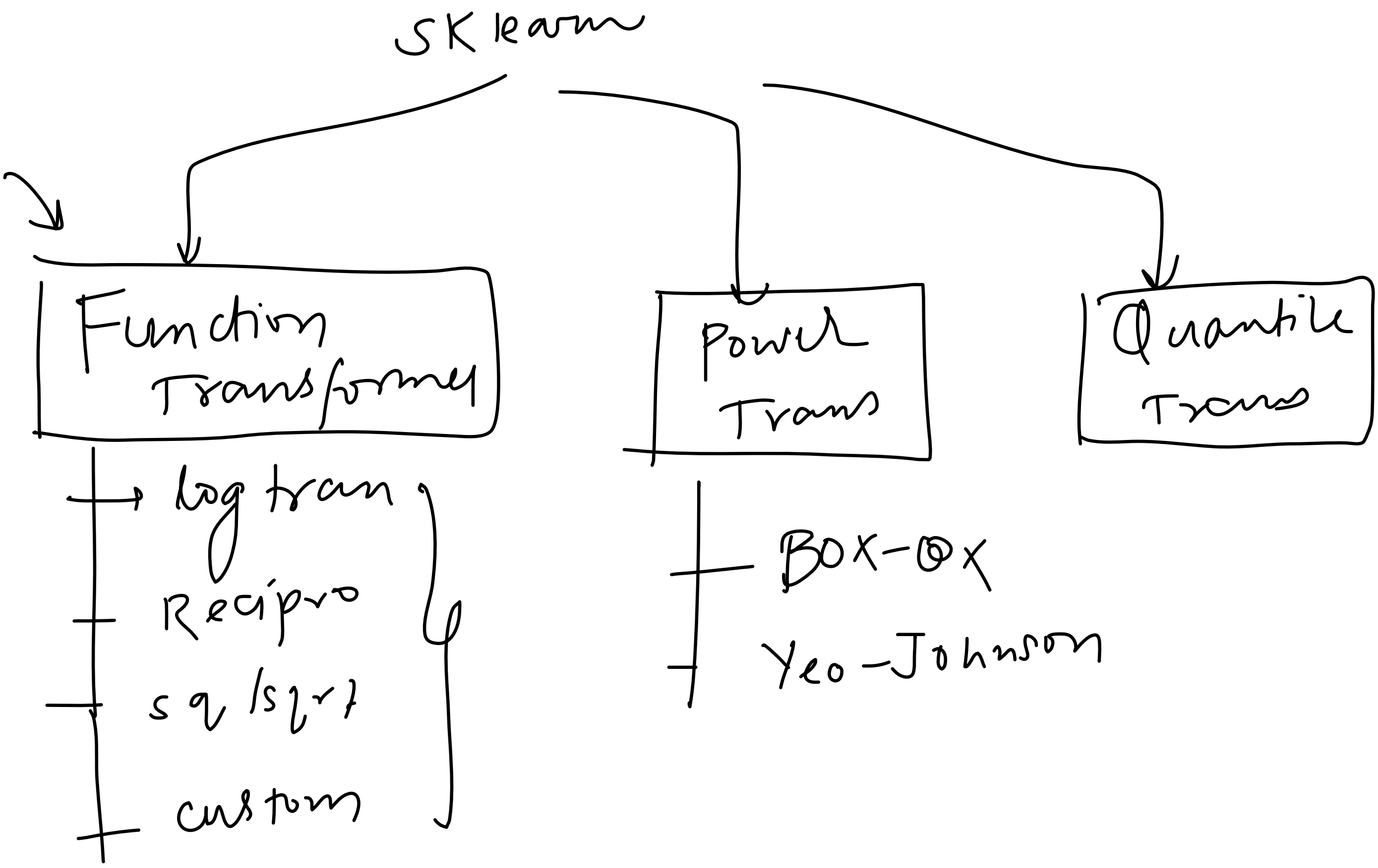
1) Box-Cox
2) Yeo-Johnson

Statistics

↳ ML → Linear Reg
Logistic

Function Transformer

Friday, April 16, 2021 5:22 PM

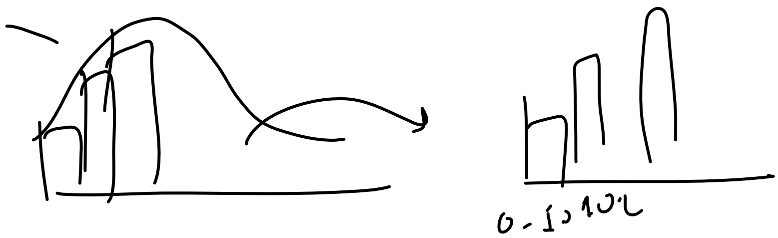


2. Discretization

Discretization is the process of transforming continuous variables into discrete variables by creating a set of contiguous intervals that span the range of the variable's values. Discretization is also called binning, where bin is an alternative name for interval.

Why use Discretization:

1. To handle Outliers
2. To improve the value spread



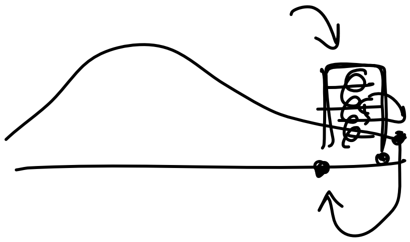
Age

23 42, 57 81 . . . 100

↓

0-10, 10-20, 20-30 . . .

5 6 10



3. Types of Discretization

Monday, April 19, 2021 3:56 PM

Binning

- Unsupervised
  - Equal width (uniform)
  - Equal frequency (quantile
  - Kmeans binning
- Supervised
  - Decision Tree Binning
- Custom Binning

4. Equal Width/Uniform Binning

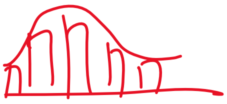
Monday, April 19, 2021 3:56 PM

1) Outliers
2) No change spread

Age

27, 32, 84, 56, . . .

$\boxed{\text{Bins} = 10}$



max 100

min 0

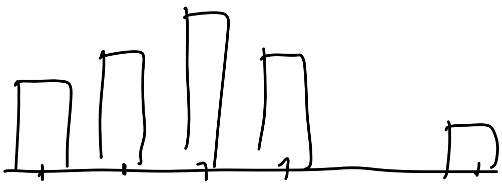
$$\frac{max - min}{bins} = \frac{100 - 0}{10} = 10$$

→ 10 bins

(0–10), (10 – 20), (20–30) . . . (90– 100)

1 2 3 4 5

5 16 17 1 1 5

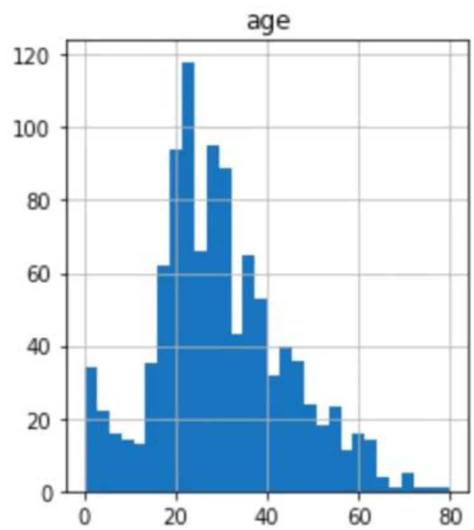


equal binning

	age	age_trf	age_labels
314	43.0	$\boxed{5.0}$	(40.21, 48.168]
523	44.0	5.0	(40.21, 48.168]
352	15.0	1.0	(8.378, 16.336]
534	30.0	3.0	(24.294, 32.252]
211	35.0	4.0	(32.252, 40.21]
530	2.0	0.0	(0.42, 8.378]
786	18.0	2.0	(16.336, 24.294]
827	1.0	0.0	(0.42, 8.378]
372	19.0	2.0	(16.336, 24.294]
518	36.0	4.0	(32.252, 40.21]

5. Equal Frequency/Quantile Binning

Monday, April 19, 2021 3:56 PM



Intervals = 10

Each interval contains 10% of total observations

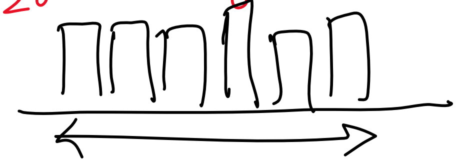
Intervals:
0-16; 16-20; 20-22; 22-25; …
50-74

1) Outliers
2) Value spread
spread

0 — ?
10th percen

0-16 | 16 — 20 | 20-22 —

10% 20% 30%

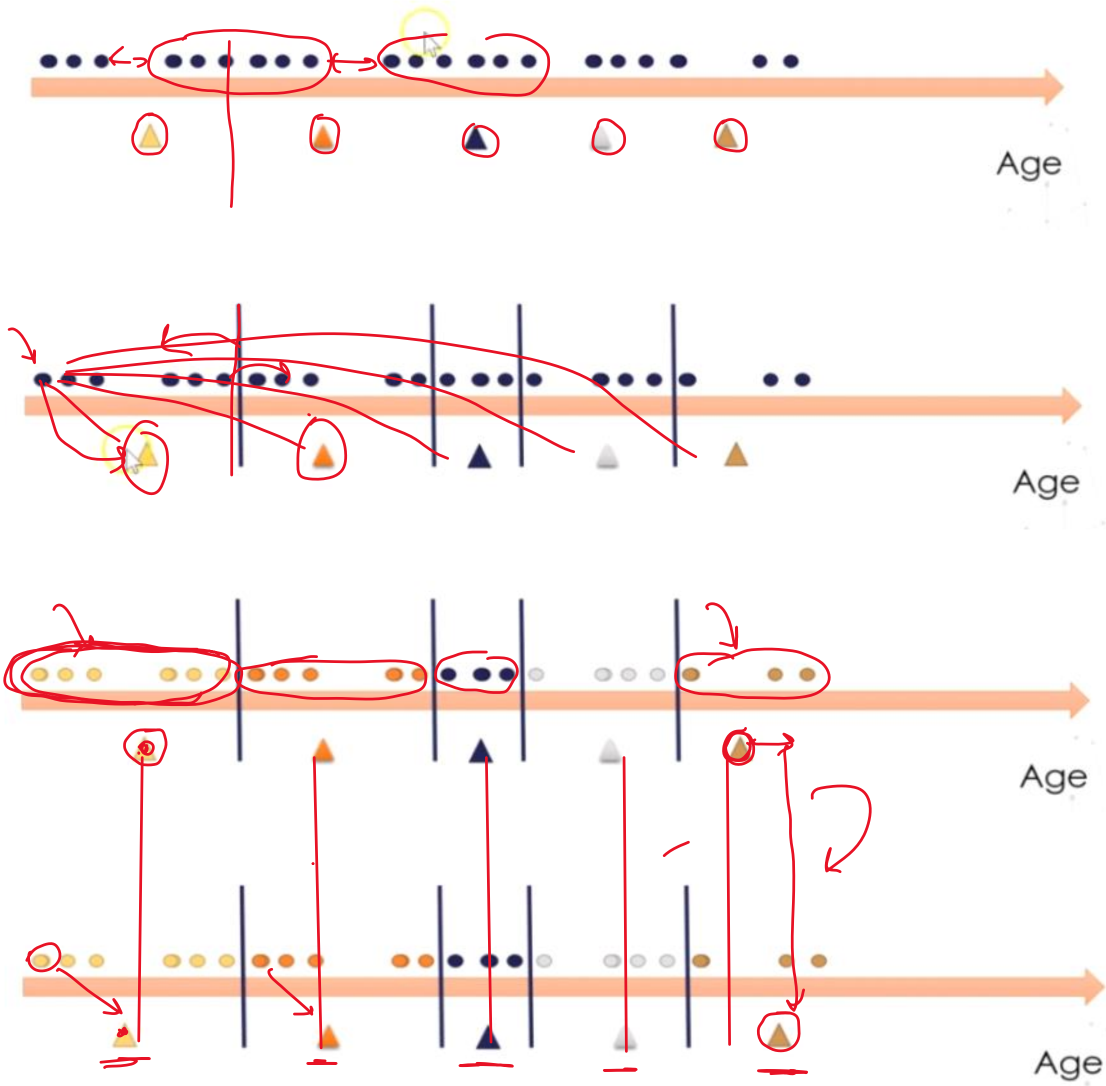


6. KMeans Binning

Monday, April 19, 2021 3:57 PM

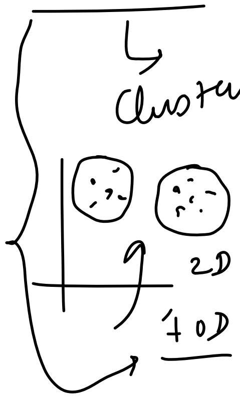
centroid

interval - 5



KMeans

↳ Clustering



7. Encoding the discretized variable

Monday, April 19, 2021 3:58 PM

SKlearn

→ KBinsDiscretizer( )

- bins = ?
- strategy
  - uniform
  - quantile
  - kmeans
- encoding
  - ordinal
  - onehotencoding

8. Example

Monday, April 19, 2021 3:58 PM

Titanic

9. Custom/Domain Based Binning

Monday, April 19, 2021 3:58 PM

[0 - 18] → Kids

(18 - 60) →

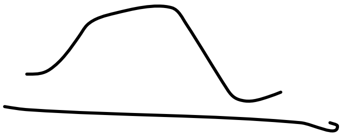
[60 - 80] →

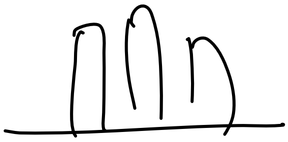
Sklearn

↳ Pandas

10. Binarization

Monday, April 19, 2021 4:17 PM





0, 1

image

0-255
color

127.5

0 1

Annual income

6L < 0

> 1

11. Example

Monday, April 19, 2021 4:17 PM

Binarizer

threshold

127.5

copy=True

false

salary → col

0

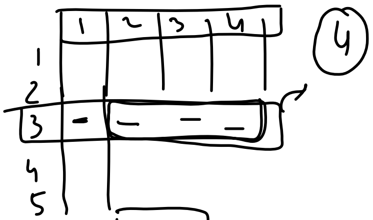
1

0

1

Handling Missing Data

Thursday, April 22, 2021 1:03 PM



CCA

Missing Values

Remove them

Impute

Simple Imputer

Univariate

num

- mean / median
- Random
- End of dis

Cat

- mode
- Missing

Multivariate

KNN imputer

Iterative imputer

MICE

→ 1 → Remove values

→ 2 → Simple Imp

3 KNNimputer

4 Iterative

5 Missing indicator

Complete Case Analysis

Thursday, April 22, 2021 12:55 PM

Complete-case analysis (CCA), also called "list-wise deletion" of cases, consists in **discarding** observations where values in **any** of the variables are missing.

row col

Complete Case Analysis means literally analyzing only those observations for which there is information in **all** of the variables in the dataset.

Assumption For CCA

Thursday, April 22, 2021 12:58 PM

1) Missing completely at random

1000, 4 → Age → 50 → 950, 4

50

MCAR

CCA

1000 → 950

Advantage/Disadvantage

Thursday, April 22, 2021 12:59 PM

Advantage

1. Easy to implement as no data manipulation required
2. Preserves variable distribution (if data is MCAR, then the distribution of the variables of the reduced dataset should match the distribution in the original dataset

Disadvantage

1. It can exclude a large fraction of the original dataset (if missing data is abundant)
2. Excluded observations could be informative for the analysis (if data is not missing at random)
3. When using our models in production, the model will not know how to handle missing data

When to use CCA?

Thursday, April 22, 2021 1:02 PM

1) MCAR ✓

2) $\boxed{5\%}$ <

$\boxed{95\%}$ remove cols

1000

~~950~~

950

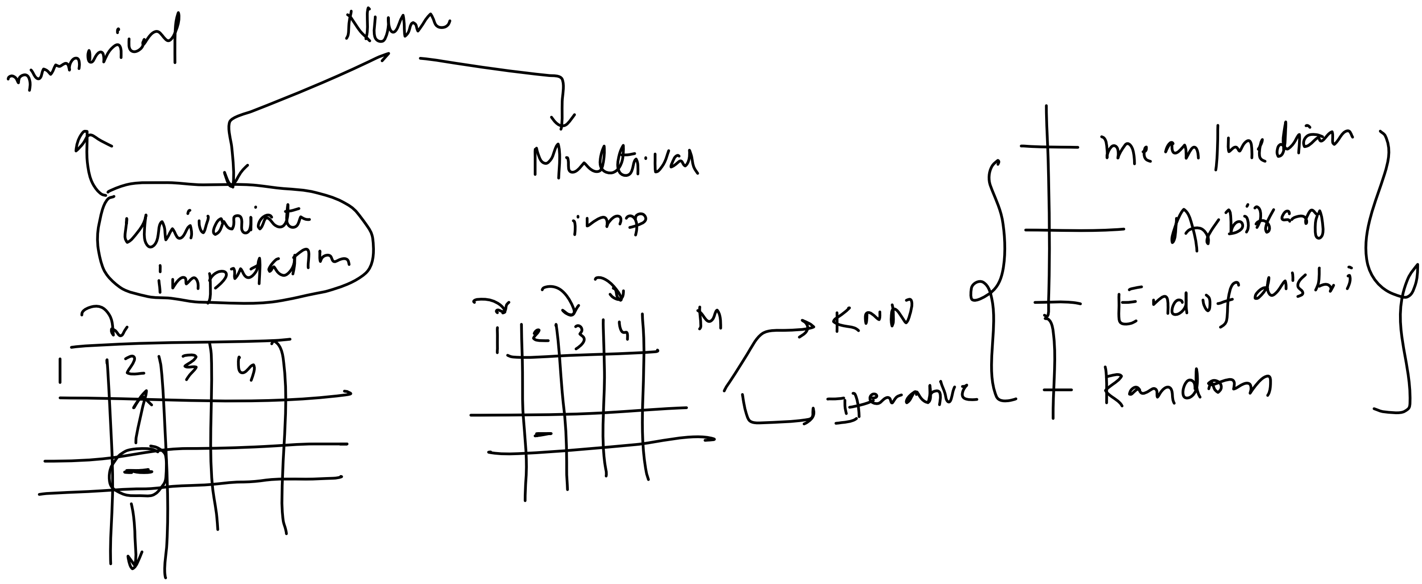
50

Example

Thursday, April 22, 2021 1:03 PM

1. Handling Missing Numerical Data

Friday, April 23, 2021 11:28 AM



2. Mean/Median Imputation

Friday, April 23, 2021 11:31 AM

Age

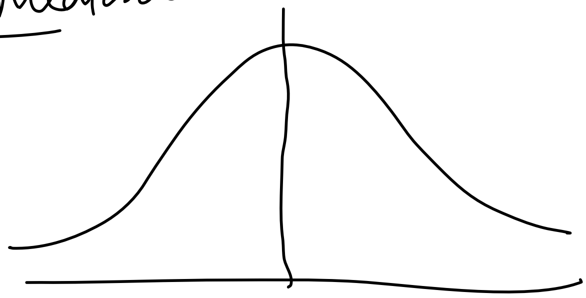
27
32
Na
27

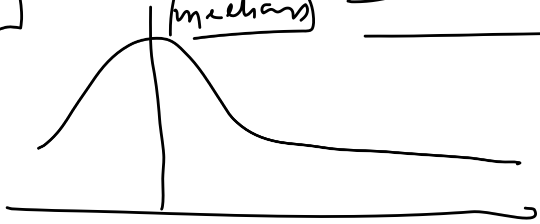
Mean/Median

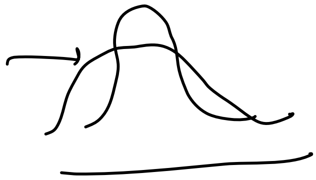
Mean

median

mean







Benefit

1) Simple
2) 5% >

Disadvantage

1) Distribution shape
2) Outliers
3) Covariance / corr changes

When to use

1) MCAR
2) 5% <

3. Arbitrary Value Imputation

Friday, April 23, 2021 11:32 AM

↳ Cat → NA → $\boxed{\text{Missing}}$

Num → −1
→ 0
→ 99,999
999 →

Data is not missing at Random

Age — 99
−1

Benefit
↳ easy to apply

Disadvantage

1) PDF distort
2) Variance
3) Covar changes

4. End of Distribution Imputation

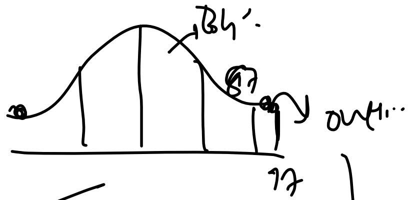
Friday, April 23, 2021 11:32 AM

99 → 999

-1

End of Distribution

99 → missing



999

Normality

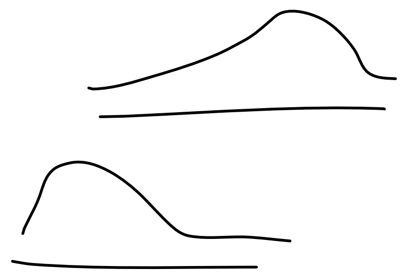
→ ($mean + 3\sigma$)

→ $mean - 3\sigma$

missingness

→ Benefit

easy



Skewed (IQR proximity)

$\boxed{Q_3 - Q_1}$

25th Q_1

75th Q_3

$Q_1 - 1.5\ IQR$

$Q_3 + 1.5\ IQR$

Disad

PDF

Covar

variance

NMAR

5. Random Sample Imputation

Friday, April 23, 2021 11:33 AM

6. Automatically select best imputation technique

Friday, April 23, 2021 11:32 AM

Handling Categorical Missing Data

Saturday, April 24, 2021 5:16 PM

most_frequent

Missing

Random

Most Frequent Value Imputation

Saturday, April 24, 2021 5:17 PM

MCAR

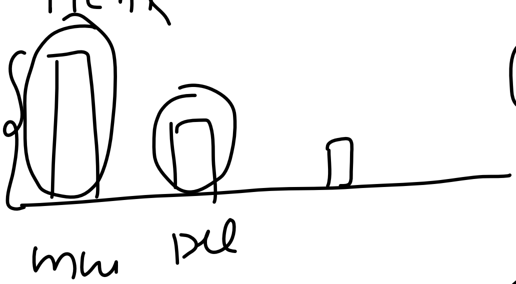
5%

$\boxed{\text{Mean / Median}}$

$\boxed{\text{Mode}}$ → frequency

Numerical

MCAR



City

Mum
Del
Kol
Na

→ Mum
Del
Kol

easy

Data

Missing Category Imputation

Saturday, April 24, 2021 5:17 PM

City

M

D

K

$\boxed{10\%<}$

$\boxed{99}$

$\boxed{\text{Missing}}$

Num

Arbitrary 99, 999, -1

Topics to be covered

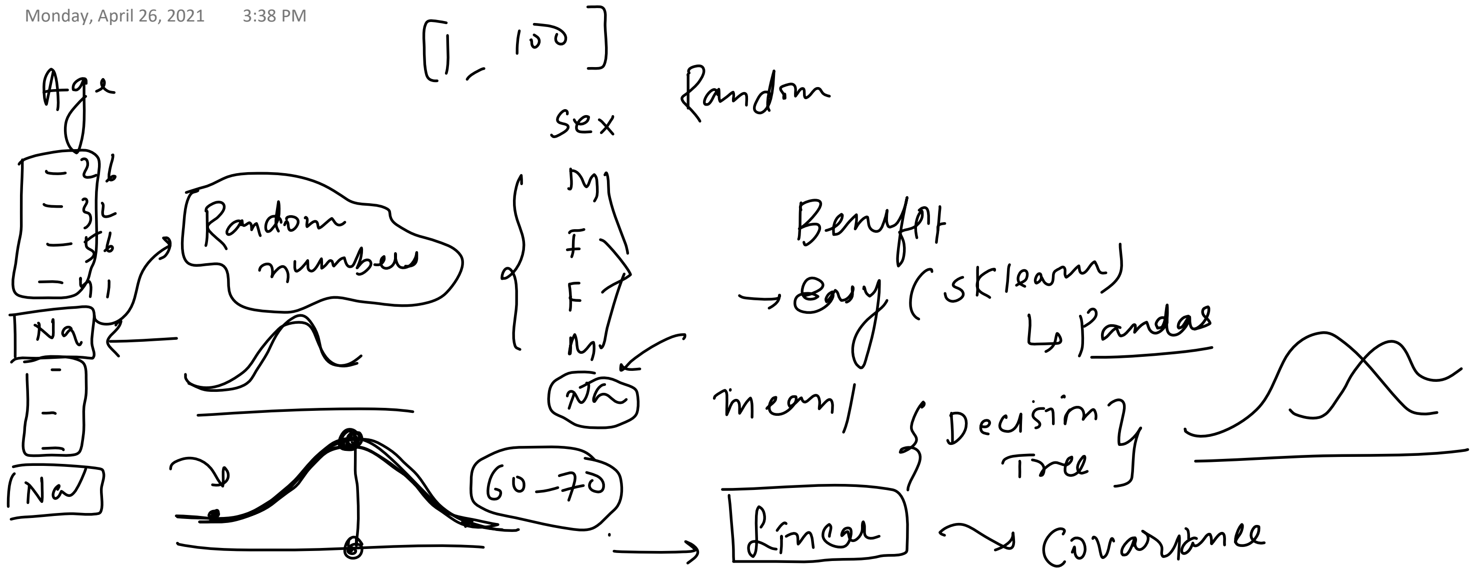
Monday, April 26, 2021 3:38 PM

6

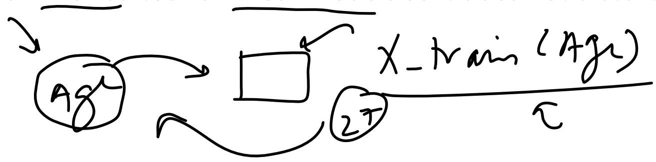
- CCA
- Numerical
- Categorical
- 4
  - Random
  - Missing Indicator
  - Automatic select
- KNN Imputer
- Iterative

Random Imputation

Monday, April 26, 2021 3:38 PM



1. Preserves the variance of the variable(But Why?)
2. Memory heavy for deployment, as we need to store the original training set to extract values from and replace the NA in coming observations
3. Well suited for linear models as it does not distort the distribution, regardless of the % of NA



Missing Indicator

Monday, April 26, 2021 3:41 PM

Age	Fare	Age_Na
27	32	F
41	35	F
Na	41	T
62	32	F

KDD

model

Automatically select value for Imputation

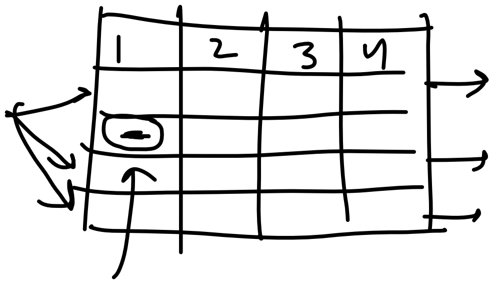
Monday, April 26, 2021 3:41 PM

$\boxed{\text{grid search cv}}$

KNN Imputer

Tuesday, April 27, 2021 12:19 PM

mean



KNN Imputer

Iterative Imputer

Working

Tuesday, April 27, 2021 7:44 AM

$\boxed{K=2}$

S NO	Feature 1	Feature 2	Feature 3	Feature 4
1 → 3	33	-----	67	21
2 →	----- 31.5	45	68	12
3 → 2	23	51	71	18
4 → 1	40	-----	81	------
5	35	60	79	------

Distance from Pt 1

$$d = \sqrt{\frac{3}{2}\left((68-67)^2 + (12-21)^2\right)}$$

$$= \sqrt{1.5(4+81)} = \sqrt{127.5} = \boxed{11.29}$$

Distance from pt 3

$$d = \sqrt{\frac{3}{3}\left((51-45)^2 + (71-68)^2 + (18-12)^2\right)}$$

$$= \sqrt{36+9+36} = \boxed{9}$$

Distance from pt 4

$$d = \sqrt{\frac{3}{1}\left((81-68)\right)^2}$$

$$d = \sqrt{3(9)} = \sqrt{27} = \boxed{5.19}$$

Distance from pt 5

$$d = \sqrt{\frac{3}{2}\left((60-45)^2 + (79-68)^2\right)}$$

$$= \sqrt{1.5(25+121)} = \sqrt{219} = \boxed{14.79}$$

Advantage & Disadvantage

1) More accurate ←

2) More no. of calculations →

3) Prediction → $\boxed{X\_train}$ →

Tuesday, April 27, 2021 12:44 PM

$K = 2$

1 →	70			
2 →				
3 →	–			
4 →	50			

$$\boxed{\frac{70+50}{2}} \rightarrow 60$$

3

1   4

10   50

$$\frac{1}{dis}$$

$$\frac{\frac{1}{10} \times 70 + \frac{1}{50} \times 50}{2}$$

$$\frac{7+1}{2} = 4$$

distance

Iterative Imputer/MICE

Wednesday, April 28, 2021 11:40 AM

MICE stands for Multivariate Imputation by Chained Equations

Assumptions

- MCAR
- MAR
- MNAR

Advantages & Disadvantage

- accurate
- slow
- memory

How it works?

Wednesday, April 28, 2021 11:41 AM

1. Actual Dataset

	R&D Spend	Administration	Marketing Spend	Profit
21	8.0	15.0	30.0	11.0
37	4.0	5.0	20.0	9.0
2	15.0	10.0	41.0	19.0
14	12.0	16.0	26.0	13.0
44	2.0	15.0	3.0	7.0

2. Removing the target column

	R&D Spend	Administration	Marketing Spend
21	8.0	15.0	30.0
37	4.0	5.0	20.0
2	15.0	10.0	41.0
14	12.0	16.0	26.0
44	2.0	15.0	3.0

3. Introduced some fake nan values

	R&D Spend	Administration	Marketing Spend
21	8.0	15.0	30.0
37	NaN	5.0	20.0
2	15.0	10.0	41.0
14	12.0	NaN	26.0
44	2.0	15.0	NaN

Step 1 - Fill all the NaN values with mean of respective cols

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	9.25	5.00	20.00
2	15.00	10.00	41.00
14	12.00	11.25	26.00
44	2.00	15.00	29.25

Step 2 - Remove all col1 missing values

Step 3 - Predict the missing values of col1 using other cols

	R&D Spend	Administration	Marketing Spend
21	8.0	15.00	30.00
37	NaN	5.00	20.00
2	15.0	10.00	41.00
14	12.0	11.25	26.00
44	2.0	15.00	29.25

y x

	Administration	Marketing Spend
21	15.00	30.00
2	10.00	41.00
14	11.25	26.00
44	15.00	29.25

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	23.14	5.00	20.00
2	15.00	10.00	41.00
14	12.00	11.25	26.00
44	2.00	15.00	29.25

Step 4 - Remove all col2 missing values

Step 5 - Predict the missing values of col2 using other cols

	R&D Spend	Administration	Marketing Spend
21	8.00	15.0	30.00
37	23.14	5.0	20.00
2	15.00	10.0	41.00
14	12.00	NaN	26.00
44	2.00	15.0	29.25

x y

	R&D Spend	Marketing Spend
21	8.00	30.00
37	23.14	20.00
2	15.00	41.00
44	2.00	29.25

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	23.14	5.00	20.00
2	15.00	10.00	41.00
14	12.00	11.06	26.00
44	2.00	15.00	29.25

Step 6 - Remove all col3 values

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.0
37	23.14	5.00	20.0
2	15.00	10.00	41.0
14	12.00	11.06	26.0
44	2.00	15.00	NaN

x y x_test

	R&D Spend	Administration
21	8.00	15.00
37	23.14	5.00
2	15.00	10.00
14	12.00	11.06

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	23.14	5.00	20.00
2	15.00	10.00	41.00
14	12.00	11.06	26.00
44	2.00	15.00	31.56

Stage 1

Iteration 0

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	9.25	5.00	20.00
2	15.00	10.00	41.00
14	12.00	11.25	26.00
44	2.00	15.00	29.25

Iteration 1

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	[redacted]	5.00	20.00
2	15.00	10.00	41.00
14	12.00	~~11.06~~	26.00
44	2.00	15.00	~~31.56~~

Difference

	R&D Spend	Administration	Marketing Spend
21	0.00	0.00	0.00
37	13.89	0.00	0.00
2	0.00	0.00	0.00
14	0.00	-0.19	0.00
44	0.00	0.00	2.31

0

Iteration 1

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	23.14	5.00	20.00
2	15.00	10.00	41.00
14	12.00	11.06	26.00
44	2.00	15.00	31.56

Iteration 2

	R&D Spend	Administration	Marketing Spend
21	8.00	15.00	30.00
37	23.78	5.00	20.00
2	15.00	10.00	41.00
14	12.00	11.22	26.00
44	2.00	15.00	31.56

Difference

	R&D Spend	Administration	Marketing Spend
21	0.00	0.00	0.0
37	0.64	0.00	0.0
2	0.00	0.00	0.0
14	0.00	0.16	0.0
44	0.00	0.00	0.0

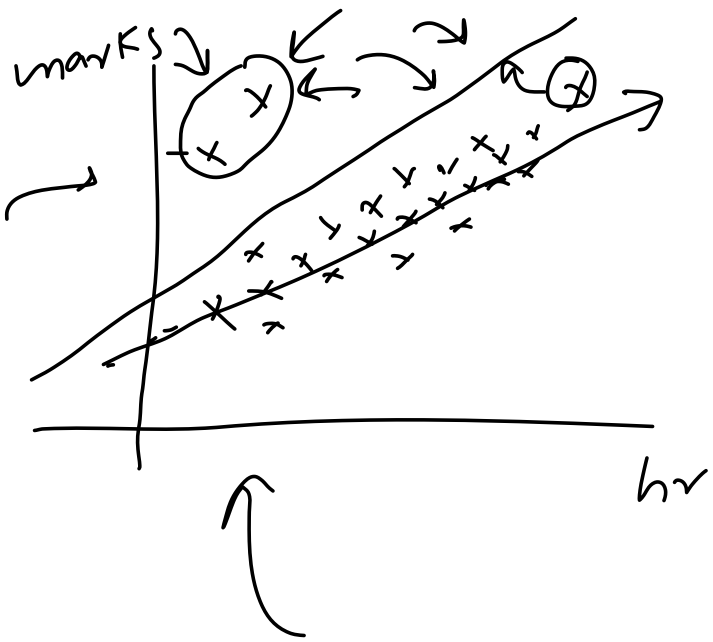
Iteration 2

Iteration 3

Difference

What are Outliers?

Thursday, April 29, 2021 3:25 PM



hr	marks

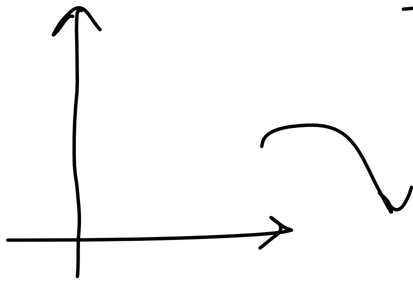
iq

When is outlier dangerous?

Thursday, April 29, 2021 3:26 PM

Age

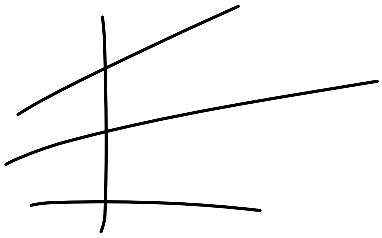
$\boxed{300}$



Anamoly detection → outliers

Effect of Outliers on ML algorithms

Thursday, April 29, 2021 3:27 PM

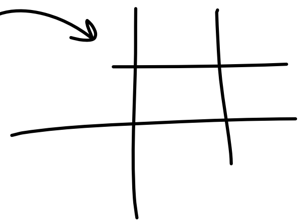


Linear Reg
Logistic Reg

Adaboost
Deep Learning

weights

Tree based



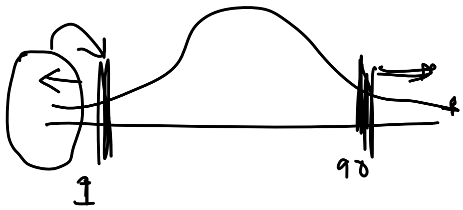
How to treat Outliers?

Thursday, April 29, 2021 3:27 PM

Trimming

- thin
- fast

Capping



- missing value
- Discretization

Numerical

90 100

90–100

How to detect Outliers?

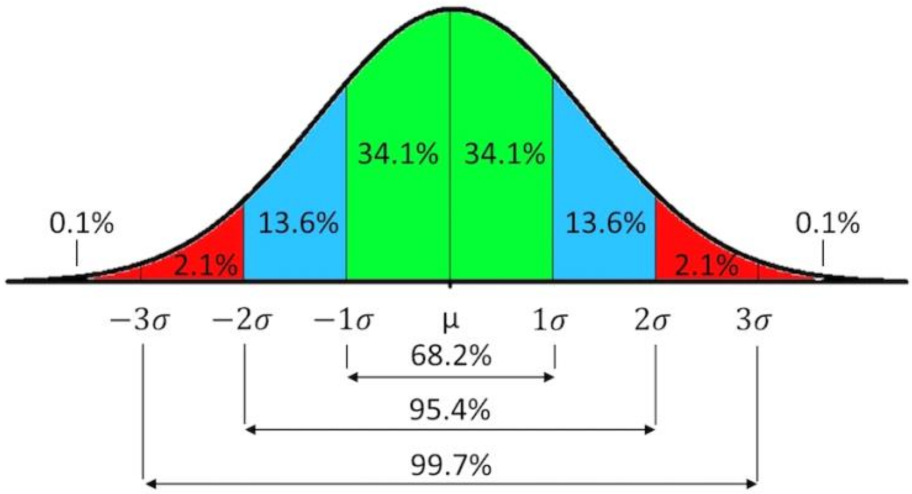
Thursday, April 29, 2021 3:27 PM

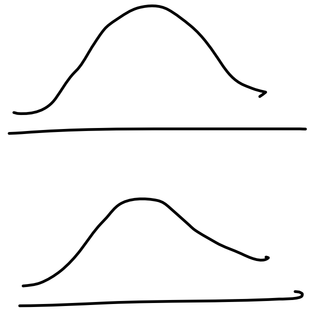
$(\mu + 3\sigma) >$

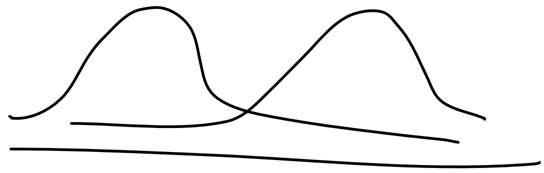
$(\mu - 3\sigma) <$

No

1. Normal Distribution



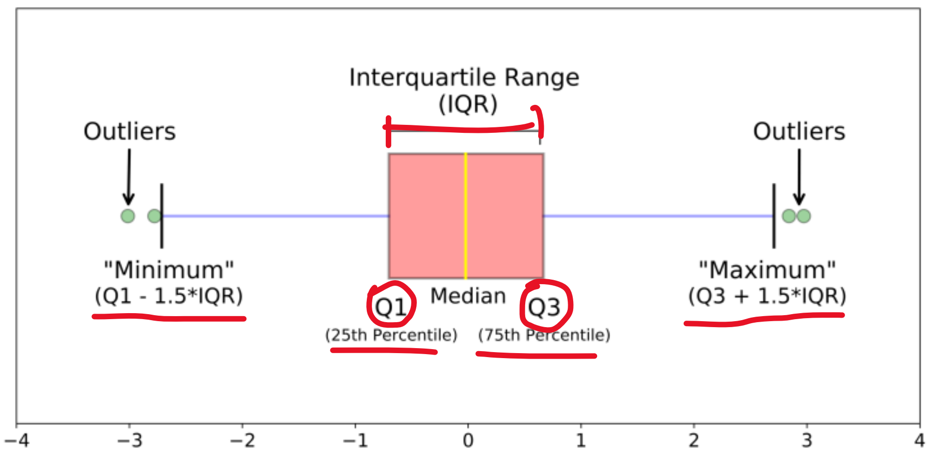




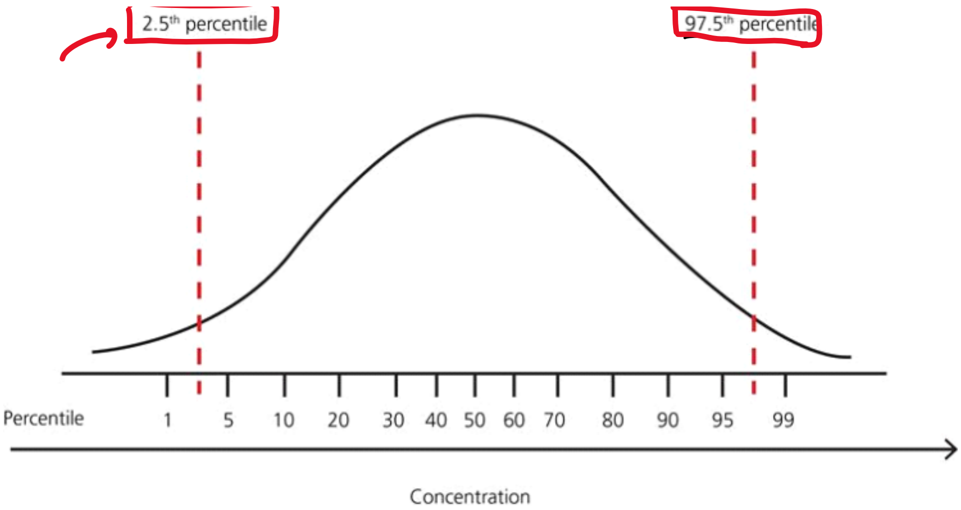
2. Skewed Distribution

min = $Q1 - 1.5 IQR$

$Q_3 + 1.5 IQR$



3. Other Distributions


2.5th percentile
97.5th percentile
Percentile
1
5
10
20
30
40
50
60
70
80
90
95
99
Concentration

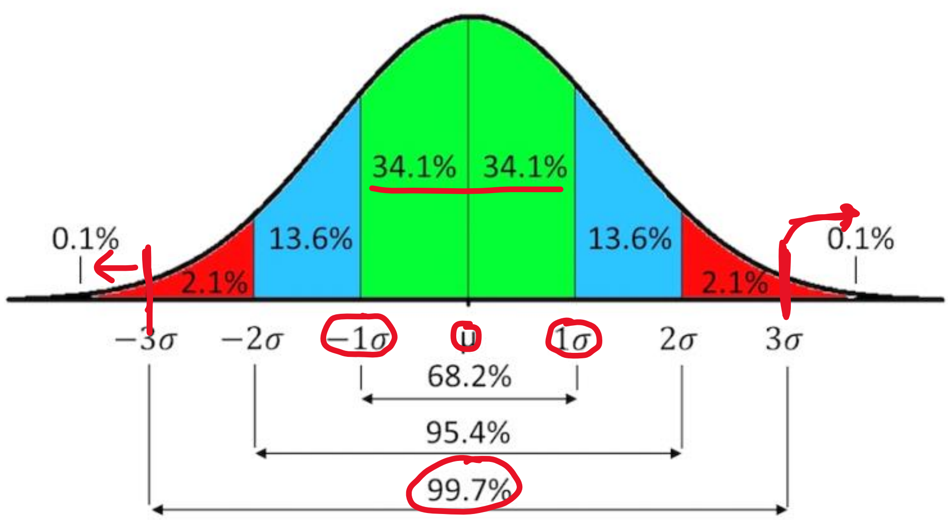
Techniques for Outlier Detection and Removal

Thursday, April 29, 2021 3:35 PM

- Z score treatment
- IQR based filtering
- Percentile
- Winsorization

Outlier removal using Z score

Friday, April 30, 2021 1:33 PM



~~μ + 1σ~~

$\left\{ \begin{matrix} \mu + \sigma \\ \mu - \sigma \end{matrix} \right\}$ 68%

$\left\{ \begin{matrix} \boxed{\mu + 2\sigma} \\ \mu - 2\sigma \end{matrix} \right\}$ 95%

Z score

$$\boxed{x_i' = \frac{x_i - \mu}{\sigma}}$$

3

1000

5

995

Age

27

32

$\boxed{-3 \text{ to } 3}$

Outlier Treatment

$\boxed{\text{Trimming}}$

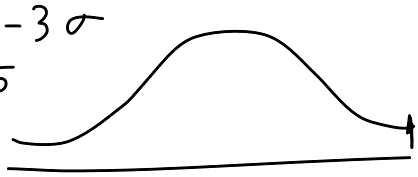
μ + 3σ

80

μ − 3σ

5

$\boxed{\text{Capping}}$



5

3 -3

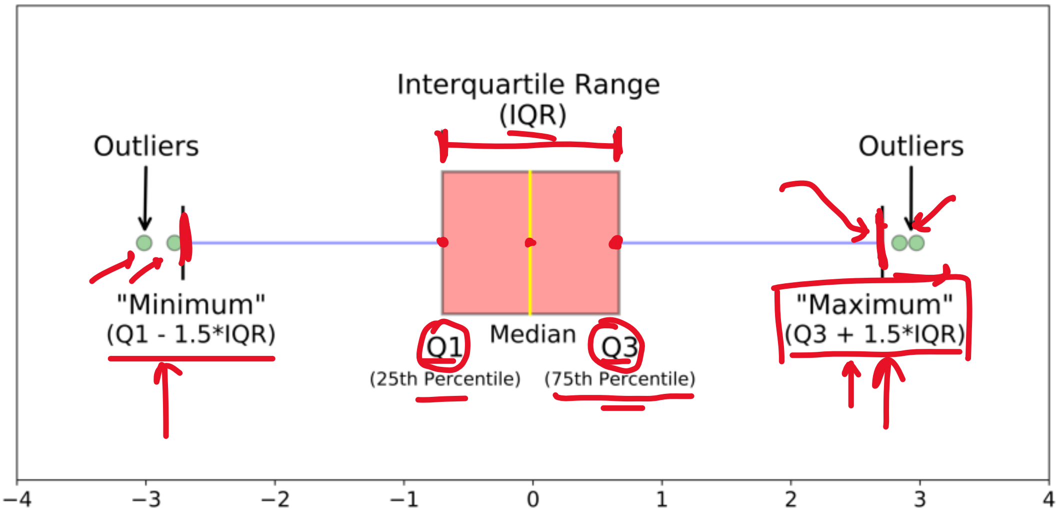
85, 3, 90

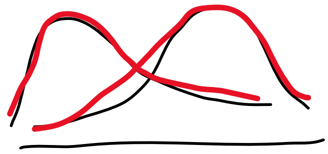
$\boxed{80 \quad 5 \quad 80}$

Outlier Detection using IQR

Friday, April 30, 2021 3:48 PM

[Boxplot
IQR]





percentiles

25
50 Median
75
100

Age
72
89
15
16

100% → 89 → max

0% → min 16

25th { ±IQR Proximity Rule }

Curse of Dimensionality

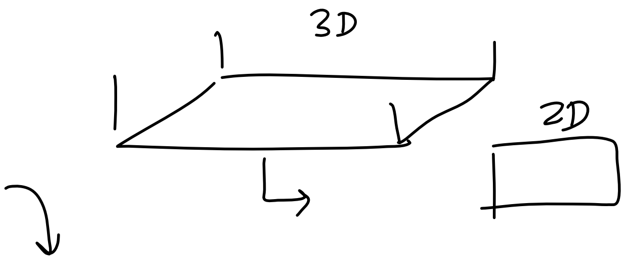
Wednesday, May 5, 2021 1:02 PM

1. Introduction

Wednesday, May 5, 2021 1:02 PM

feature extraction → CoD ↓

↳ PCA → f ↓

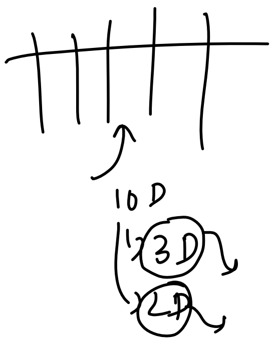


784 → 3D

Benefits

1) faster exe of algo

2) Visualization

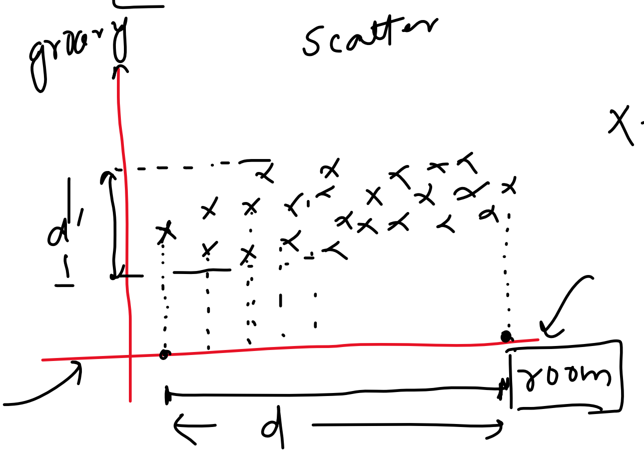


2. Geometric Intuition

Wednesday, May 5, 2021 1:02 PM

PCA → feature selection

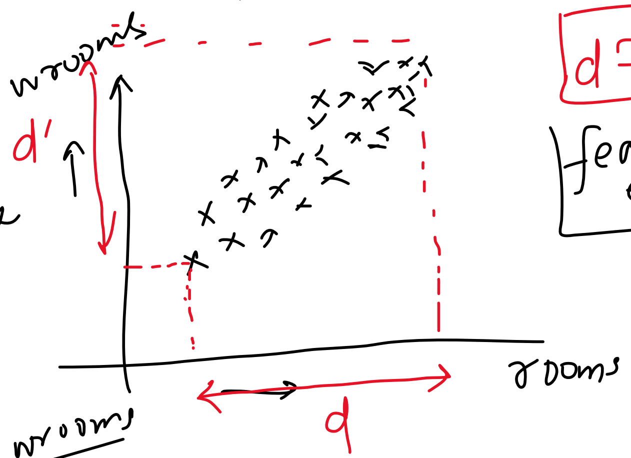
No. of rooms	No. of grocery shops	price (L)
3	2	60
4	0	130
5	6	170
2	10	90



X → rooms
variance

d > d'

No. of rooms	No. of washrooms	Price



$\boxed{d = d'}$

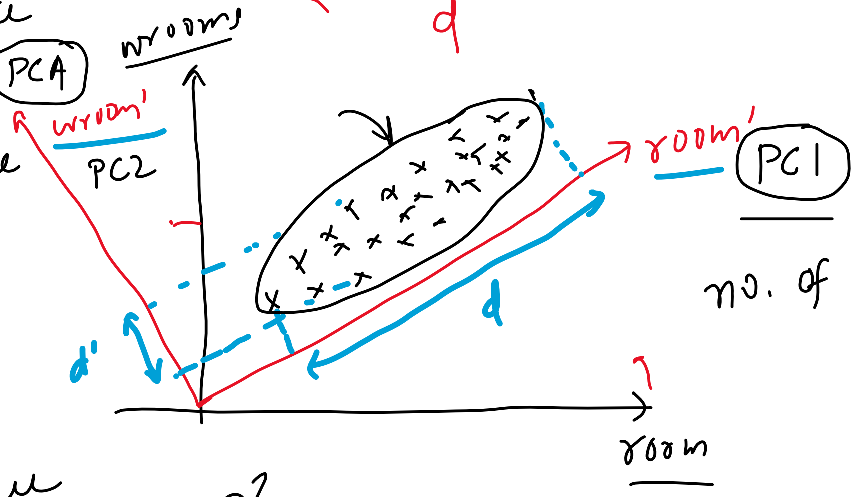
feature extraction

1D

size	price

2D

Room	wrooms	price



PCA

d > d'

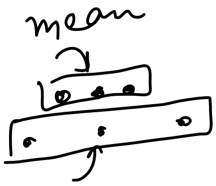
no. of PC ≤ n

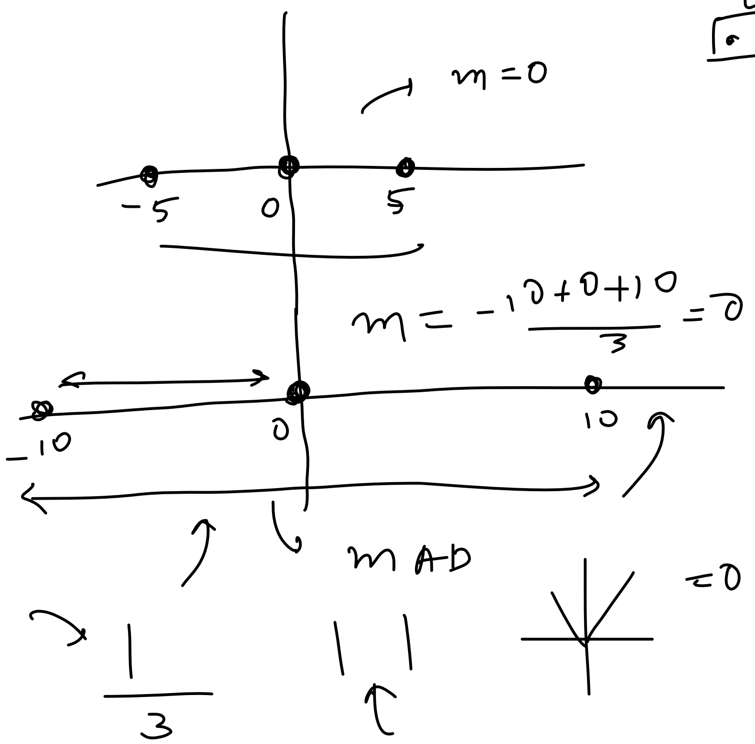
→ Variance spread → imp?

3. Why Variance is Important?

Wednesday, May 5, 2021 1:03 PM

mean





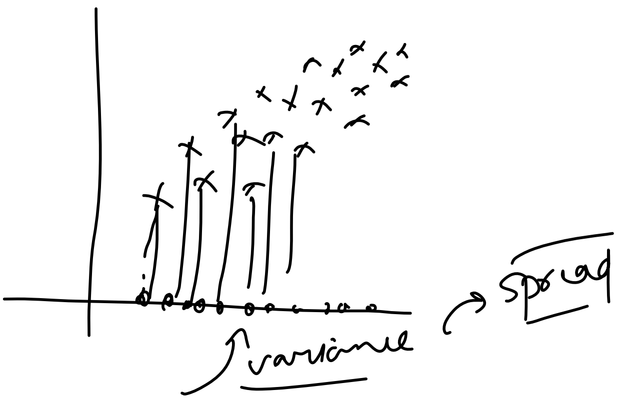
$$\frac{|\ \ |}{3}$$

variance

$$\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n} = \underline{\text{variance}}$$

$$v = \frac{25 + 0 + 25}{3} = \frac{50}{3}$$

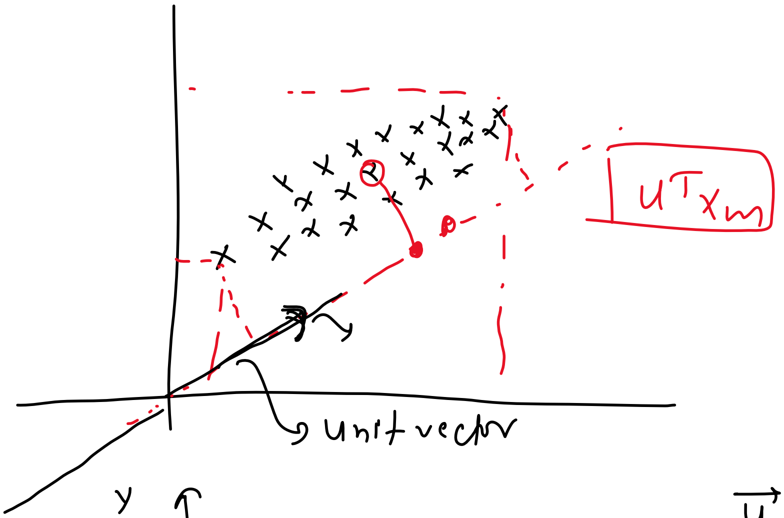
$$v = \frac{100 + 0 + 100}{3} = \sqrt{\frac{200}{3}} = sd$$



4. Problem Formulation

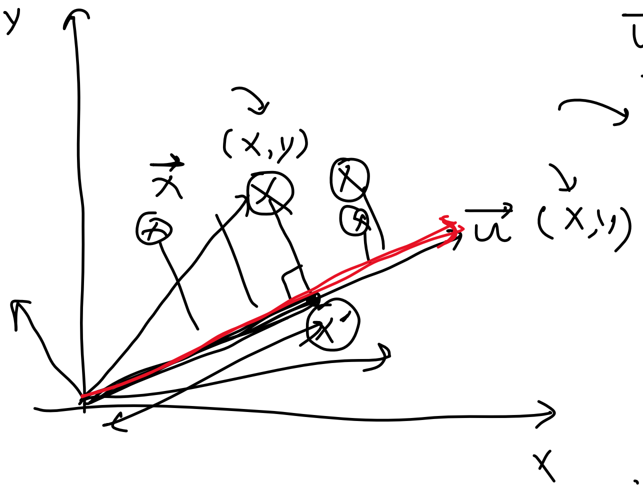
Wednesday, May 5, 2021 1:03 PM

2D → 1D



u

$$\frac{\vec{u}\cdot\vec{x}}{|u|=1} = \vec{u}\cdot\vec{x} = \boxed{u^T x}$$



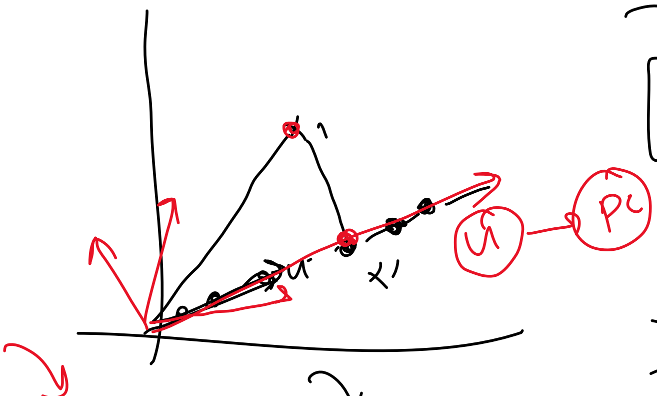
variance maxi

[x y] [x y]

×

$$[x_1\ x_1]\begin{bmatrix}x_2\\ y_2\end{bmatrix}$$

$$= \boxed{x_1x_2 + y_1y_2}$$ — scalar



$[u^Tx_1]$ $[u^Tx_2]$ . . $[u^Tx_n]$

$$\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

$$\frac{\sum_{i=1}^{n}\left(u^Tx_i - u^T\bar{x}\right)^2}{n}$$ max = variance

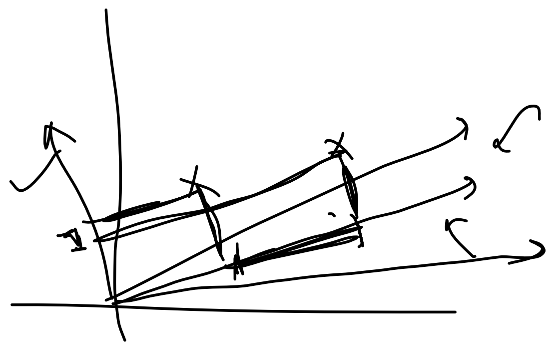
$\vec{u}$

→ optimization

cov mat

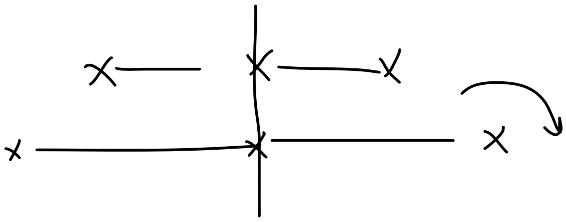
cov mat

eigen



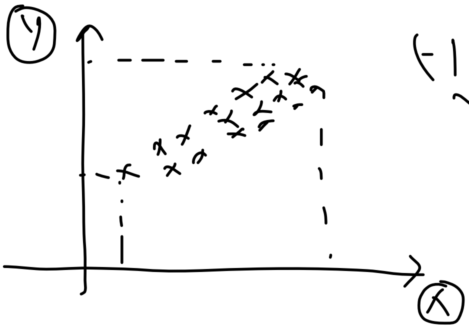
Covariance and Covariance Matrix

Thursday, May 6, 2021 7:02 AM



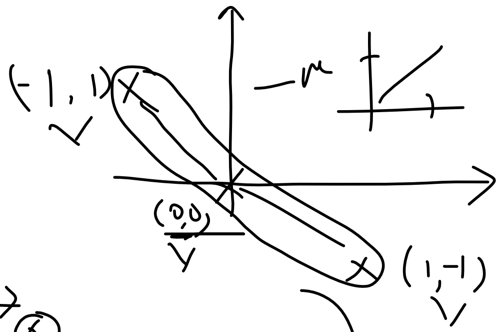
variance

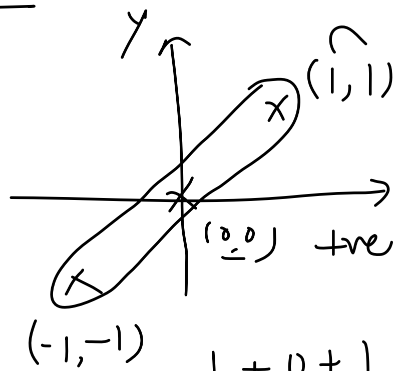
Covariance



Corelation

$\boxed{-1 \text{ to } 1}$

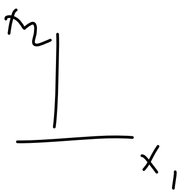




$$\frac{1+0+1}{\cancel{3}} \qquad -\frac{2}{3} \qquad \frac{-1+0-1}{3}$$

$$\frac{1+0+1}{3} = \frac{2}{3}$$

x_1	x_2



cov mat

$$\begin{bmatrix} 2 \times 2 \end{bmatrix}$$

$$\begin{array}{c|cc} X_1 & cov(x_1,x_1) & cov(x_2,x_1) \\ X_2 & cov(x_1,x_2) & cov(x_2,x_2) \\ \hline & X_1 & X_2 \end{array}$$

-ve

$\boxed{\text{cov mat}}$

$cov(x_1, x_1)$
$\downarrow$
$var(x_1)$

$$\begin{bmatrix} var\ X_1 & cov(X_2, X_1) \\ cov(X_1, X_2) & var\ X_2 \end{bmatrix} \rightarrow \text{square symmetric}$$

$cov(a,b) = cov(b,a)$

$$\begin{bmatrix} V_x & C_{xy} & C_{xz} \\ C_{xy} & V_y & C_{yz} \\ C_{xz} & C_{xy} & V_z \end{bmatrix}$$

x | y | z

X	V_x	$C_{x,y}$	$C_{x,z}$

cov matrix

$$
\begin{array}{cc}
\begin{array}{c} x \\ y \\ z \end{array} &
\begin{bmatrix} V_x & C_{x,y} & C_{x,z} \\ C_{y,x} & V_y & C_{y,z} \\ C_{z,x} & C_{z,y} & V_z \end{bmatrix} \\
 & \begin{array}{ccc} x & \quad y & \quad z \end{array}
\end{array}
\rightarrow \text{cov matrix}
$$

eigen decomposition of covariance matrix

Linear Transformation, Eigen Vectors and Eigen Values

Thursday, May 6, 2021 7:03 AM

$\begin{bmatrix}1 & 0\\ 1 & 0\end{bmatrix}$

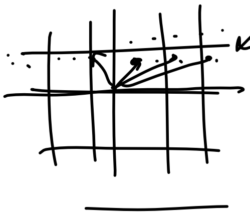
Matrices

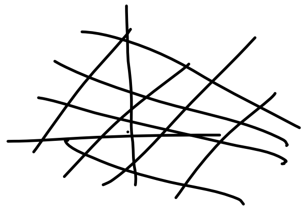
LT

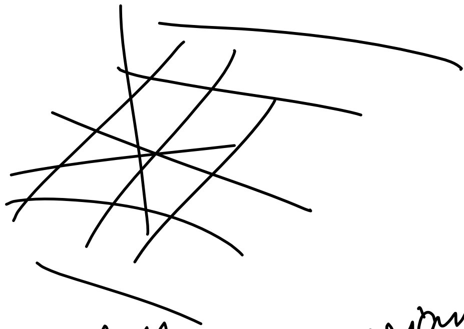
$\boxed{A\vec{v} = \lambda\vec{v}}$

PCA

2D







Eigenvectors

$A\vec{v} = \lambda\vec{v}$

Matrix

eigen val

eigenvalues

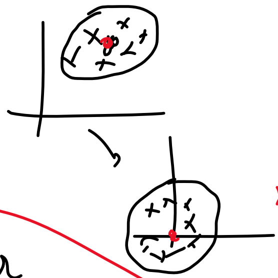
covariance matrix

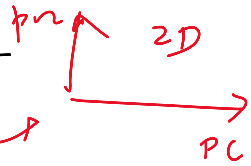
5. Step By Step Solution

Wednesday, May 5, 2021 1:03 PM

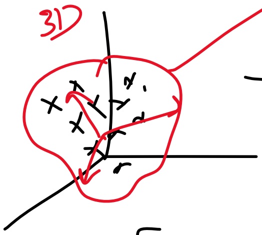
f1 | f2 | f3 | tra

$(\bar{X})$





3D



1) mean center

2) find cov matrix

3) find the eigen value/vector

$$
\begin{array}{c}
f_1 \\ f_2 \\ f_3
\end{array}
\begin{bmatrix}
V(f_1) & C(f_1,f_2) & C(f_1,f_3) \\
C(f_1,f_2) & V(f_2) & C(f_1,f_3) \\
C(f_3,f_1) & C(f_3,f_2) & V(f_3)
\end{bmatrix}
\to \text{cov mat}
$$

$$\quad f_1 \qquad f_2 \qquad f_3$$

(np.cov)

1D

np.linalg.eig()

Cov Matrix

3D → 3 vector

λ_1 λ_2 λ_3

PC1 PC2 PC3

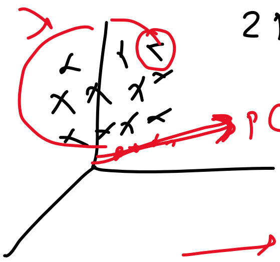
2D

How to transform points?

Thursday, May 6, 2021   12:31 PM

1000

2D, 1D

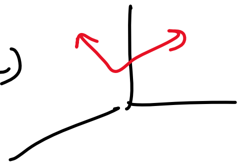


f1 | f2 | f3 | f4

(1000,4)

PC1 | fra.

(1000, 2)



PC1 (1,3)

$\to u^T . x$

(3,1)

(1000,3)

(1000,3) . (3,1)

(1000,1)

PC1 | PC2 | Target

(1000,3) (2,3)

1000,3 (3,2)

(1000,2)

6. Coding the Steps

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7. Practical Example on MNIST Dataset

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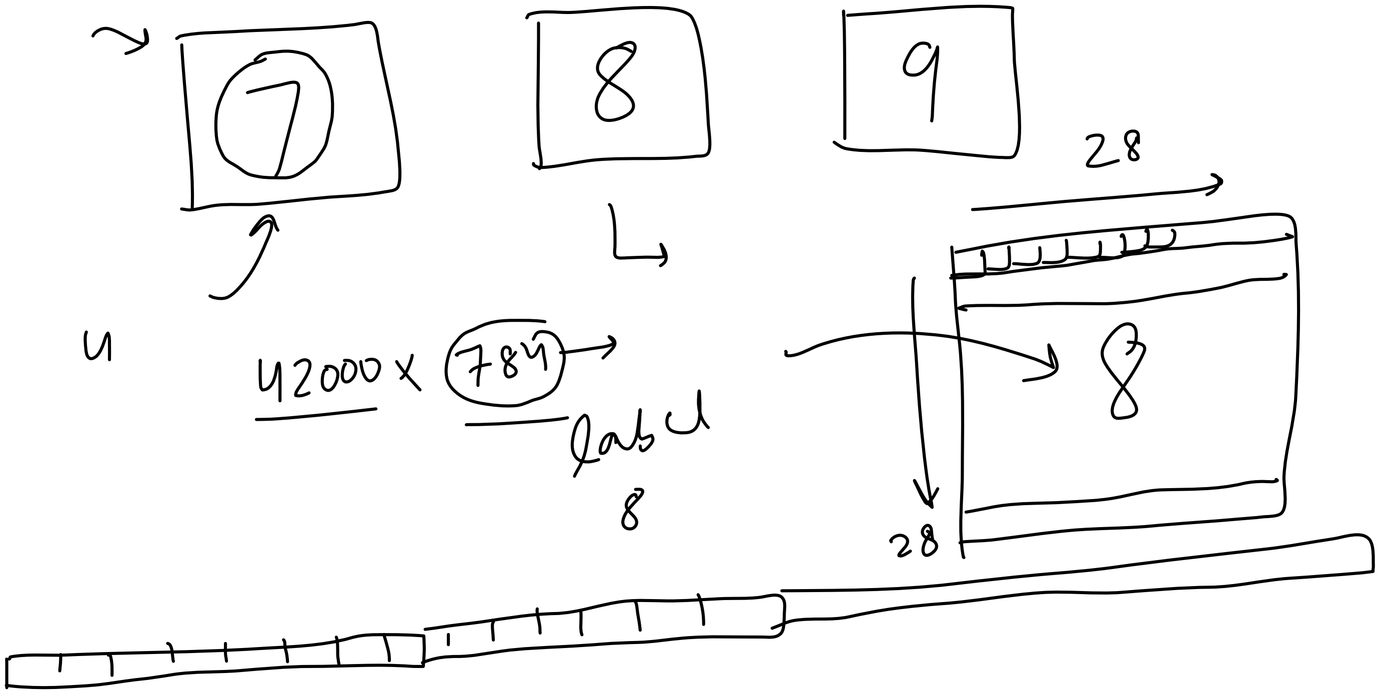
MNIST image

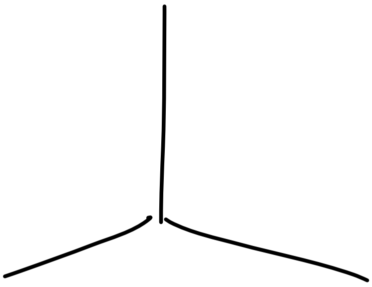


8. Visualization of MNIST Dataset

Wednesday, May 5, 2021 1:04 PM

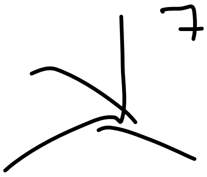
9. Explained Variance

Wednesday, May 5, 2021 1:04 PM



(x, y, z)

2 ev → (2, 3)



784 dim

3 (3, 784)

100

(100, 784)

10. Finding optimum number of Principle Components

Wednesday, May 5, 2021 1:05 PM

$ev = \lambda$

784 $\longrightarrow$ $\lambda_1, \lambda_2, \lambda_3 \dots \lambda_{784}$

$\boxed{90\%}$

$$\left(\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3 + \dots \lambda_{784}}\right) \times 100 \rightarrow \text{percentage}$$

15 $\lambda_1 - \lambda_{15}$

λ_1, λ_2, λ_3 λ_4 λ_5 λ . .

30% 25 15 10 5

90%

λ_{15}

11. When PCA does not work

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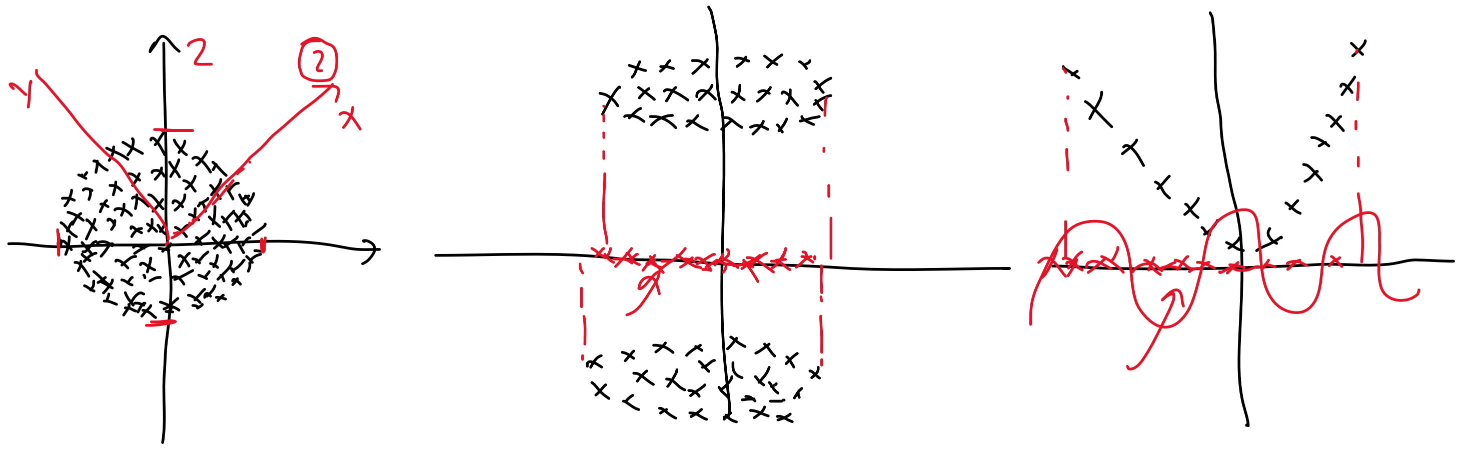
$y = x^2$



Simple Linear Regression

Monday, May 10, 2021 3:54 PM

cgpa	package
7.1	3.5
4.7	1.2
8.9	4.2
8.1	3.9
. . .	. . .

cgpa → $\boxed{\text{model}}$ → package

$\boxed{3.5 \text{ Lpa}}$

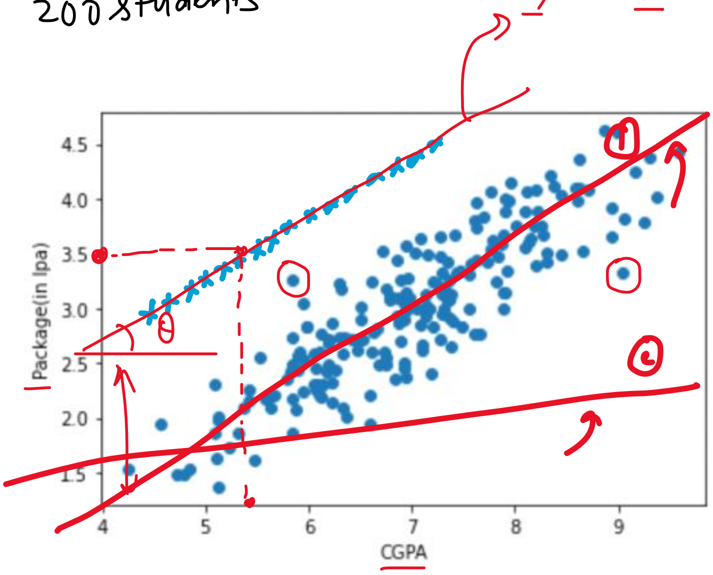
sort of linear

$\boxed{m, b}$

$y = mx + b$

package, cgpa

200 students



$\boxed{\text{Why?}}$

↳ Real word dataset

Best fit line

{ m → slope
b – y intercept }

LR

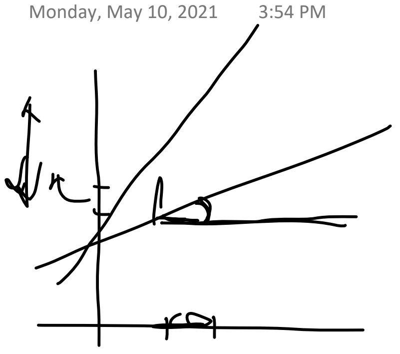
→ $\boxed{m, b}$

Code Example

Monday, May 10, 2021 3:54 PM

Intuition

Monday, May 10, 2021 3:54 PM



$y = mx + b$

$$\boxed{package = m \times cgpa + b}$$

b

$\boxed{b = 0}$

m / b

$package = m \times cgpa$

$exp \to 0$

$m \to$ weightage

$m \downarrow \uparrow$

b

$\boxed{package}$

exp	package

offset

$package = \boxed{m \times exp} + \boxed{b} = 0$

$\boxed{p = 0}$

$p = m \times exp$

How to find m and b?

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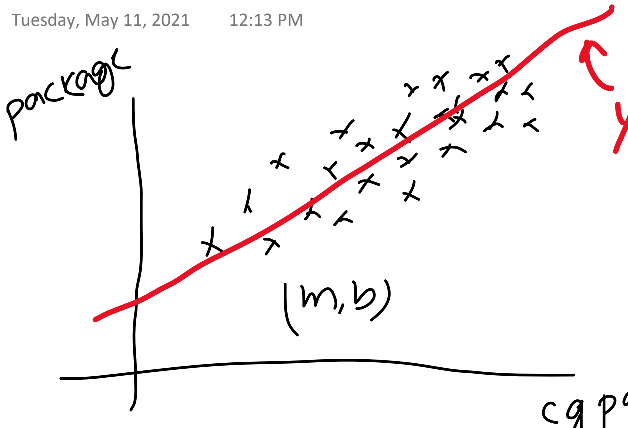


$y = mx + b$

$(m,b) \rightarrow$ Higher 1-D

LinearReg

↳ SubRegression

direct formula

Closed form solution

Non-closed form

OLS

Gradient Descent

$y \rightarrow$ pack

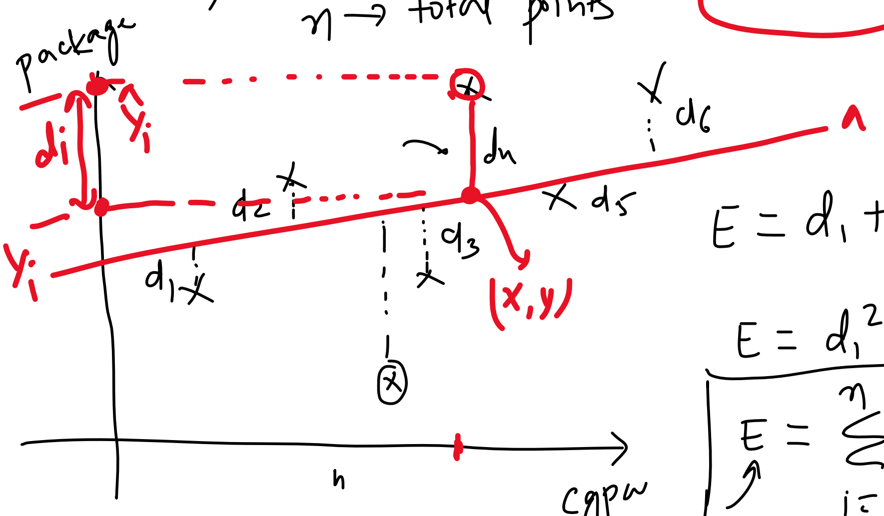
$x \rightarrow$ cgpa

$$\boxed{b = \bar{y} - m\bar{x}}$$

$$m = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$\bar{x}$, $\bar{y}$ $\rightarrow$ mean (m,b)

$n \rightarrow$ total points



$E = d_1 + d_2 + d_3 + \dots + d_n$

$E = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$

$$\boxed{E = \sum_{i=1}^{n} d_i^2}$$

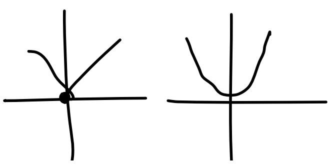
Error function J

(m,b)

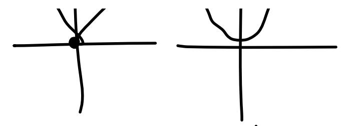
$E = |d_1| + |d_2| + |d_3| + \dots$

R1

R2 →

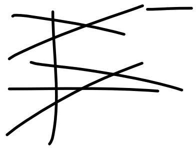


$d_i = (y_i - \hat{y}_i)$

R2 → 

$d_i = (y_i - \hat{y}_i)$

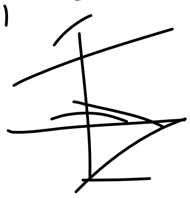
(m, b)



$$\boxed{E = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

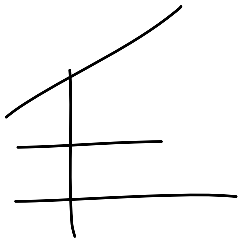
(m, b)

$\hat{y}_i$ $\quad mx_i + b$

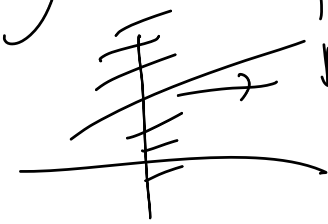


(m, b)

$$\boxed{\hat{y}_i = m x_i + b}$$



$$E(m, b) = \sum_{i=1}^{n} (y_i - m x_i - b)^2$$



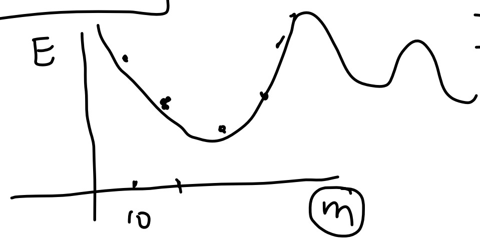
b ↑ m = 0 → $E(m, b)$

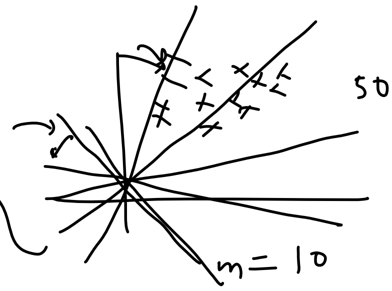
y = f(x)

$$E(m, b) = \boxed{\sum_{i=1}^{n} (y_i - m x_i - b)^2}$$

b = 0

minimum





y = f(x)

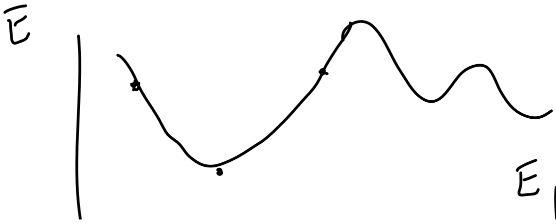
(m, b)

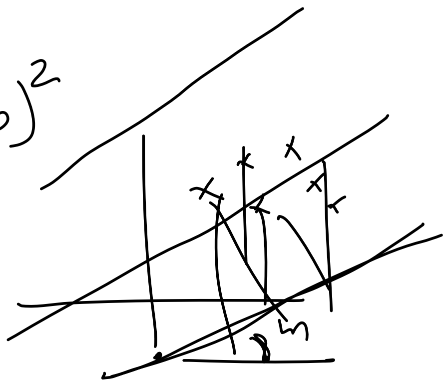
$\boxed{b = 0}$

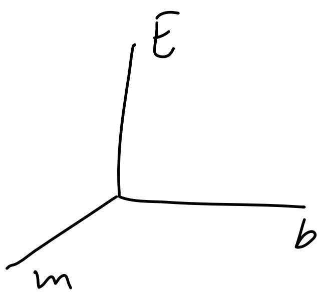
m = 1

$$E(m) = \sum_{i=1}^{n} (y_i - m x_i)^2$$

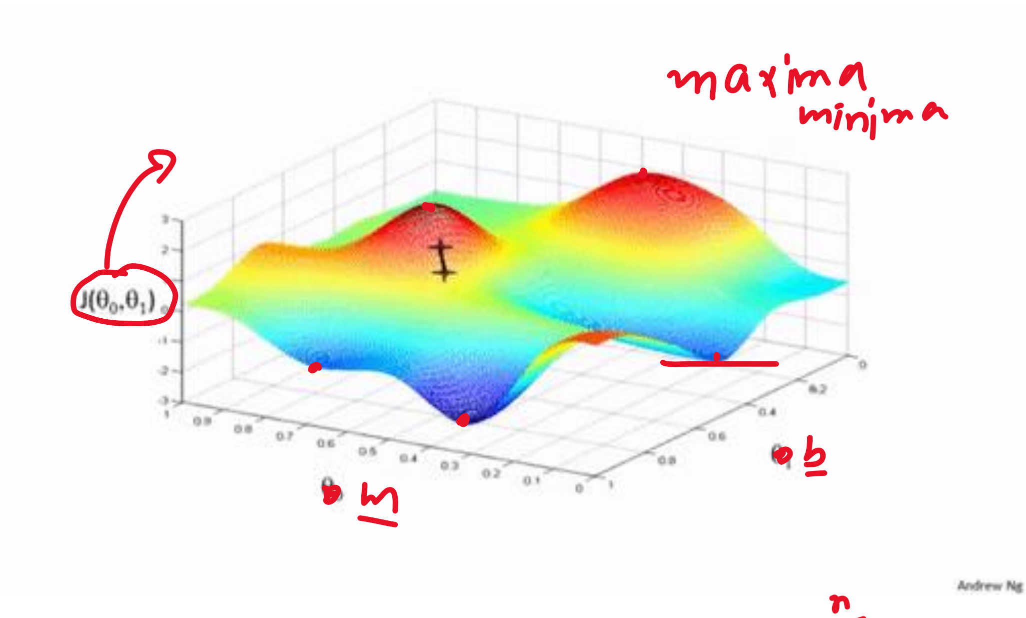
$$E(b) = \sum_{i=1}^{n} (y_i - x_i - b)^2$$







E E



$\rightarrow E(x)$

$$\frac{dE}{dx} = 0$$

$f(x,y)$

$$\frac{\partial E}{\partial m} = 0, \quad \frac{\partial E}{\partial b} = 0$$

(m, b) n

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b}\sum_{i=1}^{n}(y_i - m x_i - b)^2 = 0$$

$$= \sum \frac{\partial}{\partial b}(y_i - m x_i - b)^2 = 0$$

-1

$$\Rightarrow \sum -2(y_i - m x_i - b) = 0$$

$$\Rightarrow \sum(y_i - m x_i - b) = 0$$

$$\underbrace{b + b + b + b + \dots + b}_{n \text{ time}} = nb$$

$$\frac{\sum_{i=1}^{n} y_i}{n}$$

$$\frac{\sum y_i}{n} - \frac{\sum m x_i}{n} - \frac{\sum b}{n} = \frac{0}{n}$$

$$\bar{y} - m\bar{x} - \frac{\cancel{n}b}{\cancel{n}} = 0$$

$$\bar{y} - m\bar{x} = b$$

$$\boxed{b = \bar{y} - m\bar{x}}$$

$$E = \sum(y_i - m x_i - \bar{y} + m\bar{x})^2$$

$$\frac{\partial E}{\partial m} = \sum \frac{\partial}{\partial m}(y_i - m x_i - \bar{y} + m\bar{x})^2 = 0$$

$$\Rightarrow \sum 2(y_i - m x_1 - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$= \sum -2\,(y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$= \sum \underline{(y_i - mx_i - \bar{y} + m\bar{x})}(x_i - \bar{x}) = 0$$

$$= \sum \left[(y_i - \bar{y}) - m(x_i - \bar{x})\right](x_i - \bar{x}) = 0$$

$$= \sum \left[(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2\right] = 0$$

$$= \sum (y_i - \bar{y})(x_i - \bar{x}) = m \sum (x_i - \bar{x})^2$$

$$\boxed{m = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}}$$

Code from scratch

Tuesday, May 11, 2021 12:14 PM

Regression Metrics

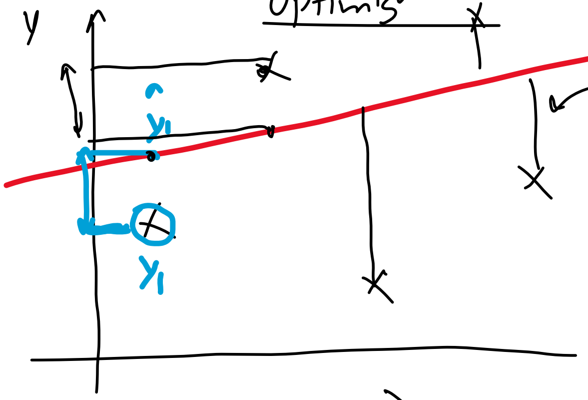
Thursday, May 13, 2021 11:56 AM

1) MAE
2) MSE
3) RMSE
4) R2 score
5) Adjusted R2 score

MAE

Thursday, May 13, 2021 11:56 AM

Optimization cgpa^{x} | $\underline{Y}$ pack (lpa)



$$\frac{|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \dots + |y_n - \hat{y}_n|}{n}$$

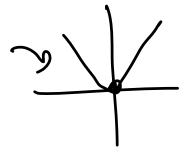
$$\boxed{mae = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}}$$

1.5 lpa

Advantage

1) same unit
2) Robust outliers

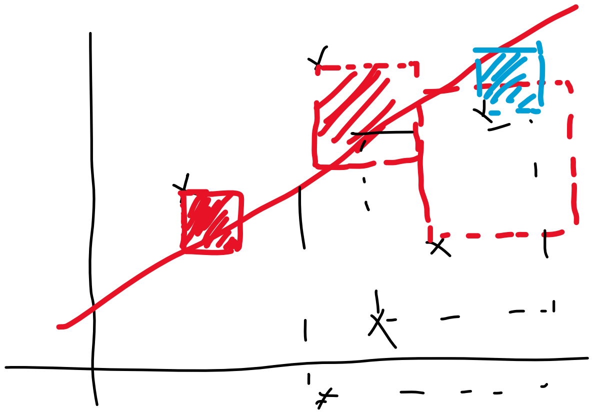
Disadvantage



MSE

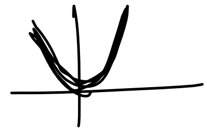
Thursday, May 13, 2021 11:56 AM

mean squared error cgpa X | y package (lpa)



11.25

$$mae = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}$$



mse

↳ function

$(y_i - \hat{y}_i)^2$

Advantage

→ L function

Disadvantage

→ Robust to outliers

$$mse = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$$

y_lpa

mse – $(lpa)^2$

RMSE

Thursday, May 13, 2021 11:56 AM

$$Rmse = \sqrt{mse}$$

mse

$$= \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}$$

rsme → loss functions

0.72

rmse - lpa → benefit

y - lpa

disadvantage

Robust outliers

R2 score

y

1.5

1.5 lpa

mse mae

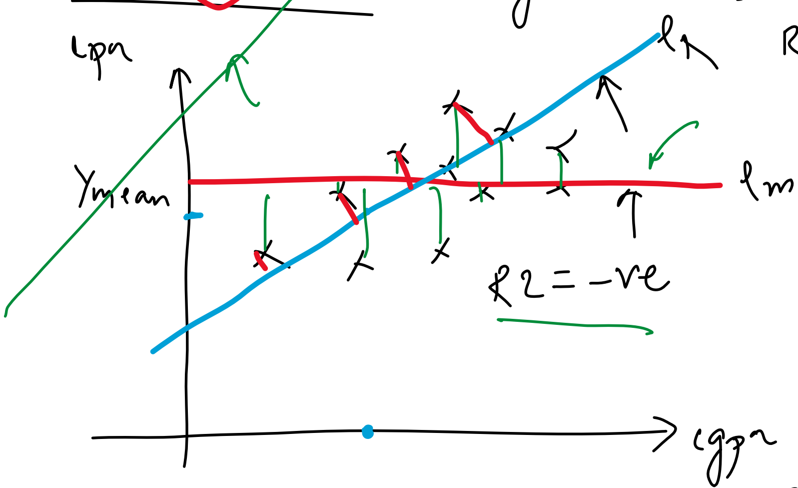
R2 Score

Thursday, May 13, 2021 11:56 AM

$\boxed{100} \rightarrow$ mean $\rightarrow \boxed{3.4}$

$\boxed{\text{cgpa}}$	package(lpa)

coeff of deter
→ goodness of fit



R2 = −ve

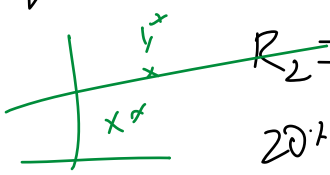
$$R2 = 1 - \frac{SS_R}{SS_M} \quad >1 \quad =1 \quad =0$$

$$R2 = 1 - \frac{\left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right]_{Reg}}{\left[\sum_{i=1}^{n} (y_i - \bar{y}_i)^2\right]_{m}}$$

→ 0

$R_2 = 0$

Sum of squared error
Reg



$R_2 = 1$

20%

$0 \rightarrow R2 \rightarrow 1$

$\boxed{SSR > SSm}$

80%

$\boxed{R2 \rightarrow 0.80}$ → $\boxed{\text{cgpa}}$ explain 80% of variance in lpa

cgpa | lpa ↑ 20%

cgpa | iq | lpa $\boxed{\text{lpa}}$

Adjusted R2 score

Thursday, May 13, 2021 11:57 AM

0.80 0.90

R2 score

cgpa | lpa

iq

temp

$R_2 \uparrow$

Adjusted R2

$1 - \uparrow$ $\downarrow$ $[1 - \downarrow] \uparrow$

R2 →

$n \rightarrow$ no. of rows

K = independent

K=1, K=2, K=3

$$R^2_{adj} = 1 - \left[\frac{(1-R2)(n-1)}{(n-1-K)} \right]$$

Multiple LR

temp

Adj R2 ↓

iq

$R^2_{adj} \uparrow$

Multiple Linear Regression

Friday, May 14, 2021 4:31 PM

LR → Simple LR, Multiple LR, polynomial

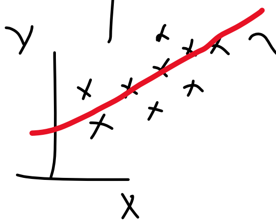
$\boxed{\text{Simple LR}}$ → Multiple LR → $x_1 \mid x_2 \mid x_3 \mid y$ → cgpa | ~~gender~~ | iq | lpa

$x_1 \mid x_2 \mid y$

cgpa | iq | lpa

3D

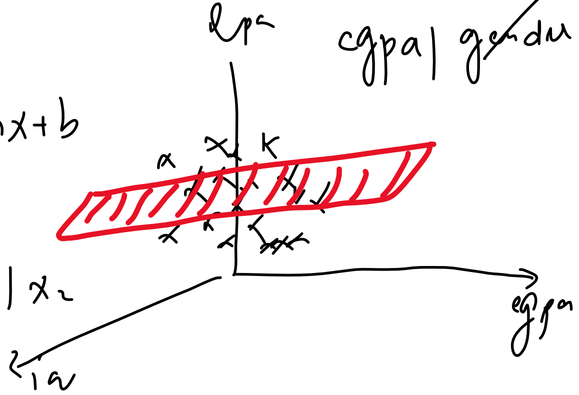
x | Y Best fit



$y = mx + b$

(m, b)

cgpa | lpa



$x_1 \mid x_2$

3D → plane

4D → hyperplane

4D

(m, b)

$$\boxed{y = mx + b}$$

β_1 β_0

β_2

$$y = mx_1 + nx_2 + b$$

$$\boxed{y = \beta_0 + \beta_1 x_1 + \beta_2 x_2}$$

$[\beta_0, \beta_1, \beta_2]$

$[\beta_0, \beta_1, \beta_2, \beta_3]$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

$$\boxed{y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i}$$

$n = 1$

$y = \beta_0 + \beta_1 x_1$

3D

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$x_1 \to$ cgpa

$x_2 \to$ iq

$y \to$ lpa

$\beta_2 \downarrow$

$\beta_1 \downarrow$

$$\boxed{lpa = \beta_0 + \beta_1 \times cgpa + \beta_2 \times iq}$$

weights β_2

$\beta_0 \to$ offest

$\beta_1 > \beta_2$

$\boxed{\text{hyperplane}}$

Code Example

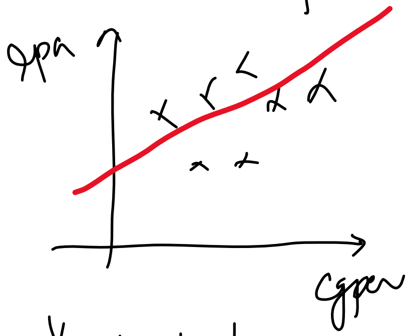
Friday, May 14, 2021 4:31 PM

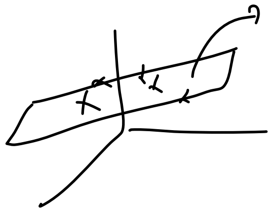
Mathematical Formulation

Saturday, May 15, 2021 4:02 PM

LR

- Simple
- Multiple
- $\boxed{\text{polynomial}}$





→ 2D plane

$$\boxed{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

$$\boxed{\beta_0, \beta_1 \dots \beta_n}$$

$y = mx + b$

$\hookrightarrow (m, b)$

cgpa | iq | gender | lpa

x_1 x_2 x_3 Y → actual

4D

100 students

(100, 4)

$\cancel{y = mx + b}$

$y = \beta_0 + \beta_1 X$

$\beta_0, \beta_1, \beta_2, \beta_3$

Y

$\hat{y}$ — X_1 X_2 X_3 predicted

$$\boxed{\hat{y}} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

$$\hat{y}_1 = \beta_0 + \beta_1 X_{11}$$

1 row

$$\hat{Y} = \begin{bmatrix} \boxed{\hat{y}_1} \\ \boxed{\hat{y}_2} \\ \vdots \\ \hat{y}_{100} \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 X_{11} & \beta_2 X_{12} & \beta_3 X_{13} \\ \beta_0 & \beta_1 X_{21} & \beta_2 X_{22} & \beta_3 X_{23} \\ \vdots & & & \\ \beta_0 & \beta_1 X_{1001} & \beta_2 X_{1002} & \beta_3 X_{1003} \end{bmatrix}$$

100 rows → $\boxed{n \text{ rows}}$

4 cols → (3) col → m cols

A B = AB

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 X_{11} & \beta_2 X_{12} & \beta_3 X_{13} & \dots & \beta_m X_{1m} \\ \beta_0 & \beta_1 X_{21} & \beta_2 X_{22} & \beta_3 X_{23} & \dots & \beta_m X_{2m} \\ \vdots & & & & & \end{bmatrix}$$

$$\begin{bmatrix} \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \vdots \\ \beta_0 & \beta_1 x_{n1} & \beta_2 x_{n2} & \beta_3 x_{n3} & \cdots & \beta_m x_{nm} \end{bmatrix}$$

$$= \overset{A}{\begin{bmatrix} 1 & X_{11} & X_{12} & X_{13} & \cdots & X_{1m} \\ 1 & X_{21} & X_{22} & X_{23} & \cdots & X_{2m} \\ \vdots \\ 1 & X_{n1} & X_{n2} & X_{n3} & \cdots & X_{nm} \end{bmatrix}} \overset{\beta}{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}}$$

$$\boxed{\hat{Y} = X\beta} \quad \text{— (1)}$$

$\boxed{X}$

$\beta \quad \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$

cgpa | gender | iq

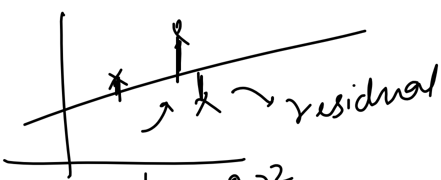
$X \to$

	x_1	x_2	x_3
1	x_{11}	x_{12}	x_{13}
1	x_{21}	x_{22}	x_{23}
1	x_{31}	x_{32}	x_{33}

$\boxed{df} \to X, y$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad e = Y - \hat{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

$$\boxed{e} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$



$\to (y_1 - \hat{y}_1)^2$

Single LR

$\sum^{n} (y_i - \hat{y}_i)^2 \qquad \ldots \qquad e^T \ldots \begin{bmatrix} y_1 - \hat{y}_1 \end{bmatrix}$

single LR

$$\boxed{E = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \to E = e^T e$$

$$\begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = e$$

$$\left[(y_1 - \hat{y}_1) \ (y_2 - \hat{y}_2) \ (y_3 - \hat{y}_3) \ldots (y_n - \hat{y}_n) \right]$$

$(1 \times n)$

$$\begin{bmatrix} (y_1 - \hat{y}_1) \\ (y_2 - \hat{y}_2) \\ (y_3 - \hat{y}_3) \\ \vdots \\ (y_n - \hat{y}_n) \end{bmatrix} \to 1 \times 1$$

$(n \times 1)$

$$(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + \ldots + (y_n - \hat{y}_n)^2$$

$$\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$(A + B)^T = A^T + B^T$$
$$(A - B)^T = A^T - B^T$$

$$E = e^T e = (y - \hat{y})^T (y - \hat{y})$$
$$= (y^T - \hat{y}^T)(y - \hat{y})$$
$$= \left[y^T - (X\beta)^T \right] (y - X\beta)$$

$$E = y^T y - y^T X \beta - (X\beta)^T y + (X\beta)^T X \beta$$

$$E = y^T y - 2 y^T X \beta + \beta^T X^T X \beta$$

matrix differ

$$\frac{dE}{d\beta} = \frac{d}{d\beta} \left[y^T y - 2 y^T X \beta + \beta^T X^T X \beta \right] = 0$$

$$\frac{dL}{d\beta} = \frac{d}{d\beta}\left[\underbrace{y^T y}_{0} - 2y^T X\beta + \beta^T X^T X \beta \right] = 0$$

$$= 0 - 2y^T X + \frac{d}{d\beta}\left[\beta^T X^T X \beta \right] = 0$$

$$\frac{dy}{dA} = 2XA^T$$

$$y = A^T X A$$

$$= -2y^T X + 2X^T X \beta^T = 0$$

$$= \cancel{2} X^T X \beta^T = \cancel{2} y^T X \quad \frac{1}{[X^T X]}$$

$$\beta^T = y^T X (X^T X)^{-1}$$

$$(\beta^T)^T = \left[y^T X (X^T X)^{-1} \right]^T$$

$$\beta = \left[(X^T X)^{-1} \right]^T (y^T X)^T$$

$$X^T X$$

$$X = n \times (m+1)$$

$$(m+1) \times n \quad n \times (m+1)$$

$$(m+1) \times (m+1)$$

$$\beta = \left[(X^T X)^{-1} \right]^T X^T y$$

$$X \rightarrow X\_train$$

$$\beta = (X^T X)^{-1} X^T y$$

$$Y = y\_train$$

$$\boxed{\beta = (X^T X)^{-1} X^T y}$$

$$(X^T X)^{-1}$$

$$n \times (m+1)^T$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$$

$$[(m+1) \times (m+1)][(m+1) \times n][n \times 1]$$

$$(m+1) \times n$$

$$\boxed{X^T X} \quad m \rightarrow \text{col's}$$

$$(m+1) \times 1$$

β_n -

$(m+1) \times 1$

$\uparrow$

β

$[(m+1) \times n]\,\underbrace{[n \times 1]}$

$\boxed{(m+1) \times 1}$

$\boxed{1000000000}$

999

$n = 1000$

$\boxed{X^T X}$ $m \rightarrow$ ...

$(m+1) \times (m+1)$

$\boxed{1000 \times 1000}$

n^3

$n \times 1$

Detour

Saturday, May 15, 2021 4:56 PM

$(AB)^T = \underline{B^T A^T}$ $\quad Y = A \quad X\beta = B \quad A^{T^T} = A$

$\underline{\boxed{Y^T X\beta}} = \underline{(X\beta)^T Y}$

$(A^T B)^T = \underline{B^T A}$ — (11)

RHS

$A^T B = \underline{B^T A}$ — (1)

RHS

$\underline{A^T B} = \underline{(A^T B)^T} \longrightarrow \underline{A^T B} = C$

$\boxed{C = C^T}$ $\quad A^T B \quad C = \boxed{Y^T X\beta}$

(X) n rows m cols

n students

$\underline{(Y^T X\beta)^T} = Y^T X\beta$

$Y = \begin{bmatrix} - \\ - \\ - \\ - \\ - \\ - \end{bmatrix}$

$(n\times 1) \to (1\times n)$

$\underbrace{(1\times n)\ (n\times(m+1))}\ [(m+1)\times 1]$

$n \times (m+1)$

$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$ m+1

$(m+1)\times 1$

$1\times(m+1) \quad (m+1)\times 1$

$1\times 1 = [7]^T$

$[7]$

Why Gradient Descent

Saturday, May 15, 2021 5:35 PM

Sklearn

→ $\boxed{\text{OLS}}$ →

→ Gradient Descent

OLS

LinearRegression

SGDRegressor

Code From Scratch

Monday, May 17, 2021 12:15 PM

(100,3) (100)

100 (100,4)

$$\beta = (X^T X)^{-1} X^T Y$$

$X \rightarrow$ matrix

cgpa | iq | gender lpa

1	—	—	—
1	—	—	—

$X \rightarrow$ X_train

$Y \rightarrow$ y train

diabetes → SKlearn → R2 → same

diabetes → My class → R2 → same

X test β_0 $\beta_1 \rightarrow \beta_{10}$

3

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$$= \beta_0 + \text{np.dot}(\beta, \text{X\_test})$$

X_test coeff

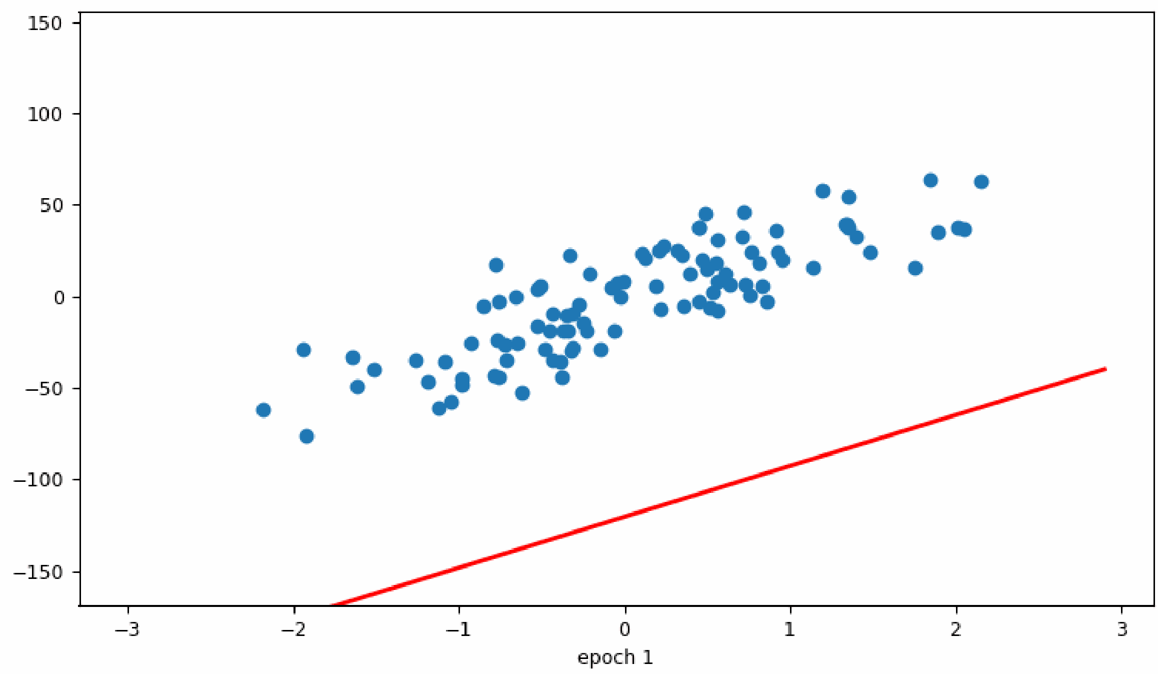
(89,10) (10,1)

$(89,1) + \beta_0$

89 → 1

(89,1)

$\boxed{\text{y pred}}$



$$\sum (y_i - m x_i - b)^2$$

$$\sum -2 (y_i - m x_i - b) x_i$$

-2

What is Gradient Descent?

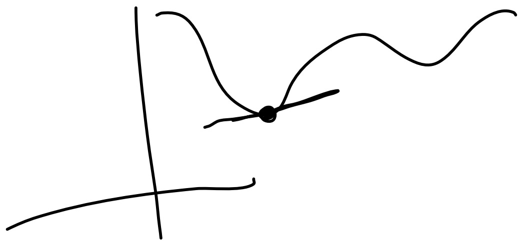
Thursday, May 20, 2021 1:46 PM

Linear Reg

Logistic Reg

Tsne

Deep Learning



The Plan

Thursday, May 20, 2021 1:46 PM

5 videos

1) Gradient Descent

2 → Batch GD

3 → SGD → SGDRegres

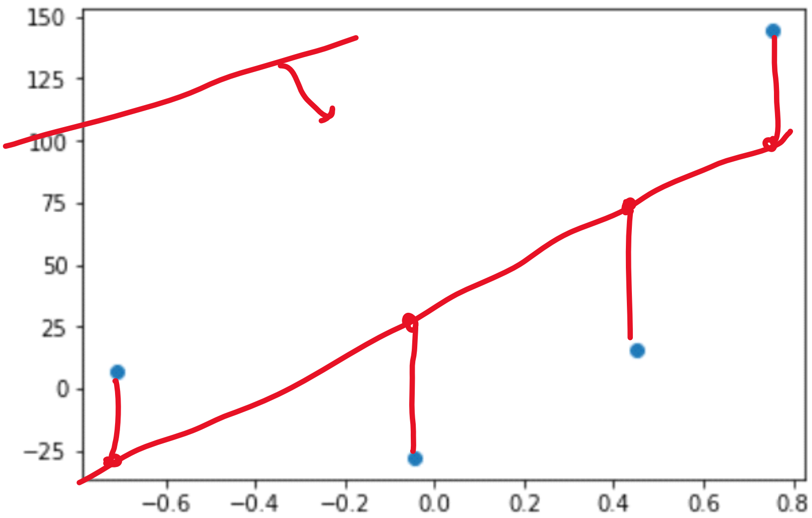
4 → MBGD

5

Intuition

Thursday, May 20, 2021 1:46 PM

2 cols
4 rows



cgpa | lpa

OLS →

$\hat{y}_i = m x_i + b$

$m = 78.35$

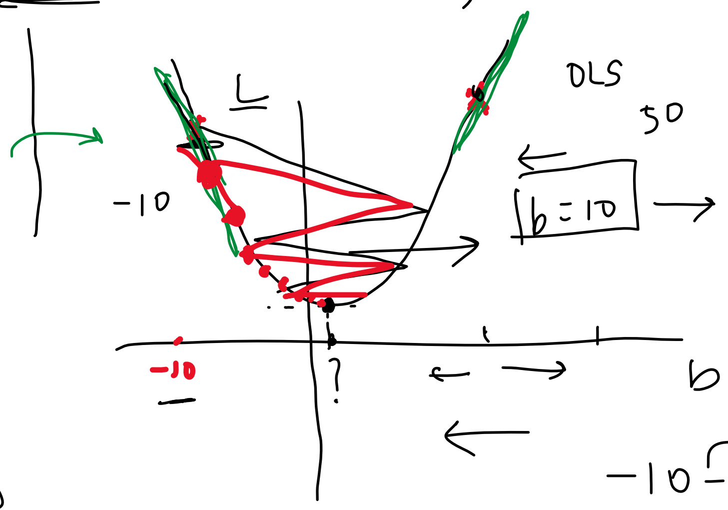
$$L = \sum_{i=1}^{4} (y_i - \hat{y}_i)^2$$

$$L = \sum_{i=1}^{4} (y_i - m x_i - b)^2$$

$$L_{min} = \sum_{i=1}^{4} (y_i - 78.35 * x_i - b)^2$$

$L \to b^2$

$L(b)$ (b)



-50

$\boxed{b = 10}$ →

Step 1 - select a random b_{min}

L_{min}

b increment

b decrement

$b = -10$

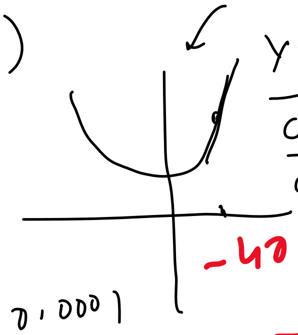
$x = 5$ →

$-10 - (-50)$

40

-40

$10 - (50)$



$y = x^2 + 2$

$\frac{dy}{dx} = 2x$

$= 10$

-9.5

$b = -10$ Slope = −ve

$$\boxed{b_{new} = b_{old} - slope}$$

1

$$\boxed{b_{new} = b_{old} - \eta \, slope}$$

$\boxed{0.01}$ 0.1, 0.0001

$*-50$

-10

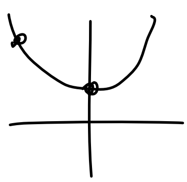
$b_{new} = -10 + (0.01 \times 50)$

$= -10 + 0.5 = -9.5$

$b_{new} = -9.5 - (0.01 * -40) = -9.5 + 0.4 = -9.1$

When to stop

1) diff b_{old} b_{new} => 0.0001



$b_{new} - b_{old} = 0.0001$

=> 0.0001

2) Iteration → $\boxed{1000}$, 100 ,

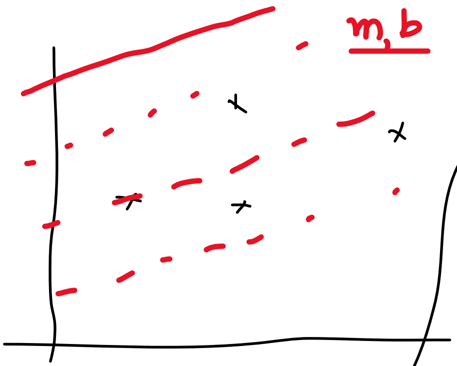
epochs

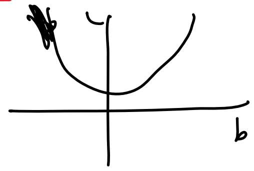
$\boxed{b_{new} - b_{old} = 0}$

Mathematical Formulation

Thursday, May 20, 2021 1:47 PM

m = 78.35





Step → Start with a random $b = b$

1000, 100

$\boxed{\eta = 0.01}$ $\boxed{b = 0}$

i = 0   n, η,

for i in epochs:

$$b_{new} = b_{old} - \eta * slope\,(b = 0)$$

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\frac{dL}{db} = \frac{d}{db}\left(\sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right) = 2\sum_{i=1}^{n} (y_i - mx_i - b)(-1)$$

b = 0

$$\frac{d}{db}\sum_{i=1}^{n} (y_i - mx_i - b)^2$$

−1

$$slope = \boxed{-2\sum_{i=1}^{n} (y_i - mx_i - b)}$$

$$= -2\sum_{i=1}^{n} (y_i - 78.35 * x_i - 0)$$

i = 1

$\boxed{\text{epoch}}$

slope (b=0)

b new

$$b_{new} = b_{old} - \boxed{\eta\, slope}_{\,b = b_{old}}$$

stepsBC

Example

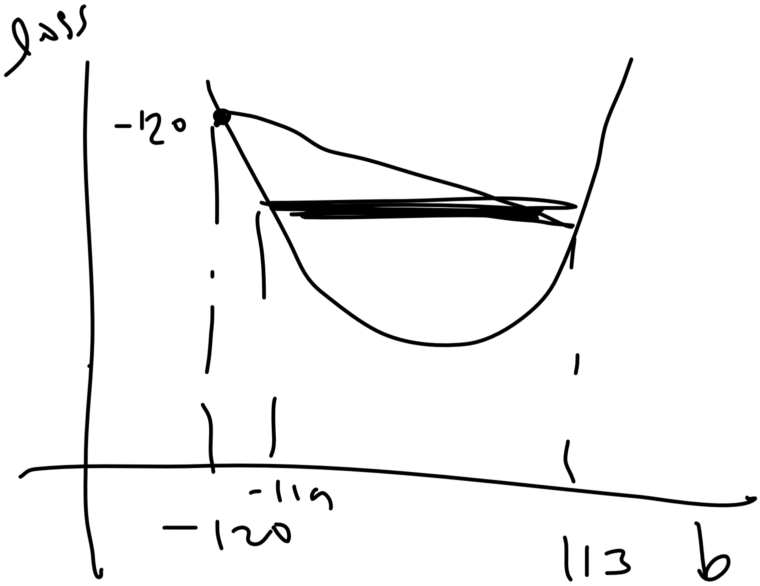
Thursday, May 20, 2021 1:47 PM

Code from Scratch

Thursday, May 20, 2021 1:48 PM

Visualization 1

Thursday, May 20, 2021 1:52 PM



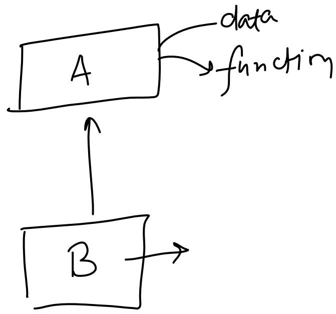
→ Inheritance
→ Super
→ Abstraction
→ Static method
→ MCQs

→ DS Algo

Society

Biology

→ OOP



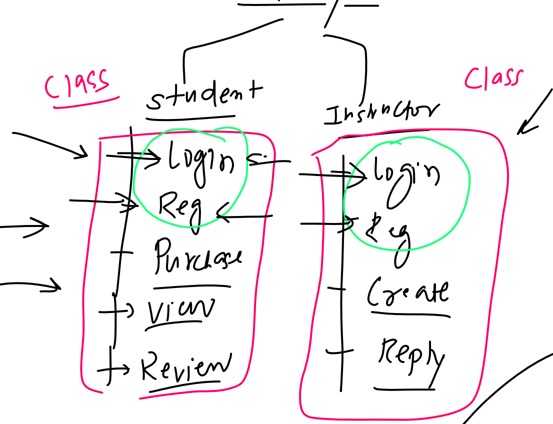
database

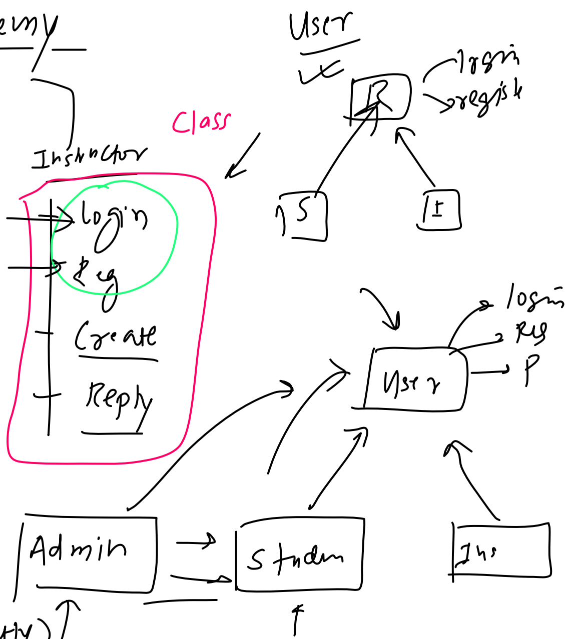
→ DRY

Child IS A Parent

mobile is A product

Udemy



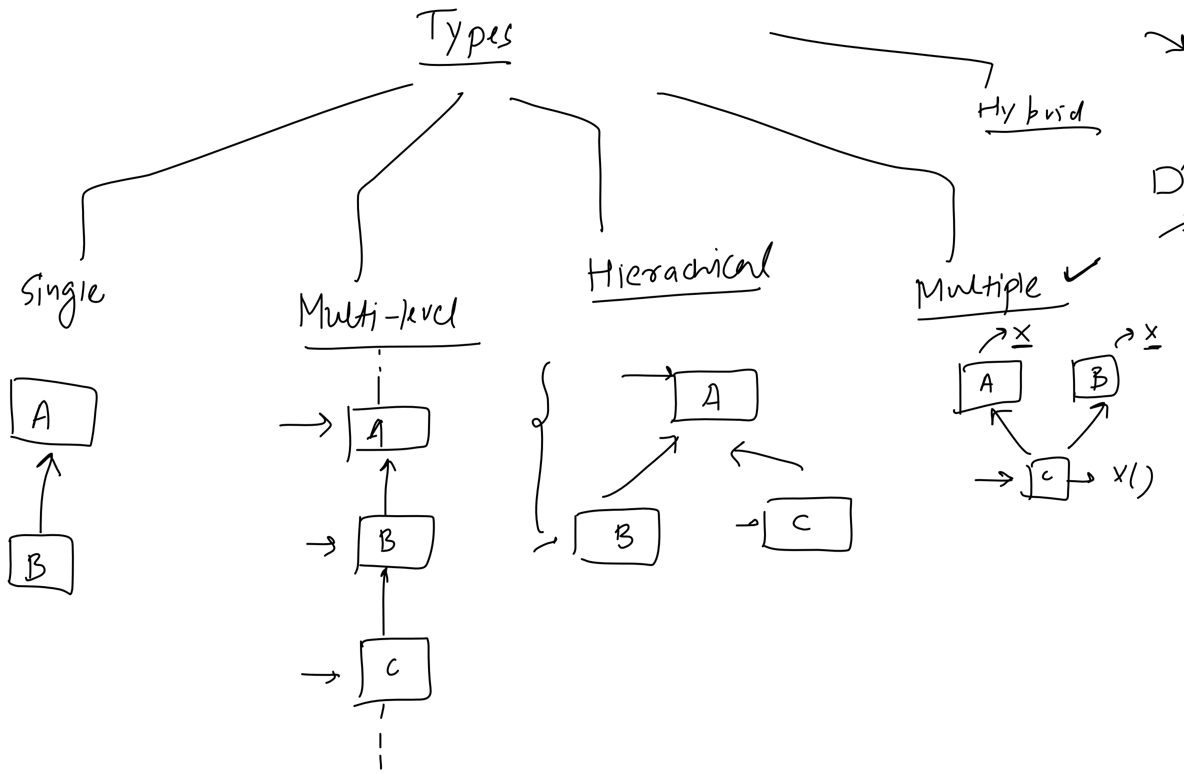


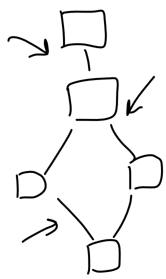
Student is a user

can be used
→ data (property)
→ function
→ constructor

can't be used
→ private members (directly)

Types





Product

Phone

Smart

Product

Phone

Smart

iPhone

smart

featur

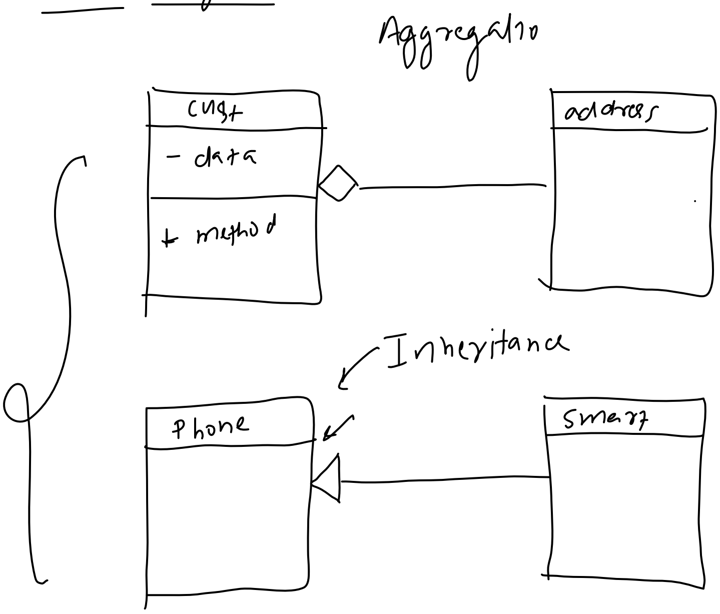
Relationships

Inheritance
(is A)

Aggregation
(has A)

Customer has A Address

Class diagram



Construct
static method
inst vs sta
super

OOP

Class
object

Encapsulation

Polymor

Inheri

Abst

Class
object

Encapsulation

20 → 9 → DS Algo → 8 classes

{ 17 18 19 20 }
SQL

Class 9 → Time complexity
class 10 → Dynamic Arrays (Lists)
class 11 → LL
class 12 → Stacks + Queues
class 13 → Hashing → searching and sorting
Class 14 → Tree + Graph
Class 15 → Dynamic + greedy
Class 16 → Exam + MCQ + qns

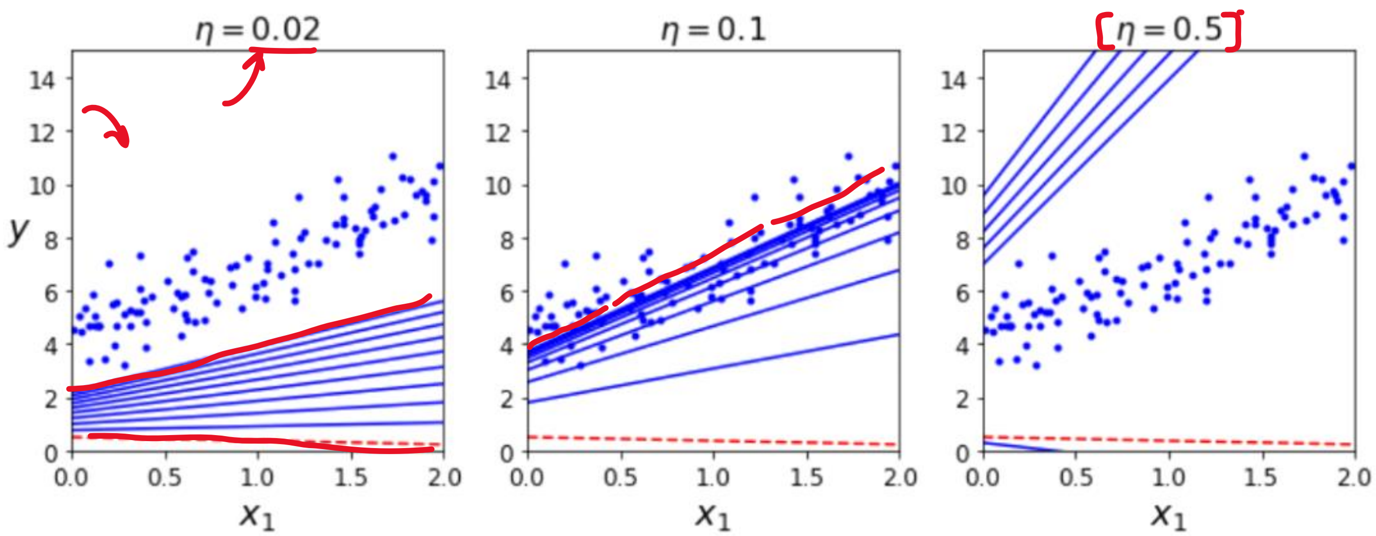
Python → exception, file, 4 lams

Few Discussions

Thursday, May 20, 2021 4:47 PM

1. Effect of Learning rate
2. The universality of Gradient Descent

epoch = 10



$b = 0$

$$\boxed{b = b_{old} - \eta \; slope}$$

$$\frac{dL}{db} = [\quad]$$

$$\boxed{\Sigma(y_i - \hat{y}_i)^2}$$

LR

Deep →

LOR → [function

Adding m into the mix

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Steps

1) init random vals for m and b

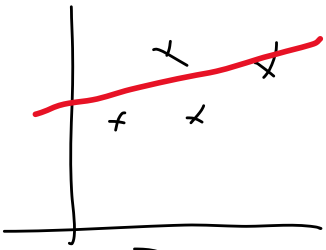
$m = 1$ and $b = 0$

2) epochs = 100, lr = 0.01

for i in epochs:

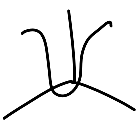
$b = b - \eta \boxed{\text{slope}}$

$m = m - \eta \boxed{\text{slope}}$



$z = f(x, y)$

$f(x)$



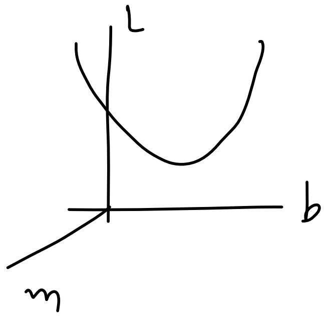
$b = 0$

$$\underline{L} = \sum (y_i - \hat{y}_i)^2 = \sum (y_i - m x_i - b)^2$$

$L(m, b)$ $\quad \frac{dL}{db}$

$$\text{b\_slope} = \boxed{\frac{\partial L}{\partial b}}$$

∂



$$\text{m\_slope} = \frac{\partial L}{\partial m}$$

2D $\quad$ parabola (m, b)

$$\sum (y_i - m x_i - b)^2$$

$$\frac{\partial L}{\partial b} = -2 \sum (y_i - m x_i - b)$$

$0 \quad 0 \quad -1$

$$= \boxed{-2 \sum (y_i - m x_i - b)}$$

$=$ slope_b at $b = 0$

$$\frac{\partial L}{\partial m} = 2 \sum (y_i - m x_i - b)$$

$0 \quad -x_i \quad 0$

$$= -2 \sum (y_i - m x_i - b) x_i$$

slope_m at $\boxed{m = 1}$

Code

Thursday, May 20, 2021 1:48 PM

Visualization 2

Thursday, May 20, 2021 1:52 PM

Effect of Learning Data

Thursday, May 20, 2021 1:53 PM

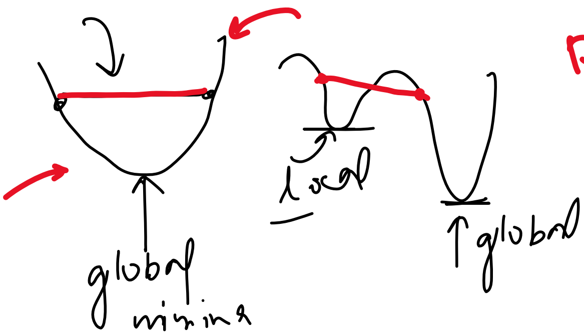
Effect of Loss Function

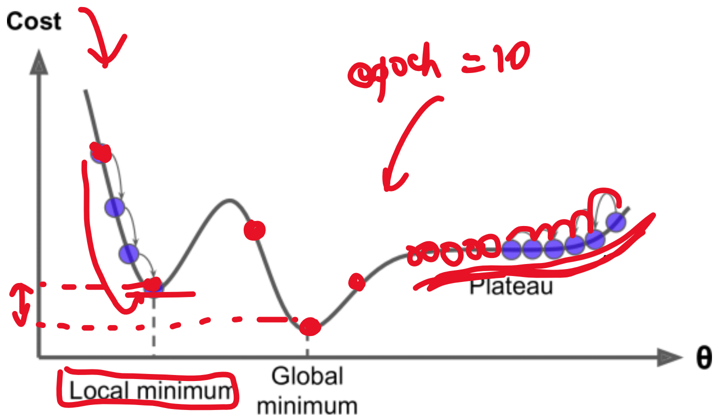
Thursday, May 20, 2021 1:53 PM

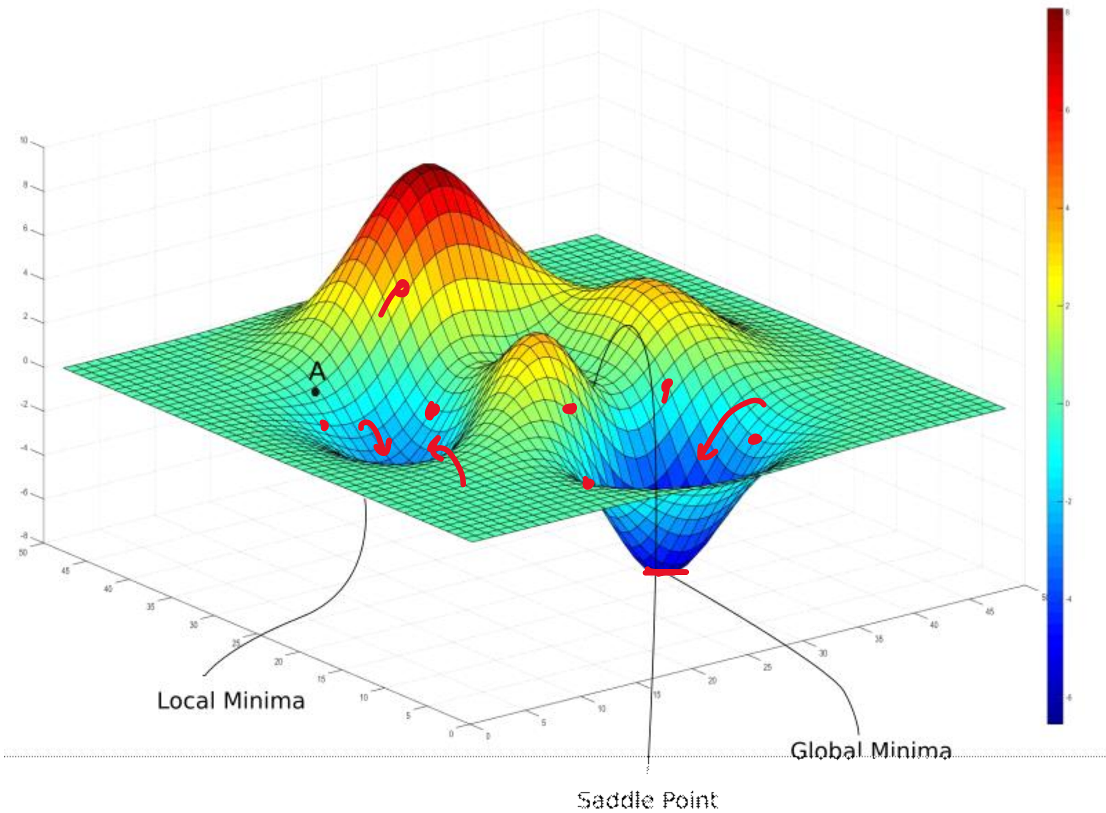
$$L = \sum (y_i - \hat{y}_i)^2 \rightarrow \text{convex function}$$

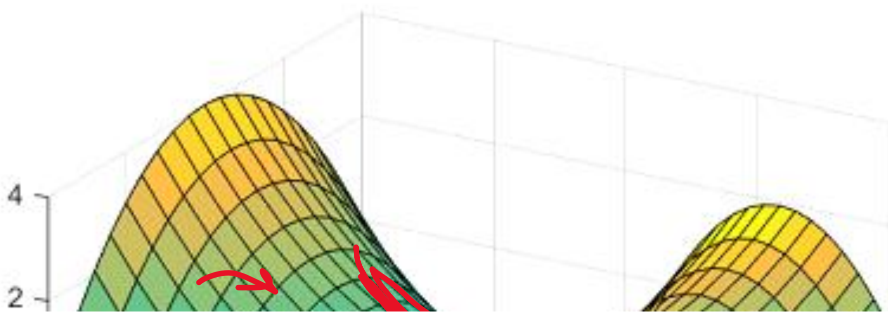
10 cols

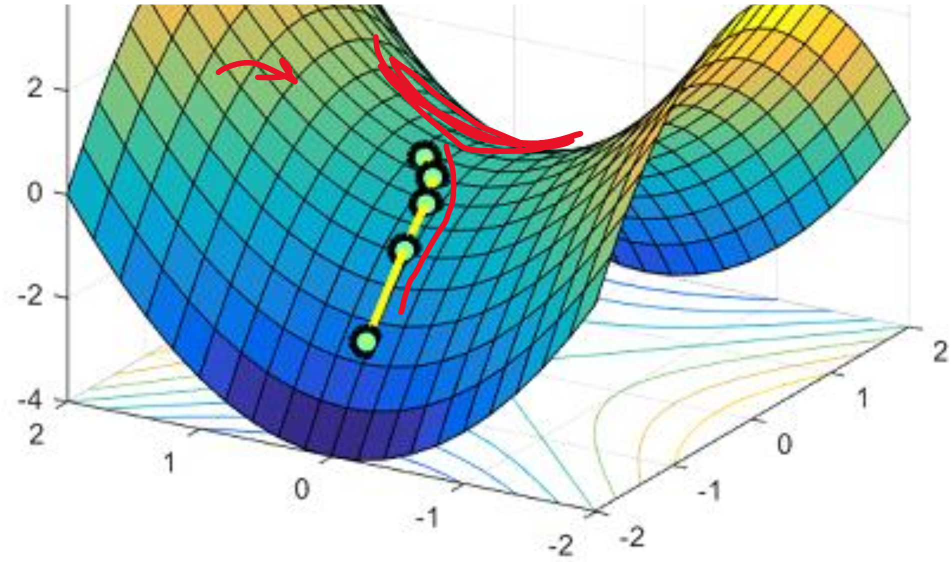
Deep learn





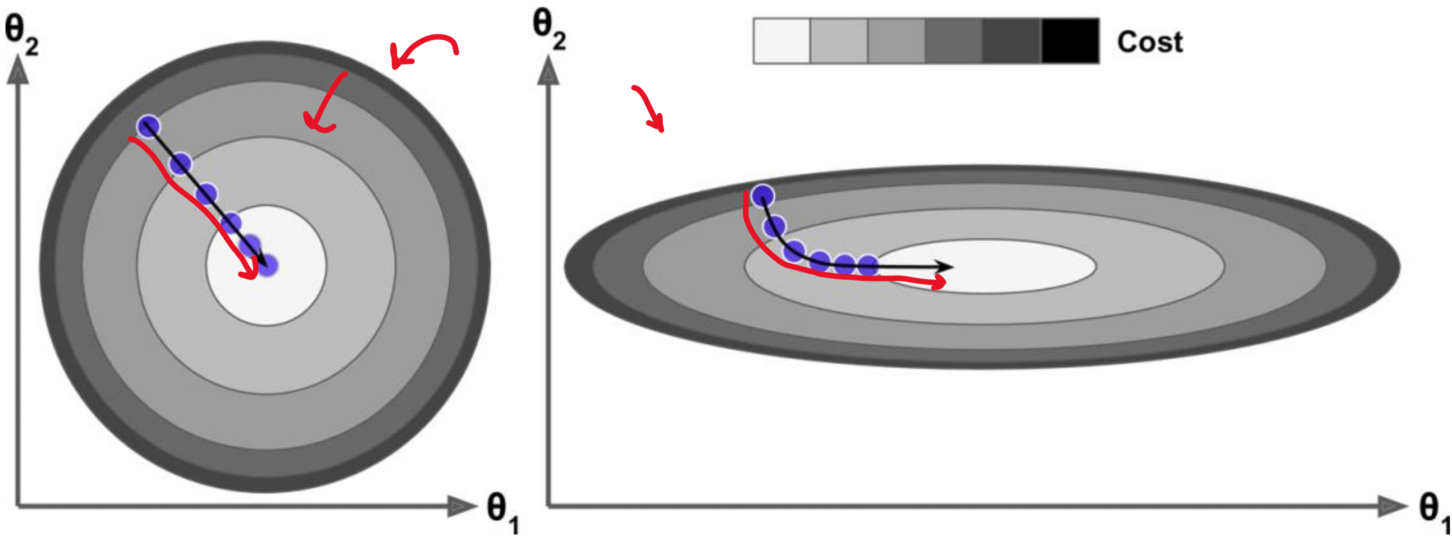





2
0
-2
-4
2
1
0
-1
-2
-2
-1
0
1
2

Effect of Data

Thursday, May 20, 2021 1:53 PM

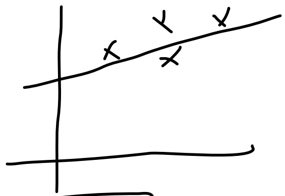


Types of Gradient Descent

Saturday, May 22, 2021 1:30 PM

Gradient Descent

Batch GD | Stochastic GD | Mini batch GD



$\to m, b$

(m, b)

$\frac{dL}{dm_0} \to \boxed{\frac{\partial L}{\partial m}}$ $\qquad \frac{dL}{db} \to \frac{\partial L}{\partial b}$

$m = 1, b = 0$

Batch gd

iterate

$\underline{m_n} = m_0 - \eta \times \boxed{(\text{slope})}_{m=0}$

$\underline{b_n} = b_0 - \eta \times (\text{slope})_{b=0}$

$\boxed{300}$ $\boxed{300}$

10 times

1 update

300 update

$\boxed{\text{Stochastic}}$

1 row

Mini

batch size = 30

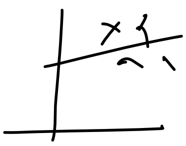
2D

x y

cgpa | lpa

| cgpa | iq | gender | lpa

m, b



$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$

$\{\beta_0, \beta_1, \beta_2, \beta_3\}$

n dim

$\{\beta_0, \beta_1, \beta_2 \ldots \beta_n\}$

Gradient Descent

Batch GD

Saturday, May 22, 2021 3:38 PM

x cgpa | iq | lpa

(2,3)

$\{m, b\}$

n-dim-dataset 3-cols

$x_{11} \mid x_{12}$

$\hat{y}_1$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \quad x_{21} \mid x$$

(lpa) (cgpa) (iq)

$\{\beta_0, \beta_1, \beta_2\}$

$L(\beta_0, \beta_1, \beta_2)$ 4D

1) Random values

$\beta_0 = 0$, $\beta_1, \beta_2 = 1$

$$\frac{\partial L}{\partial \beta_0} \quad \frac{\partial L}{\partial \beta_1} \quad \frac{\partial L}{\partial \beta_2} \quad \frac{\partial L}{\partial \beta_3} \quad \beta_n \beta_m$$

2) epoch = 100, lr = 0.1

$$\begin{cases} \beta_0 = \beta_0 - \eta \, \text{slope} \\ \beta_1 = \beta_1 - \eta \, \text{slope} \\ \beta_2 = \beta_2 - \eta \, \text{slope} \end{cases}$$

n-dim

$(n+1)$ $\beta_0 - \beta_n$

MSE

$\{$row=2, cols = 2+1$\}$

$$L = \frac{1}{2=n} \sum_{i=1}^{n=2} (y_i - \hat{y}_i)^2$$

$\hat{y}_i = \beta_0 +$

$$= \frac{1}{2}\left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2\right]$$

x_1	x_2	y
8.1	9.3	3.2
7.5	9.5	3.5

$$L = \frac{1}{2}\left[(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})^2\right]$$

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12}$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22}$$

$$\frac{\partial L}{\partial \beta_0} = \frac{1}{2}\left[2(y_1 - \hat{y}_1)(-1) + 2(y_2 - \hat{y}_2)(-1)\right]$$

$$\frac{\partial L}{\partial \beta_0} = -\frac{2}{2}\left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2)\right]$$

$$= \frac{-2}{n}\left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + (y_3 - \hat{y}_3) + \dots + (y_n - \hat{y}_n)\right]$$

y → array

(353) (y_i) → [- - -] 353 y_i

$$= \frac{-2}{n}\left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + (y_3 - \hat{y}_3) \dots \right]$$

$$\boxed{= \frac{-2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i) = \frac{\partial L}{\partial \beta_0}}$$

x_i, 353

$y_i \to [\,-\,-\,-\,]$ 353

$353 \times$ model

X_train

353 y-hat

y_i

$\hat{y}_i$

$$L = \frac{1}{2}\sum_{i=1}^{2}(y_i - \hat{y}_i)^2 \qquad \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \frac{\partial}{\partial \beta}\; -\beta_1 x_{11} = -x_{11}$$

$$L = \frac{1}{2}\left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 \right]$$

$$L = \frac{1}{2}\left[(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})^2 \right]$$

	β_1: x_1	β_2: x_2	y
1	8.1	93	3.2
2	7.5	95	3.5
3			
4			
⋮			
n			

x_{i1}: x_{11}, x_{21}, x_{31}, …, x_{n1}

$$\frac{\partial L}{\partial \beta_1} = \frac{1}{2}\left[2(y_1 - \hat{y}_1)(-x_{11}) + 2(y_2 - \hat{y}_2)(-x_{21}) \right]$$

11, 21, 31 . . . n1

$$\frac{\partial L}{\partial \beta_1} = \frac{-2}{n}\left[(y_1 - \hat{y}_1)(x_{11}) + (y_2 - \hat{y}_2)(x_{21}) + (y_3 - \hat{y}_3)(x_{31}) + \dots + (y_n - \hat{y}_n)(x_{n1}) \right]$$

$$\boxed{\frac{\partial L}{\partial \beta_1} = \frac{-2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)\, x_{i1}}$$

$x_{i1} \to$ 1 col data

$\beta_1 \to$ values of 1 col.

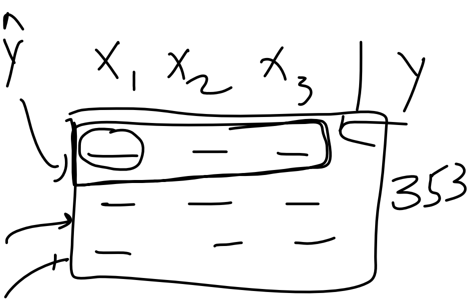
$$\frac{\partial L}{\partial \beta_2} = \frac{-2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)\, x_{i2}$$

m cols

$\beta_0 - \beta_m$

$$\frac{\partial L}{\partial \beta_m} = \frac{-2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)\, x_{im}$$

code



$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13}$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23}$$

β_0 + np.dot(X_train, coef)

(353, 10) (10, 1)

$\beta_0 + (353, 1)$

$\rightarrow (353, 1)$

$$\hat{y} = \beta_0 + \begin{bmatrix} x_{11} & x_{12} & x_{13} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$

$\boxed{\hat{y}}$ = np.dot(coef_, X_train) + β_0

y_pred

x_1	x_2	y	$\hat{y}$
1	2	5	6
3	4	7	8

2

y $\quad$ $\hat{y}$

$[5 \quad 7]$ $[6 \quad 8]$

$\boxed{y - \hat{y}} = [-1 \quad -1]$

$$\frac{\partial L}{\partial \beta_1} = -\frac{2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i) x_{i1} = -\frac{2}{n} \left[[-1 \quad -1] \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right]$$

$$\frac{\partial L}{\partial \beta_2} = [y - \hat{y}] \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$= [-1 \quad -1] \begin{bmatrix} 2 \\ 4 \end{bmatrix} \times \frac{-2}{n}$$

$[-4 + -4]$

$-2 \quad -8 = 8$

(1, 2)

$[X \; Y]$

(1,2) $\quad$ (2,2)

$$\left[[y - \hat{y}] \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \right] \times \frac{-2}{n}$$

$$\frac{\partial L}{\partial \beta_1} \cdots \frac{\partial L}{\partial \beta_{10}} = \left[\boxed{(y_i - \hat{y}_i)} \text{ X\_train} \right] \times \frac{-2}{n}$$

y_train

$y_i = 353$

353

(353, 1) (353, 10) → (1, 353)

$\downarrow$

(1, 353) (353, 10)

(1, 10) × $\boxed{\frac{-2}{n}}$ = (1, 10)

coef_der

The Problem with Batch GD

Tuesday, May 25, 2021 6:43 AM

$n = \underline{100000} \longrightarrow 10^5$

col $\rightarrow$ 5 $\longrightarrow$ ~~6 coefs~~ $\underline{10}^2$

epoch $\rightarrow$ 50 $\longrightarrow 10^3$ 6

total $\rightarrow$ $\boxed{10^{10}}$ $\rightarrow$ slow on big data

1) $\dfrac{50}{1} \longrightarrow 1000$

$6 \longrightarrow \underline{6000}$

→ Deep Learning

→ CNN → images

→ RNN → text

2) Hardware

Stochastic GD

Tuesday, May 25, 2021 6:44 AM

5, 10

$\boxed{\text{100 epochs}}$

Batch

coefs

n rows

Stochastic

row → update

1 epoch

n updates n rows

faster

1 epoch

1 update

→ random

→ faster convergence

→ Stotastic

steady soln

Code

Tuesday, May 25, 2021 6:44 AM

$$\frac{\partial L}{\partial \beta_0} = \frac{-2}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)$$

$m \rightarrow 1$

$i = idx$

$$= -2 (y_i - \hat{y}_i)$$

Time Comparison

Tuesday, May 25, 2021 12:39 PM

no. of epoch → fixed

$e = 100$

batch

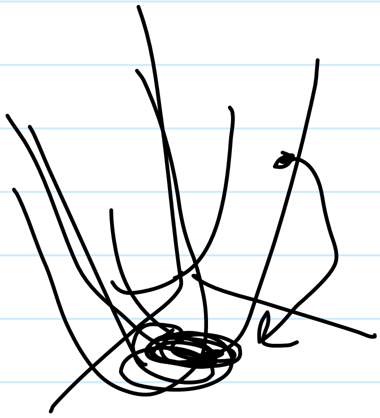
stochastic

→ 100

$100 \times n$ → no. of rows

Visualizations

Tuesday, May 25, 2021 6:44 AM



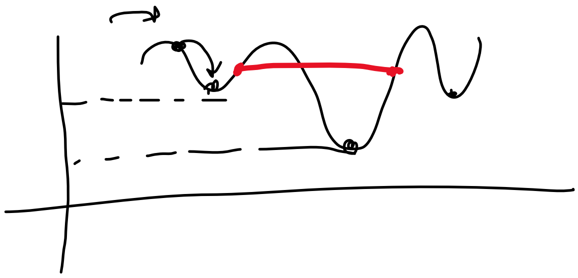
When to use Stochastic GD

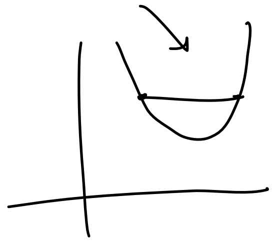
Tuesday, May 25, 2021 6:44 AM

1) Big data → sgd

2) Non convex function

non convex

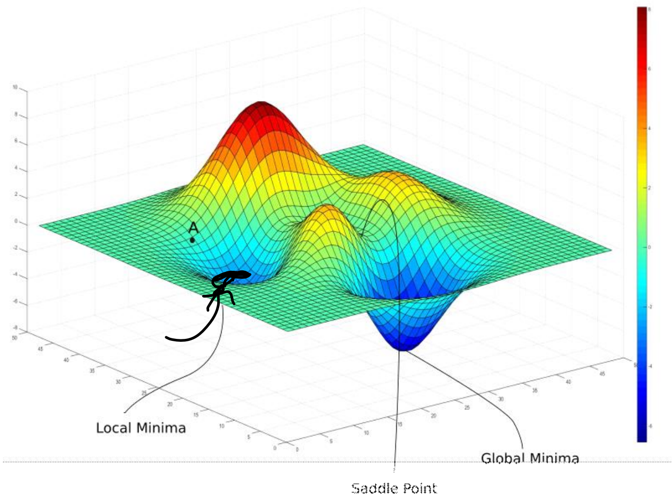


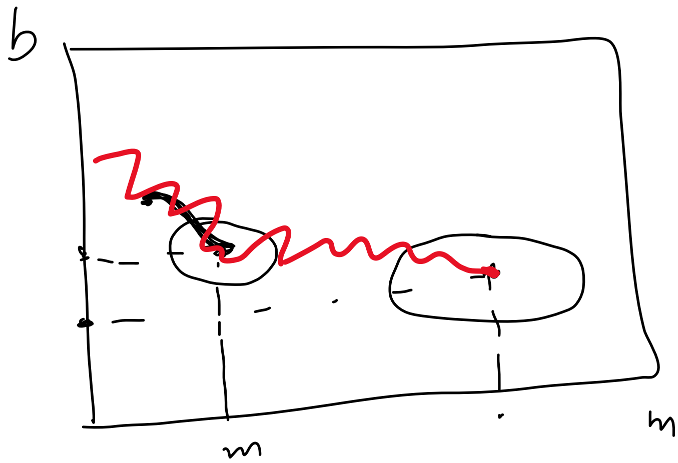


convex

↳ global minima

learning

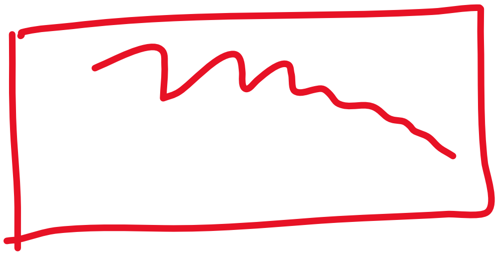




Learning Schedules

Tuesday, May 25, 2021 7:00 AM

DL



$n = 100$

epoch = 1 → lr = 0.1 → lr = 0.03

```
t0,t1 = 5,50
def learning_rate(t):
    return t0/(t + t1)

for i in range(epochs):
    for j in range(X.shape[0]):

        lr = learning_rate(i * X.shape[0] + j)
```

n

Sklearn Implementation

Tuesday, May 25, 2021 2:16 PM

Mini-Batch Gradient Descent

Wednesday, May 26, 2021 4:47 PM

Batch
1 update / epoch
{slow}
{small, convex data}

{Stochastic (n rows)
n updates / epoch
{fast
[bigdata, non convex} → random

n rows
batch_size = n
1 / epoch
batch_size = 1

$\boxed{\text{Mini batch}}$

batch → group of rows
$\boxed{n}$ = 1000
↳ batches → 100 → 10 batches

1000
100
10 update / epoch

batches update / epoch

Code

Wednesday, May 26, 2021 4:47 PM

Visualization

Wednesday, May 26, 2021 4:47 PM

Sklearn Implmentation

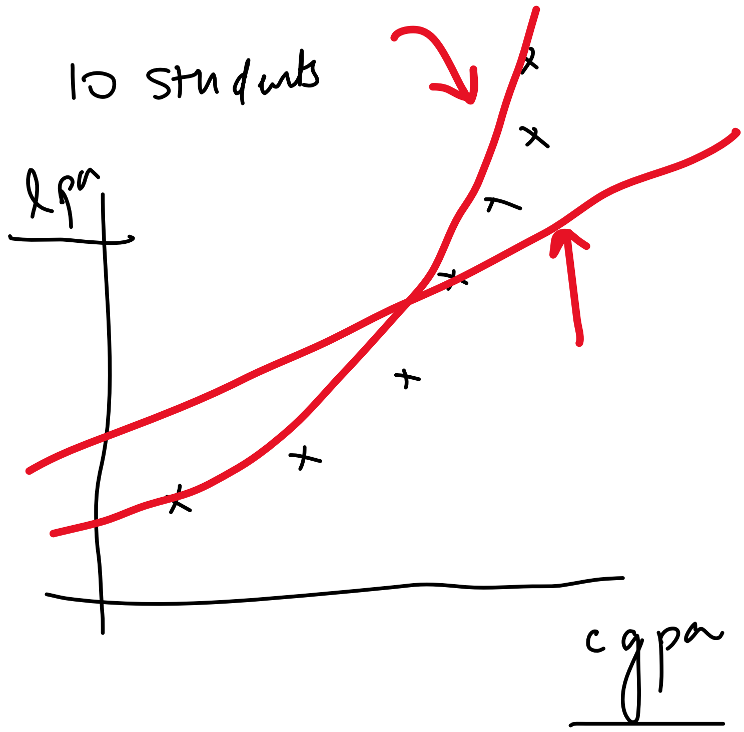
Wednesday, May 26, 2021 4:47 PM

Introduction

Friday, May 28, 2021 1:47 PM

cgpa | lpa

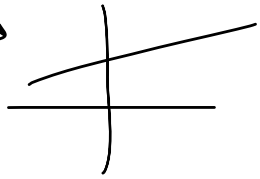
10 students

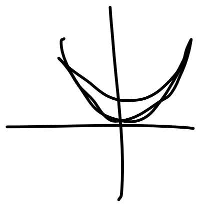
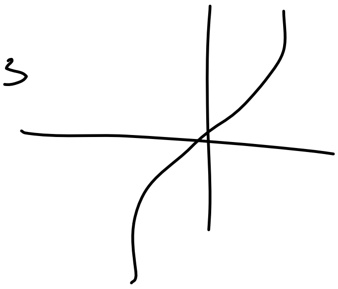


Intuition

Friday, May 28, 2021 1:47 PM

polynomials

$y = mx + b$ 

$y = x^2$ →  $y = x^3$ 

$x^3 + x^2$

$y = ax^4 + bx^3 + cx^2 + dx + 2$

degree

x	y
1	
2	
3	

→ $y = mx + b$

transform polynomial → degree = 2

3

x^0	x^1	x^2
1	2	4
1	3	9

$Y = \beta_0 + \beta_1 x^0 + \beta_2 x^2$

Multiple Polynomial Regression

Friday, May 28, 2021 1:48 PM

Why is it linear

Friday, May 28, 2021 1:48 PM

When to use Polynomial Regression

Friday, May 28, 2021 1:48 PM

Bias And Variance

Tuesday, June 1, 2021 12:20 PM

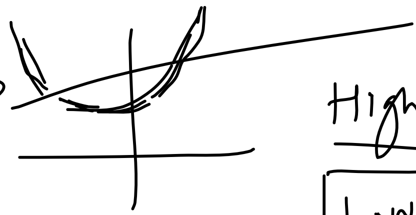
Bias

x	y
2	4
6	36
10	100

$\boxed{y = mx + b}$ → bias

$\boxed{y = x^2}$

non-biased



High bias

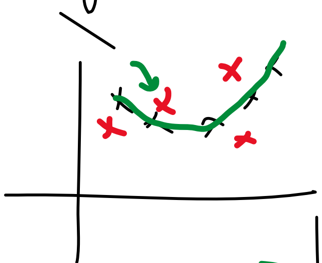
$\boxed{\text{Low bias}}$

Variance → high variance → Low variance

$\boxed{\text{Overfitting}}$

Low bias

High variance



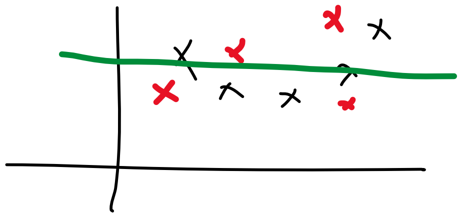
{LB, LV}

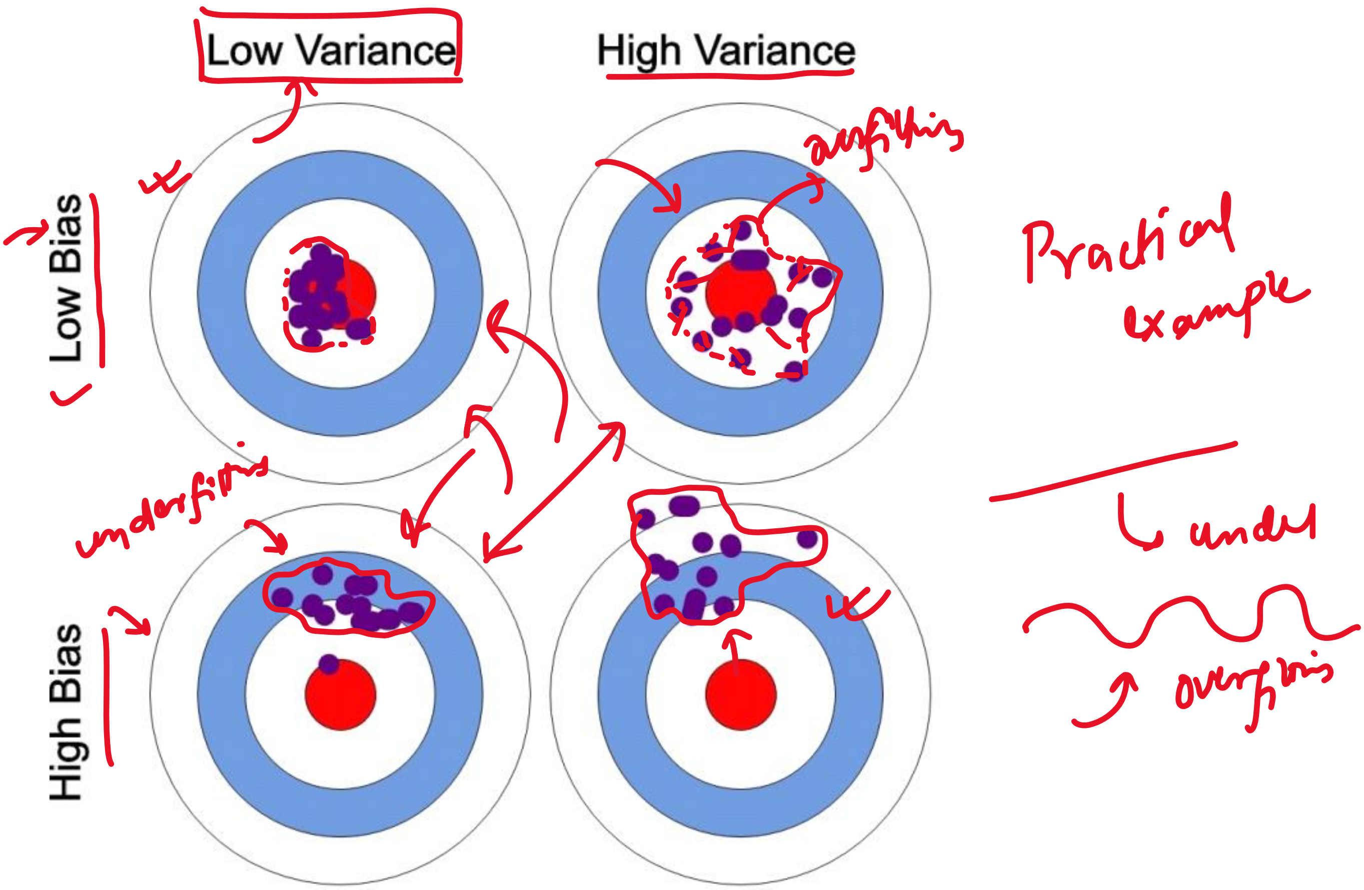
High variance

underfitting

High bias

Low variance





How to calculate Bias and Variance

Tuesday, June 1, 2021 1:44 PM

model → Bias
model → Variance
model → Total error
↳ gen. error

→ datasets
↳ sampling

$$\boxed{Bias = E[\hat{\theta}] - \theta}$$

Expected value → Probability

X → input, y → output

X → (300, 5)
y → 300

x_1 x_2 x_3 x_4 x_5 y (300, 6)

Step 1 – train-test split
[X_train, y_train (250, 5) (250)] ⟷ [X_test, y_test] [(50, 5) (50,)]

50 times → 1

X_train, y_train

sampling with replacement

50 datasets

X_train, y_train

5 input → 6

Step 2 → model → degree → 1
50 times

training your model on different data → 2

m1 – m50

Step 3 50 models X_test, y_test (50 rows)

m1 ← X test → y_pred 1 [ ]
m2 ← X_test → y_pred 2 [ ]
[ ]

50 values

(50, 50)

→ [y_pred1]
[y_pred2]
→ [y_pred4]
⋮
[y_pred50]

col wise mean

m2 ← X_test → y_pred2

50

50 row

[y_pred50]

mean_prediction ← 50 values

$$\frac{(\text{mean\_prediction} - \text{y\_test})^2}{50} = \boxed{\text{bias}}$$

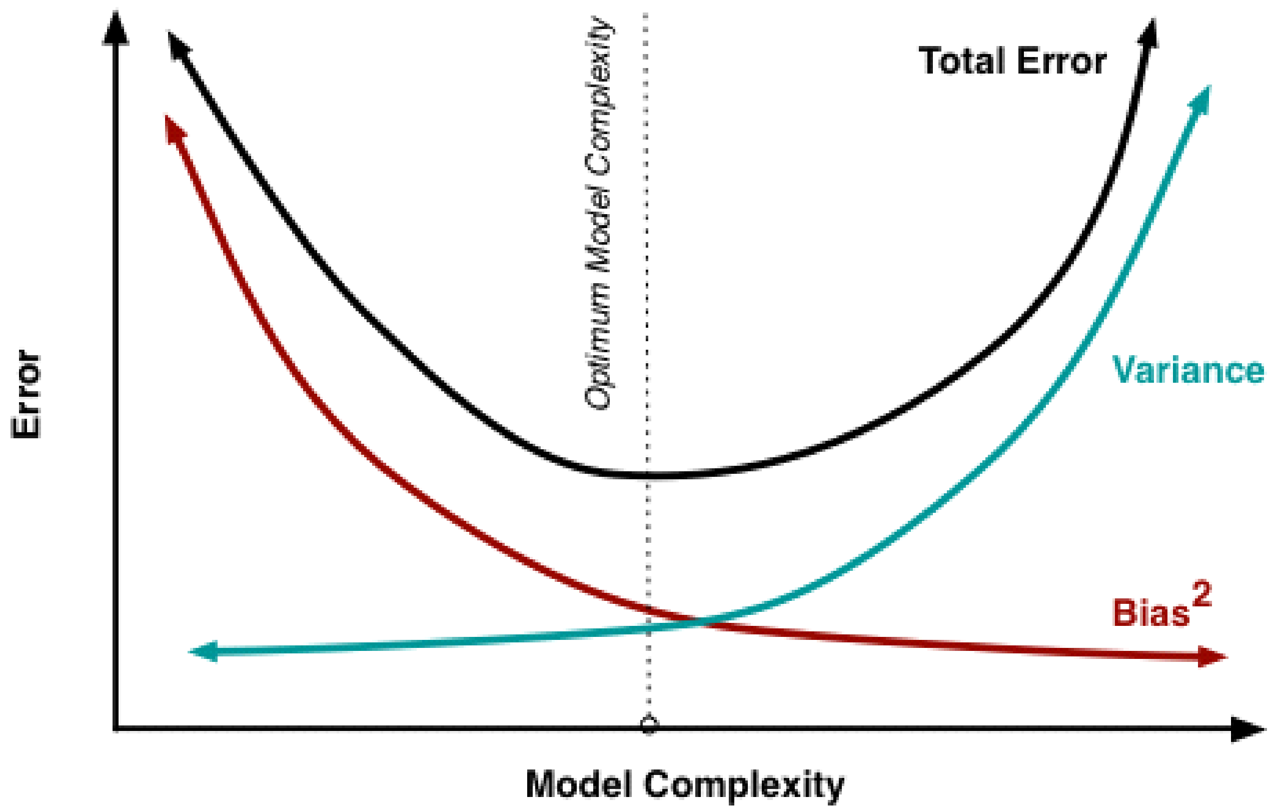
Variance

Code

Monday, May 31, 2021 9:26 AM

Bias Variance Decomposition for Squared Error

Monday, May 31, 2021 9:26 AM

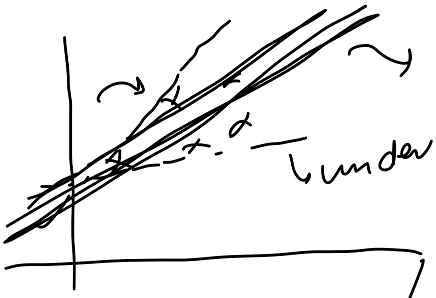


Relationship between Bias and Variance

Monday, May 31, 2021 9:26 AM

Ridge Regression

Wednesday, June 2, 2021 11:13 AM



bias ↑ variance ↓

under

$$y = mx + b$$

2D lines

m slope

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda m^2$$

m

$$\boxed{\frac{\partial L}{\partial b} = 0} \rightarrow b$$

$$\boxed{\frac{\partial L}{\partial m} = 0} \rightarrow m$$

$$L = \sum_{i=1}^{n} (y_i - m x_i - b)^2 + \lambda m^2$$

$$b = \bar{y} - m\bar{x}$$

$\bar{y} \rightarrow$ y_mean

$\bar{x} \rightarrow$ x_mean

$m \rightarrow$ slope

$$L = \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x})^2 + \lambda m^2$$

$$\frac{\partial L}{\partial m} = 2 \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) + 2\lambda m = 0$$

$$= -\cancel{2} \sum_{i=1}^{n} (y_i - \bar{y} - m x_i + m\bar{x})(x_i - \bar{x}) + \cancel{2}\lambda m = 0$$

$$= \lambda m - \sum_{i=1}^{n} \left[(y_i - \bar{y}) - m(x_i - \bar{x}) \right] (x_i - \bar{x}) = 0$$

$$= \lambda m - \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) - m (x_i - \bar{x})^2 = 0$$

$$= \lambda m - \underline{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})} + m \sum_{i=1}^{n} (x_i - \bar{x})^2 = 0$$

$$= \lambda m + m \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

$$= \boxed{m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2 + \lambda}}$$

λ → hyperparameter, alpha

$\lambda = 0 = \lambda = 10{,}100$

$$\boxed{m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}}$$

m → $b = \bar{y} - \underline{m\bar{x}}$

Ridge Regression for 2D data

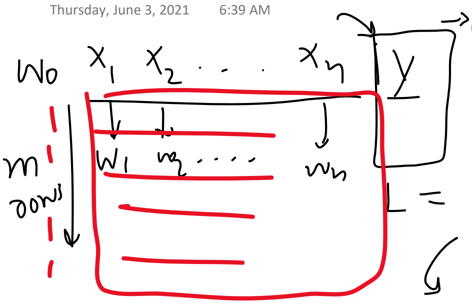
Thursday, June 3, 2021 6:38 AM

Code

Thursday, June 3, 2021 6:39 AM

Ridge Regression for nD data

Thursday, June 3, 2021 6:39 AM



$\to$ Ridge

$$L = \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

$$= (XW - Y)^T (XW - Y)$$

m vals

$$Y = \begin{bmatrix} - \\ - \\ - \\ - \\ - \end{bmatrix} \quad W = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ 1 & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$

$(n \times 1)$

Normal LR $\to$ Ridge

$$L = \boxed{(XW - Y)^T (XW - Y) + \lambda \|W\|^2}$$

$\lambda w_0^2 + \lambda w_1^2 + \lambda w_2^2 + \dots + \lambda w_n^2$

$$[w_0\ w_1\ w_2 \dots w_n] \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \quad W^T W \to \lambda (w_0^2 + w_1^2 + w_2^2 \dots w_n^2)$$

1 $(a-b)^T = a^T - b^T$

$\frac{dL}{dW}$ $(ab)^T = b^T a^T$

$\boxed{Y^T W X \to W X^T Y^T}$

2 $L = (XW - Y)^T (XW - Y) + \lambda W^T W$

3 $L = [(XW)^T - (Y)^T] (XW - Y) + \lambda W^T W$

4 $= (W^T X^T - Y^T)(XW - Y) + \lambda W^T W$

5 $= W^T X^T X W - W^T X^T Y - Y^T X W + Y^T Y + \lambda W^T W$

$$L = W^T X^T X W - 2 W^T X^T Y + Y^T Y + \lambda W^T W$$

$$\frac{dL}{dW} = \cancel{2} X^T X W - \cancel{2} X^T Y + 0 + \cancel{2} \lambda W = 0$$

$$X^T X W + \lambda W = X^T Y$$

3 (4×4)

$X^TXw + \lambda w = X^Ty$

$3 \quad (4 \times 4)$

$(X^TX + \lambda I)w = X^Ty$

$(n \times 1, n \times 1)$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{w = (X^TX + \lambda I)^{-1} X^Ty}$$

$w = (X^TX)^{-1} X^Ty$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Code

Thursday, June 3, 2021 6:39 AM

Ridge Regression using Gradient Descent

Friday, June 4, 2021 2:17 PM

Vector form loss

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$L = (XW - Y)^T (XW - Y) + \lambda ||W||^2$$

$$\boxed{L = (XW - y)^T (XW - y) + \lambda W^T W}$$

$x_1 \, x_2 \dots x_n$ y

m rows

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & x_{23} & \dots & x_{2n} \\ 1 & x_{m1} & x_{m2} & x_{m3} & \dots & x_{mn} \end{bmatrix} \quad \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \quad \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \to (n+1)$$

$w_0, w_1, \dots w_n$ (parameters)

$$w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}, \quad w_1 = w_1 - \eta \frac{\partial L}{\partial w_1} \quad \dots \quad w_n = w_n - \eta \frac{\partial L}{\partial w_n}$$

$$W_{new} = W_{old} - \eta \boxed{\frac{\partial L}{\partial W}} \to \text{gradient} \begin{bmatrix} \partial L / \partial w_0 \\ \partial L / \partial w_1 \\ \vdots \\ \partial L / \partial w_n \end{bmatrix}$$

L, W

$$L = \frac{1}{2}(XW - y)^T (XW - y) + \frac{1}{2}\lambda W^T W$$

$$= \frac{1}{2}(W^T X^T - y^T)(XW - y) + \frac{1}{2}\lambda W^T W$$

$$= \frac{1}{2}\left[W^T X^T X W - \boxed{W^T X^T y} - \boxed{y^T W X} + y^T y \right] + \frac{1}{2}\lambda W^T W$$

$$= \frac{1}{2}\left[W^T X^T X W - \boxed{2 y^T W X} + y^T y \right] + \frac{1}{2}\lambda W^T W$$

$2 W^T X^T y$

$X^T y$

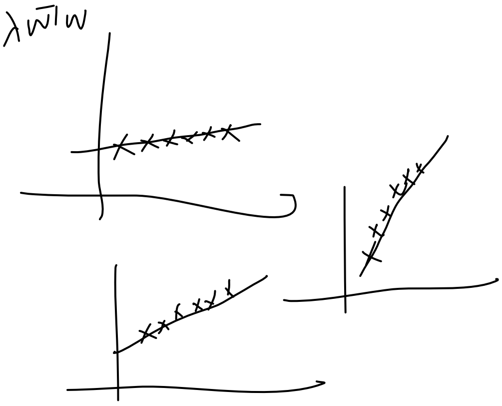
$$\frac{dL}{dW} = \frac{1}{\cancel{2}}\left[\cancel{2} X^T X W - \cancel{2} \cancel{y^T X} \right] + \frac{1}{\cancel{2}} \cancel{2} \lambda W$$

$X^T y$

$$= \boxed{X^T X W - \cancel{X^T X} + \lambda W} = \frac{dL}{dW} \quad \left(\frac{\partial L}{\partial W}\right)$$

$$W = \begin{array}{c} w_0 \quad w_1 \dots \dots w_n \\ \left[0, 1 \dots \dots 1\right] \end{array}$$

starting



$$W = \begin{array}{cccc} w_0 & w_1 & \cdots & w_n \\ \end{array}$$
$$W = [0, 1 \ldots\ldots 1]_{n+1}$$ starting

epochs

$$W = W - \eta \frac{dL}{dW}$$

$W \rightarrow$ final answer

epoch times

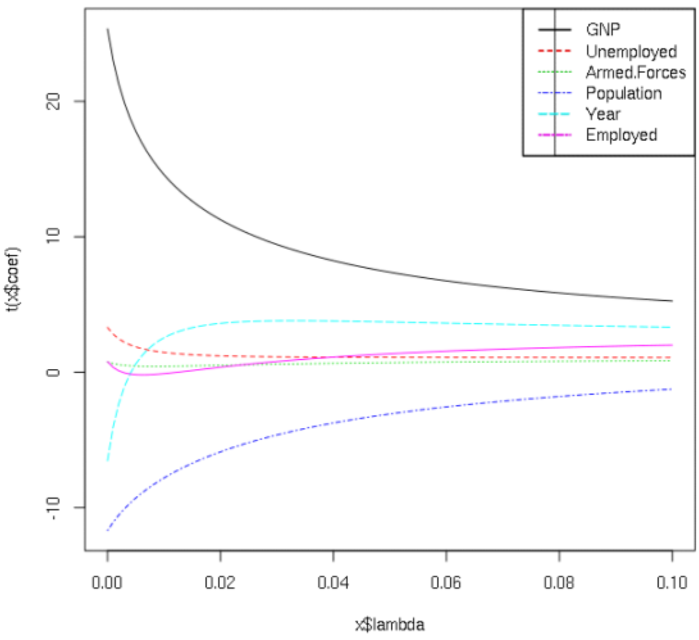
$\hookrightarrow W = W - \eta \frac{dL}{dW}$

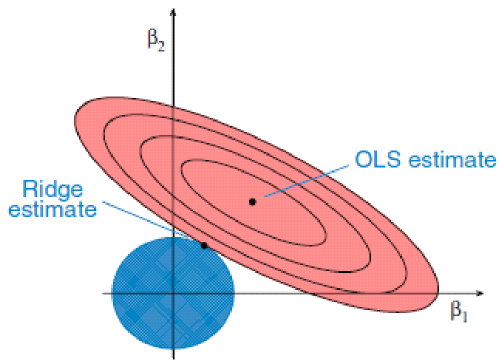
$$\boxed{\frac{dL}{dW} = X^T X W - X^T y + \lambda W}$$

Notes

Friday, June 4, 2021 4:47 PM

Why is it called ridge





5 Key Understandings

Saturday, June 5, 2021 4:20 PM

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \boxed{\lambda \|w\|^2}$$

$\lambda(w_1^2 + w_2^2 + \dots + w_n^2)^2$

Overfitting ↓

{ Shrinkage
coef

1. How the coefficients get affected?

Saturday, June 5, 2021 4:20 PM

$\lambda \uparrow$

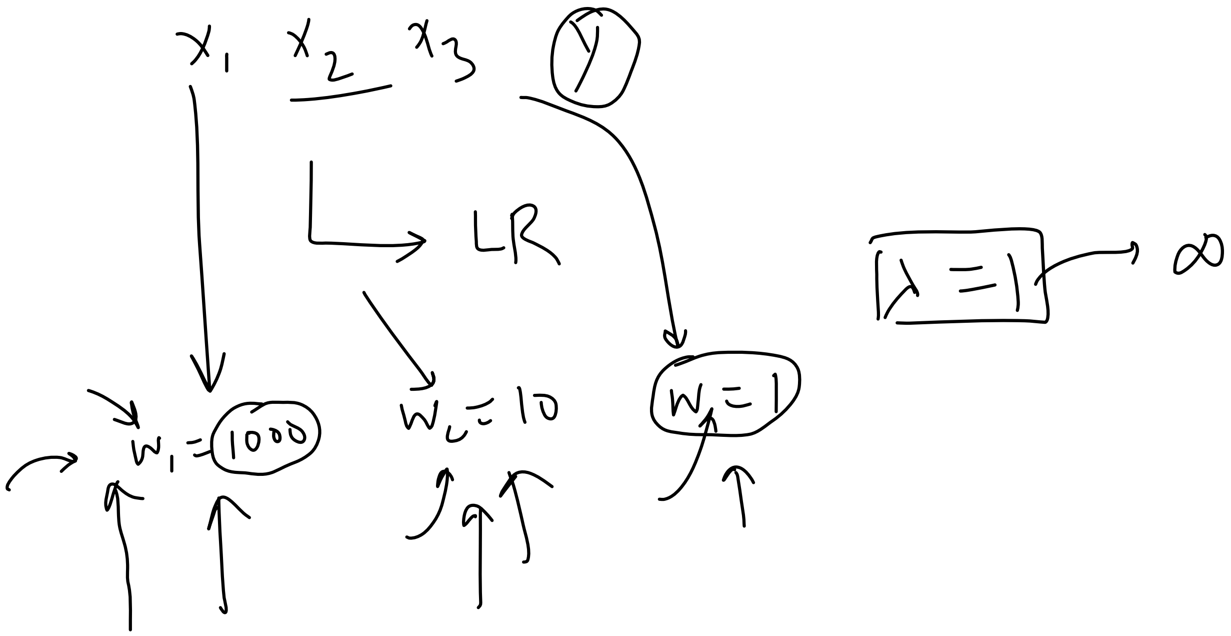
$\lambda \rightarrow 0 \rightarrow \infty$

$\boxed{w_0, w_2 \ldots w_n} \rightarrow \downarrow 0 \rightarrow 0$

2. Higher Values are impacted more

Saturday, June 5, 2021 4:21 PM

Never reaches 0



3. Bias Variance Tradeoff

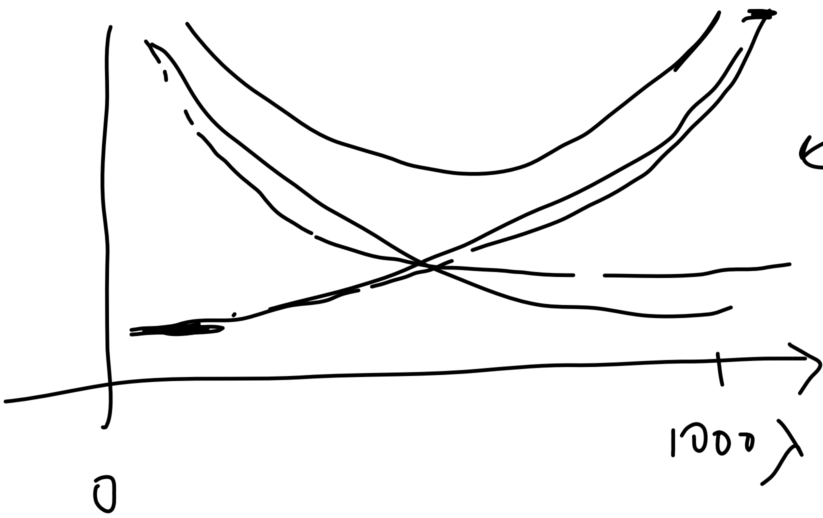
Saturday, June 5, 2021 4:21 PM

Bias Variance

$\lambda \downarrow \uparrow$

Bias ↓ overfit Variance ↑

Bias ↑ underfitting Varaince ↓



← graph

4. Impact on the Loss Function

Saturday, June 5, 2021 4:21 PM

$\lambda \rightarrow$

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \|W\|^2$$

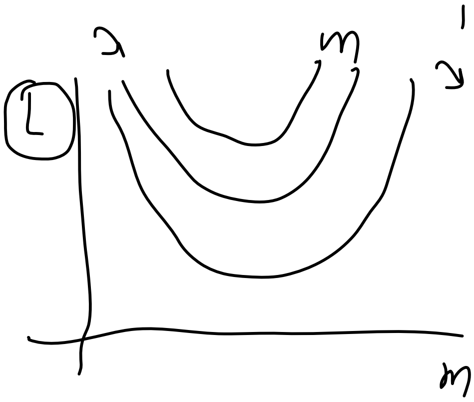
$x, y \rightarrow \boxed{m, b} \rightarrow$ b is constant

$b = -2.29$

m

$b = 0$

$$L = \sum_{i=1}^{n} (y_i - m x_i)^2 + \lambda m^2$$

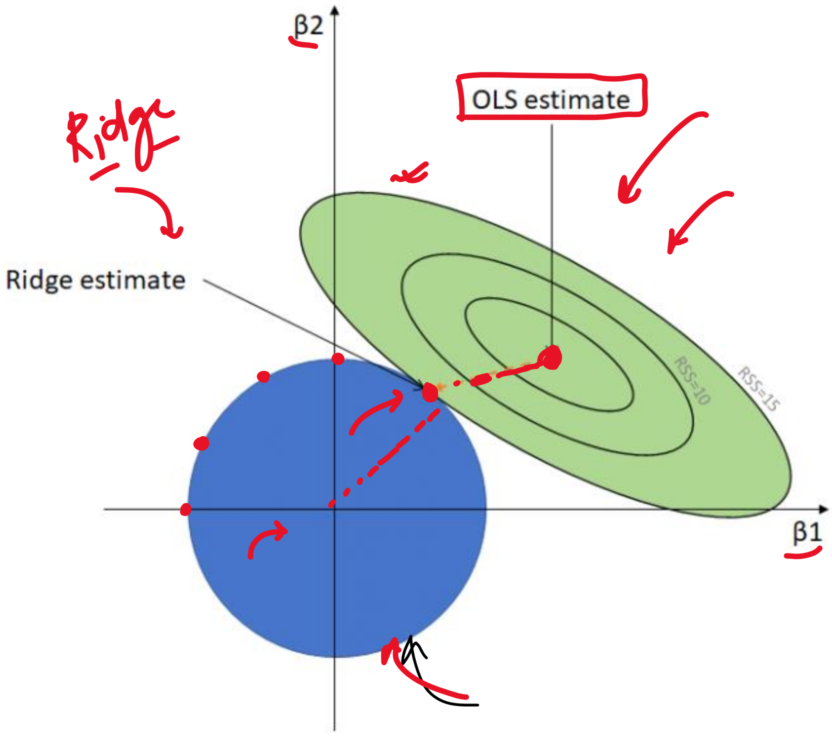


$\lambda = 1$

$\lambda =$

5. Why called Ridge

Saturday, June 5, 2021 4:22 PM



Hard constraint

Ridge constraint

2 coef, β_1 β_2 β_0

$L = \underbrace{MSE} + \underbrace{\lambda \|w\|^2}$

contour

$(y_i - \hat{y}_i)^2$

$$\sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}))^2$$

MSE

$$\boxed{\lambda(\beta_1^2 + \beta_2^2)}$$

Practical Tip

Monday, June 7, 2021 1:20 PM

Use ridge when there are more than 2 input cols

Ridge

$x_1 \quad x_2 \quad x_3 \; . \; . \; .$

coeffs

$>= 2$

$x_1 \quad y$

Lasso Regression

Thursday, June 10, 2021 6:42 AM

$y = mx + b$ $\quad$ $y = b$

$\downarrow \quad \lambda \uparrow$ $\quad$ under

L1 Regularization | overfitting $\rightarrow$ L2 reg

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda ||w||^2$$

$\lambda \geq 0$

$\hookrightarrow \lambda (w_1^2 + w_2^2 + \dots + w_n^2)$

$\rightarrow 0$ $\quad$ $w_1 \rightarrow w_n \rightarrow$ coeff

overfitting | under

alpha

$\hookrightarrow$ Lasso $\quad$ L1 norm $\rightarrow [\,|w_1| + |w_2| + |w_3| + \dots + |w_n|\,]$

$$L = MSE + \boxed{\lambda ||w||}$$

$\rightarrow$ code implement | sklearn

$\rightarrow$ Key point $\rightarrow$

$\rightarrow$ Difference Ridge Vs Lasso

underfitting

$\lambda = 0 \rightarrow$ Linear

overfitting

$x_1 \quad x_2 \quad x_3 \quad 0 \quad 0 \quad x_n \quad y$ $\quad \leftarrow$ degree

0.5 $\quad$ 0.2

ridge

lasso $\rightarrow \lambda \rightarrow \uparrow$

feature selection $\rightarrow$ 0 $\quad$ lasso

dim < $\quad$ decrease

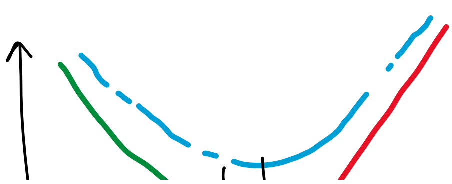
$\lambda \uparrow$

overfitting $\downarrow$ $\quad$ Bias $\uparrow$

Variance $\downarrow$

Varince $\uparrow$ $\quad$ Bias $\downarrow$

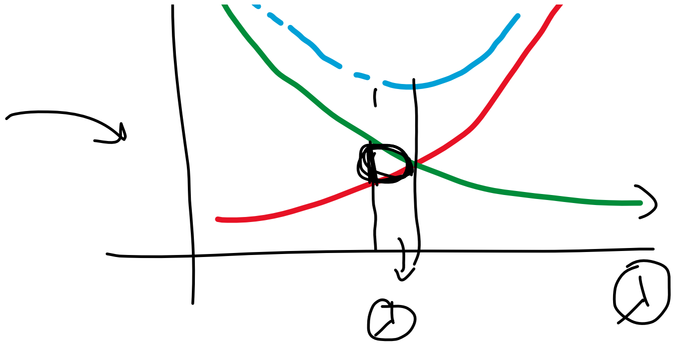
values



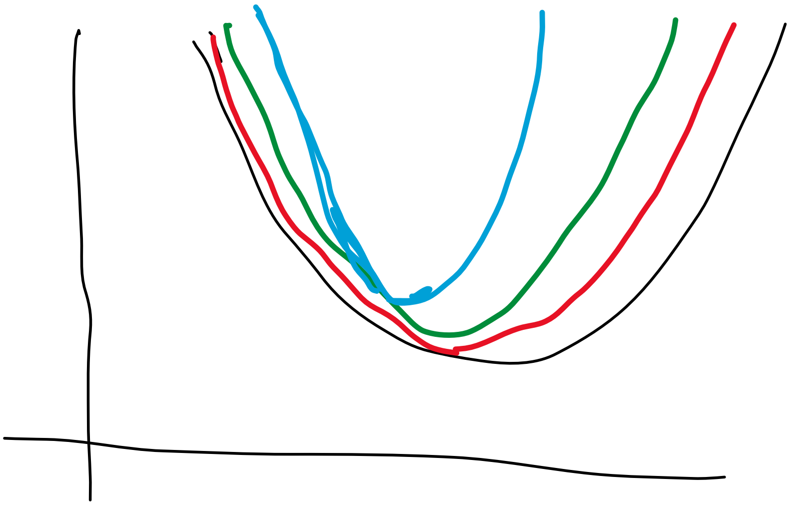
Red $\rightarrow$ bias

green $-$ variance

blue $-$ total error



blue - total error



Understanding Sparsity

Friday, June 11, 2021 11:21 AM

Lasso sparsity

$\lambda \uparrow \quad w \to 0$

Ridge

alpha/λ	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6
0.0000	-9.160885	-205.462260	516.684624	340.627341	-895.543609	561.214533	153.884786	126.734316	861.121400	52.419828
0.0001	-9.118336	-205.337133	516.880570	340.556792	-883.415291	551.553259	148.578680	125.355917	856.480254	52.467627
0.0010	-8.763583	-204.321125	518.371729	339.975385	-787.690766	475.274718	106.786540	114.632063	819.739542	52.872100
0.0100	-6.401088	-198.669767	522.048548	336.348363	-383.709187	152.663678	-66.060583	75.611090	659.869402	55.828128
0.1000	6.642753	-172.242166	485.523872	314.682122	-72.939323	-80.590053	-174.466515	83.616653	484.363285	73.584154
1.0000	42.242217	-57.305508	282.170831	198.061386	14.363544	-22.551274	-136.930053	102.023193	260.104308	98.552274
10.0000	21.174004	1.659796	63.659772	48.493240	18.421492	12.875448	-38.915435	38.842464	61.612405	35.505355
100.0000	2.858979	0.629452	7.540604	5.849997	2.710879	2.142134	-4.834047	5.108223	7.448466	4.576129
1000.0000	0.295726	0.069290	0.769004	0.597829	0.282900	0.225936	-0.495607	0.527031	0.761497	0.471029
10000.0000	0.029674	0.006995	0.077054	0.059915	0.028412	0.022715	-0.049686	0.052870	0.076321	0.047241

single $x \mid y \rightarrow$

alpha	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6
0.0000	-9.160885	-205.462260	516.684624	340.627341	-895.543596	561.214523	153.884780	126.734314	861.121395	52.419828
0.0001	-9.071288	-205.337332	516.780313	340.539730	-888.652320	555.952271	150.585260	125.453044	858.639860	52.379002
0.0010	-8.264924	-204.213177	517.641106	339.751339	-826.653342	508.609613	120.899583	113.924518	836.314382	52.011583
0.0100	-1.361404	-192.944226	526.348511	332.649058	-430.205495	191.277876	-44.048113	68.990747	688.384976	47.939528
0.1000	$\boxed{0.000000}$	-113.976046	526.737112	292.635423	-82.691928	$\boxed{-0.000000}$	-152.691332	$\boxed{0.000000}$	551.077200	7.169852
1.0000	0.000000	0.000000	363.882636	27.278420	0.000000	0.000000	-0.000000	0.000000	336.135971	0.000000
10.0000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	-0.000000	0.000000	0.000000	0.000000
100.0000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	-0.000000	0.000000	0.000000	0.000000
1000.0000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	-0.000000	0.000000	0.000000	0.000000
10000.0000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	-0.000000	0.000000	0.000000	0.000000

$\rightarrow$ feature selection

$$m = \frac{\sum_{i=1} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1} (x_i - \bar{x})^2 + \boxed{\lambda}}$$

simple

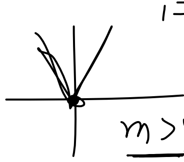
$x \mid y \rightarrow \quad y = mx + b$

$\hookrightarrow \quad b = \bar{y} - m\bar{x}$

$\bar{y} \to \text{mean}(y)$

$\bar{x} \to \text{mean}(x)$

$$m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$



$|m| = m$

$m > 0$

$b = \overline{\bar{y} - m\bar{x}}$

$m = ?$

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda |m|$$

0

$|m| = -m$

λm

$$\frac{d}{dm} \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x})^2 + 2\lambda m$$

$$\frac{dL}{dm} = \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x})^2 + 2\lambda|m| = 2\sum (y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) + 2\lambda = 0$$

$$-2\sum \left[(y_i - \bar{y}) - m(x_i - \bar{x}) \right] (x_i - \bar{x}) + 2\lambda = 0$$

$$m \sum (x_i - \bar{x})^2 = \sum (y_i - \bar{y})(x_i - \bar{x}) - \lambda$$

$$m = \sum (y_i - \bar{y})(x_i - \bar{x}) - \lambda$$

$$-\sum \left[(y_i-\bar{y})(x_i-\bar{x}) - m(x_i-\bar{x})^2 \right] + \lambda = 0$$

$$-\sum (y_i-\bar{y})(x_i-\bar{x}) + m\sum (x_i-\bar{x})^2 + \lambda = 0$$

$$\boxed{m = \frac{\sum (y_i-\bar{y})(x_i-\bar{x}) - \lambda}{\sum (x_i-\bar{x})^2}}$$

Lasso coeff sparsity

$\boxed{\text{for } m>0}$ $\lambda > 0$

$$m = \frac{\boxed{\sum (y_i-\bar{y})(x_i-\bar{x})} - \lambda}{\sum (x_i-\bar{x})^2}$$

for $m=0$

$$m = \frac{\sum (y_i-\bar{y})(x_i-\bar{x})}{\sum (x_i-\bar{x})^2}$$

for $m<0$

$$m = \frac{\sum (y_i-\bar{y})(x_i-\bar{x}) + \lambda}{\sum (x_i-\bar{x})^2}$$

$m \rightarrow$

$2, \frac{9}{5}, 1, 0, -1, 5$ $\quad 0 \rightarrow$

$\lambda \uparrow$

$$m = \frac{YX - \lambda}{X^2} \qquad \left\{ \begin{array}{l} YX = 100 \\ X^2 = 50 \end{array} \right.$$

$\lambda = 100$ $\boxed{m = 0}$ $\quad 150$

$$m = \frac{100 - \lambda}{50}$$

$\lambda = 0$, $m = 2$ | $\lambda = 10$, $m = \frac{9}{5}$

$\lambda > 100$, $m = -1$ $\quad 150$

$\lambda = 50$ $m = 1$

$$m = \frac{YX + \lambda}{X_2} = \frac{100 + \lambda}{50} = \frac{100 + 150}{50} =$$

$m = 5$

λ -ve

$m < 0$

$\lambda > 0$

$$m = \frac{\sum (y_i-\bar{y})(x_i-\bar{x}) + \lambda}{\sum (x_i-\bar{x})^2}$$

$$m = \frac{-100 - \lambda}{50} = \frac{-100 - 150}{50} = -5$$

$$m = \frac{-100 + \lambda}{50}$$

$\lambda = 0$ $m = -2$

$\lambda = 50$ $m = -1$

$\lambda = 100$ $m = 0 \rightarrow 1$

$\lambda = 150$

$\lambda = 150$, $m = -5$

$\boxed{m = 1}$

1) $0 \rightarrow$

2) stop

$$m = \frac{\sum (y_i-\bar{y})(x_i-\bar{x})}{\sum (x_i-\bar{x})^2 + \lambda}$$

0.1 $\quad$ 0 X

0

$\lambda \rightarrow$

Ridge $<<0$

$==0$

Lasso $\rightarrow \lambda \rightarrow$ numerator

$\lambda = 10000000$ $\quad$ Denominator

ElasticNet Regression

Saturday, June 12, 2021 1:04 PM

Ridge

$\lambda(w_1^2 + w_2^2 + \dots + w_n^2)$

L_2

$$L = \underbrace{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}_{mse} + \underbrace{\lambda \|w\|^2}_{overfitting}$$

$\lambda \uparrow \quad w \downarrow 0$

100 cols

Lasso

$\lambda(|w_1| + |w_2| + \dots + |w_n|)$

L_1

$$L = \underbrace{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}_{mse} + \lambda \|w\|$$

$\lambda \uparrow \quad w \to 0 \to$ feature selection

EN Reg

$$L = \sum (y_i - \hat{y}_i)^2 + a\|w\|^2 + b\|w\|$$

$\{\lambda, l1\_ratio\}$

$\boxed{l_1, \lambda}$

$\lambda = a + b$

$l1\_ratio = \dfrac{a}{a+b}$

$l1 = \dfrac{a}{\lambda}$

$a = l_1 \times \lambda$

$b = \lambda - a$

$\lambda = 1 \qquad l1\_ratio = 0.5$

$a = 0.5 \qquad b = 0.5$

$l1\_ratio > 0.9$ ↙ 90% ridge 10% lasso

Input cols → multicollinearity (ElasticNet)

$x_1 \mid x_2$

Introduction

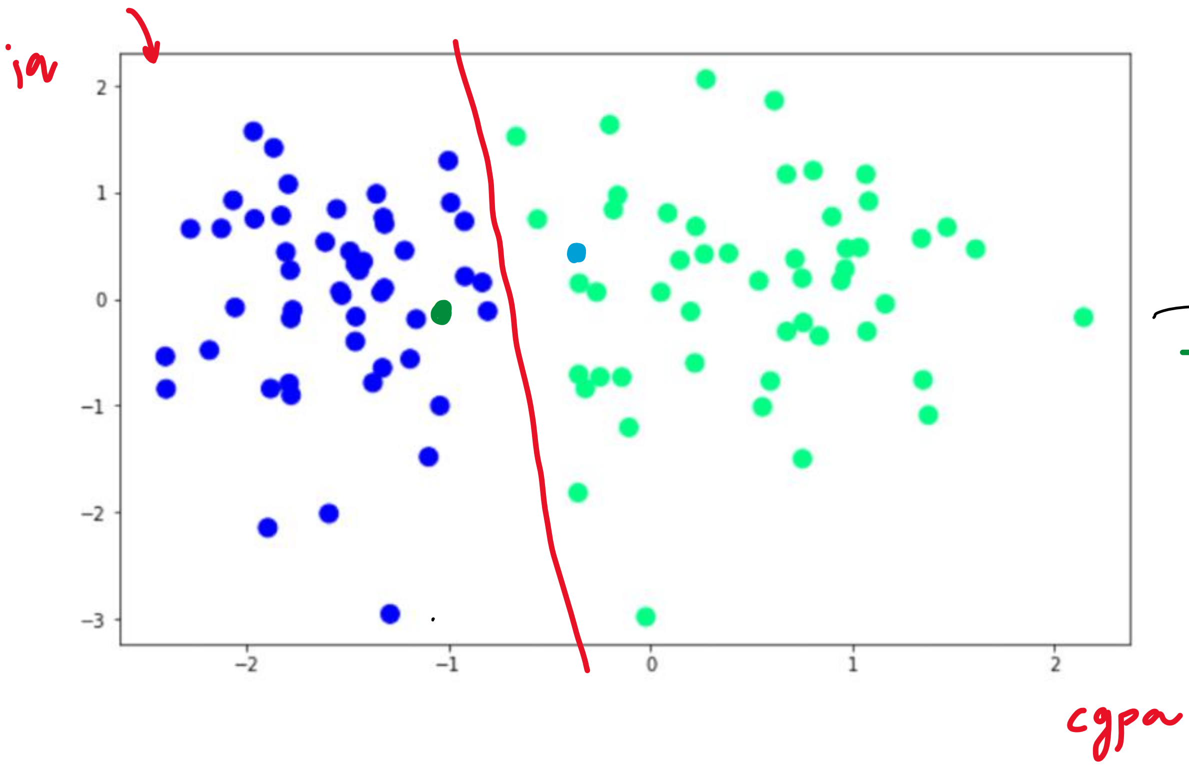
Tuesday, June 15, 2021 12:07 PM

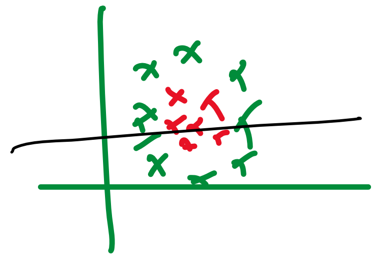
2 perspectives

- geometric
- Probability

Requirement

Tuesday, June 15, 2021 12:07 PM



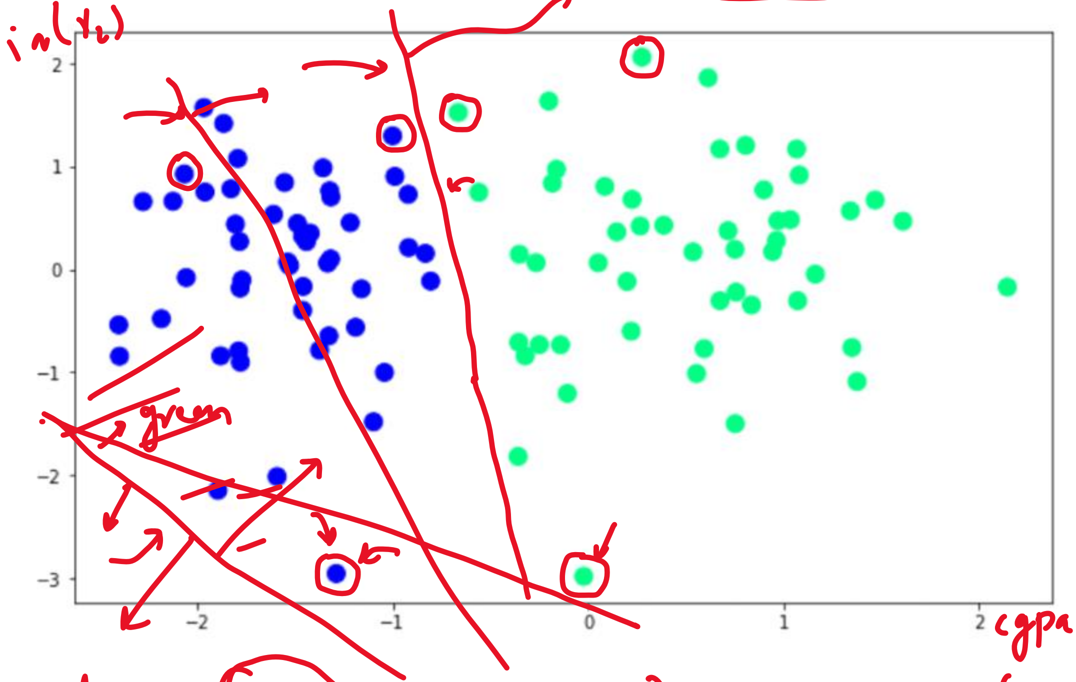


Perceptron Trick

Tuesday, June 15, 2021 12:08 PM

$y = mx + b$

$Ax + By + c = 0$



$Ax_1 + Bx_2 + c = 0$

$Ax_1 + Bx_2 + Cx_3 + d = 0$

$\boxed{A, B, C}$

$A = 1, B = 1$

$C = 0$

loop → 1000

→ random student

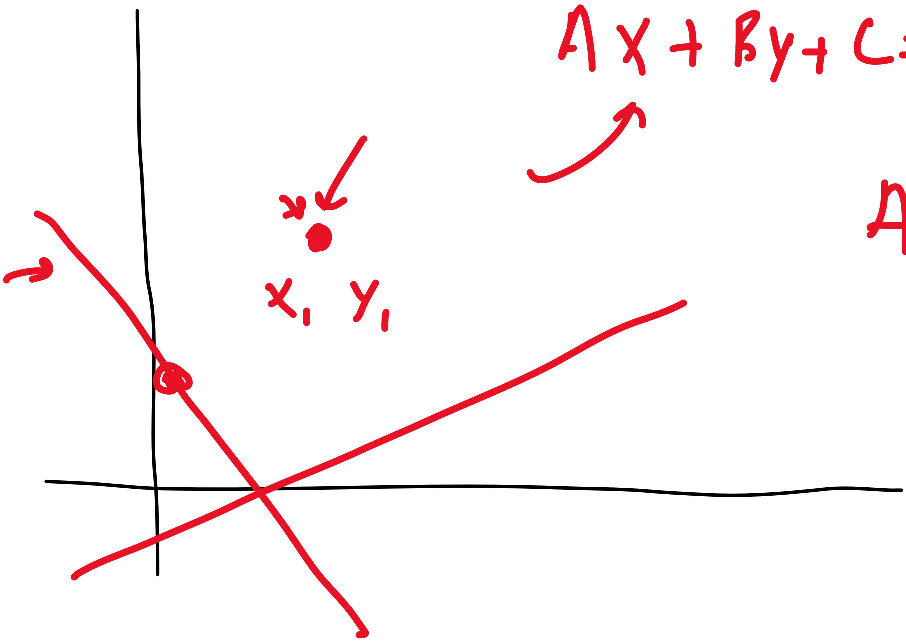
1000

while converge

How to label regions?

Tuesday, June 15, 2021 1:12 PM

$$Ax + By + C = 0$$



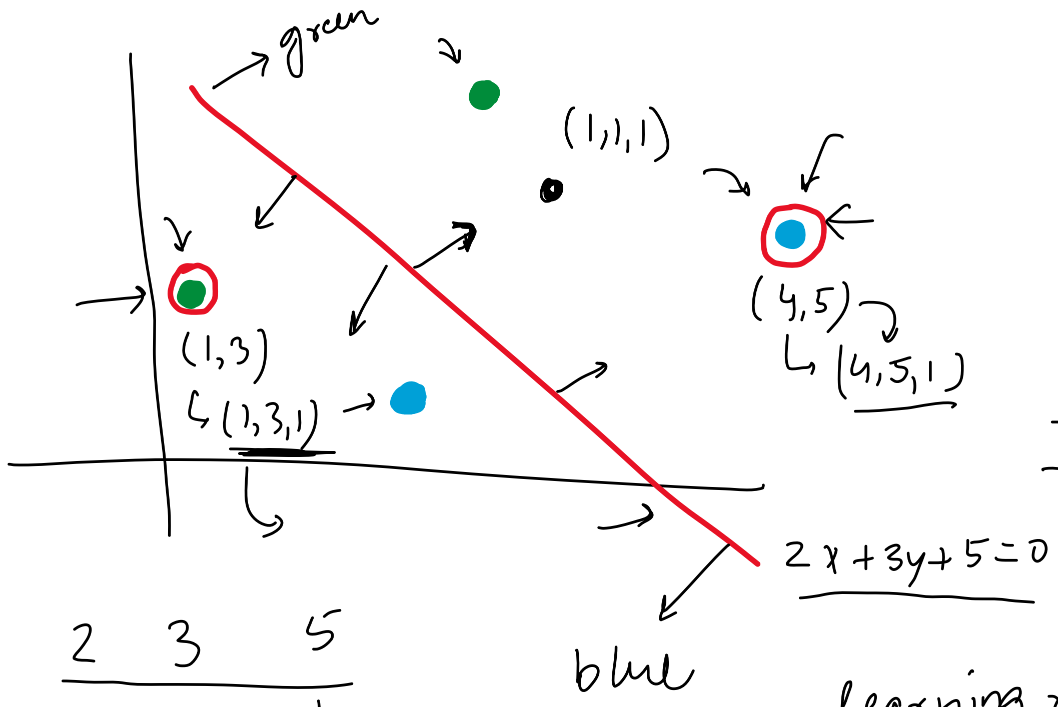
$$Ax_1 + By_1 + C > 0$$

$$<$$

$$= 0$$

Transformations

Tuesday, June 15, 2021 1:31 PM



$$\begin{array}{rrr} 2 & 3 & 5 \\ - \; 4 & 5 & 1 \\ \hline \end{array}$$

$$-2x - 2y + 4 = 0$$

$$\begin{array}{lll} 2 & 3 & 5 \\ \hline + \; 0.01 & 0.03 & 0.01 \\ - & - & - \\ \hline \end{array}$$

$$3x + 6y + 6 = 0$$

learning rate → 0.01

$$\text{coef} = \boxed{\text{coef}} - \boxed{\eta \text{ coordinate}}$$

Algorithm

Tuesday, June 15, 2021 2:31 PM

x_0	(x_1) cgpa	(x_2) iq	y placed
1	7.5	81	1
1	8.9	109	1
1	7.0	81	0

$$Ax + By + C = 0$$

$$\boxed{w_0 + w_1 x_1 + w_2 x_2 = 0} \rightarrow \sum_{i=0}^{2} w_i x_i = 0$$

$w_0 = C$, $w_1 = A$, $w_2 = B$

$$\boxed{w_0 \; w_1 \; w_2}$$

$$w_0 x_0 + w_1 x_1 + w_2 x_2 = 0$$

$$w_0 \times 1 + w_1 \times 7.5 + w_2 \times 81 \rightarrow \sum_{i=0}^{2} w_i x_i = 0$$

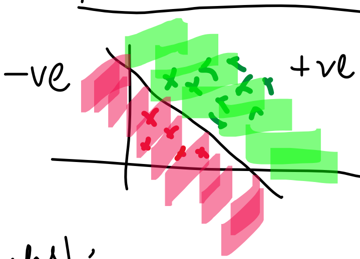
$$[w_0 \;\; w_1 \;\; w_2] \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$$

$= \quad \geq 0 \rightarrow 1$

$< 0 \rightarrow 0$

$$\boxed{w_0 x_0 + w_1 x_1 + w_2 x_2} \rightarrow 1$$

$\hookrightarrow 0$



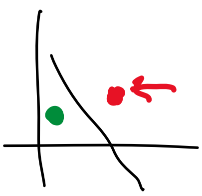
$x_i \in P$

$x_i \in N$

epoch → 1000 ; $\eta = 0.01$

for i in range(epochs):

randomly select a student



$[w_0 \; w_1 \; w_2]$

- if $x_i \in N$ and $\sum_{i=0}^{2} w_i x_i \geq 0$ { (+ve)

$$W_{new} = W_{old} - \eta x_i$$

- if $x_i \in P$ and $\sum_{i=1}^{2} w_i x_i < 0$

$$W_{new} = W_{old} + \eta x_i$$

$x_i \in N$ $\quad w_i x_i < 0$

$x_i \in P$ $\quad w_i x_i \geq 0$

Simplified Algo

Tuesday, June 15, 2021 2:44 PM

if $x_i \in N$ and $\sum w_i x_i \geq 0$

$$w_n = w_0 - \eta x_i$$

if $x_i \in P$ and $\sum w_i x_i < 0$

$$w_n = w_0 + \eta x_i$$

for i in 1000
random student

$$\boxed{w_n = w_0 + \eta (y_i - \hat{y}_i) x_i}$$

0

$\underline{\underline{w_n}} = \underline{\underline{w_0}}$

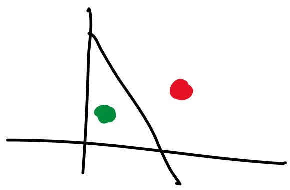
1

−1

$$w_n = w_0 + \eta x_i$$

$$w_n = w_0 - \eta x_i$$

y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
1	1	0
0	0	0
1	0	1
0	1	−1



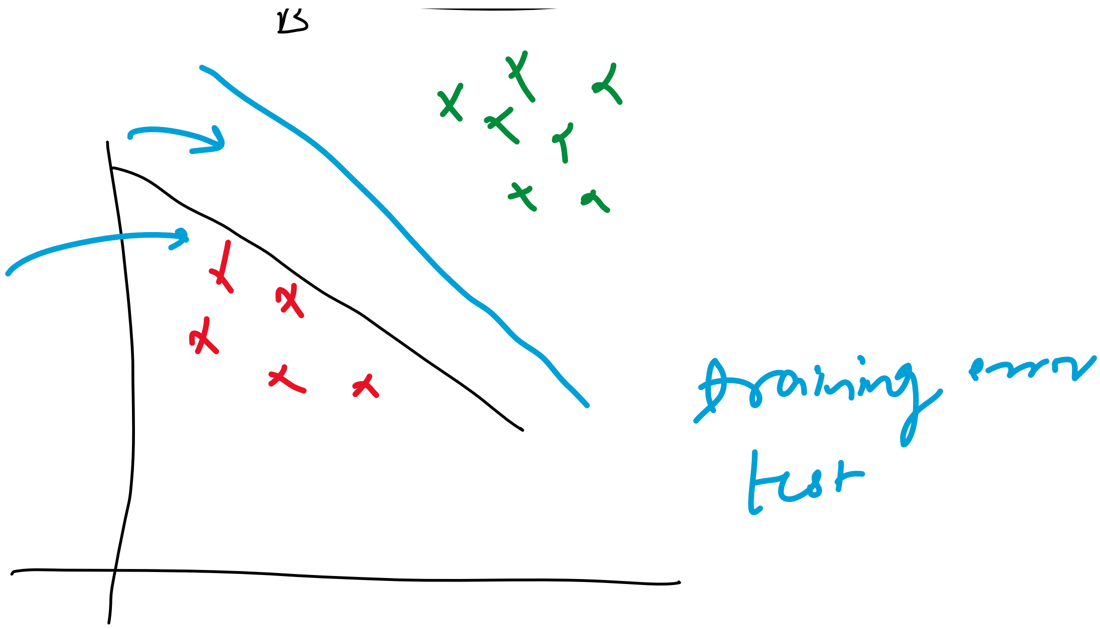
for i in range(epochs):
 select a random student (i)
 $w_n = w_0 + \eta (y_i - \hat{y}_i) x_i$

A, B, C

$Ax + By + C = 0$ → $y = mx + b$

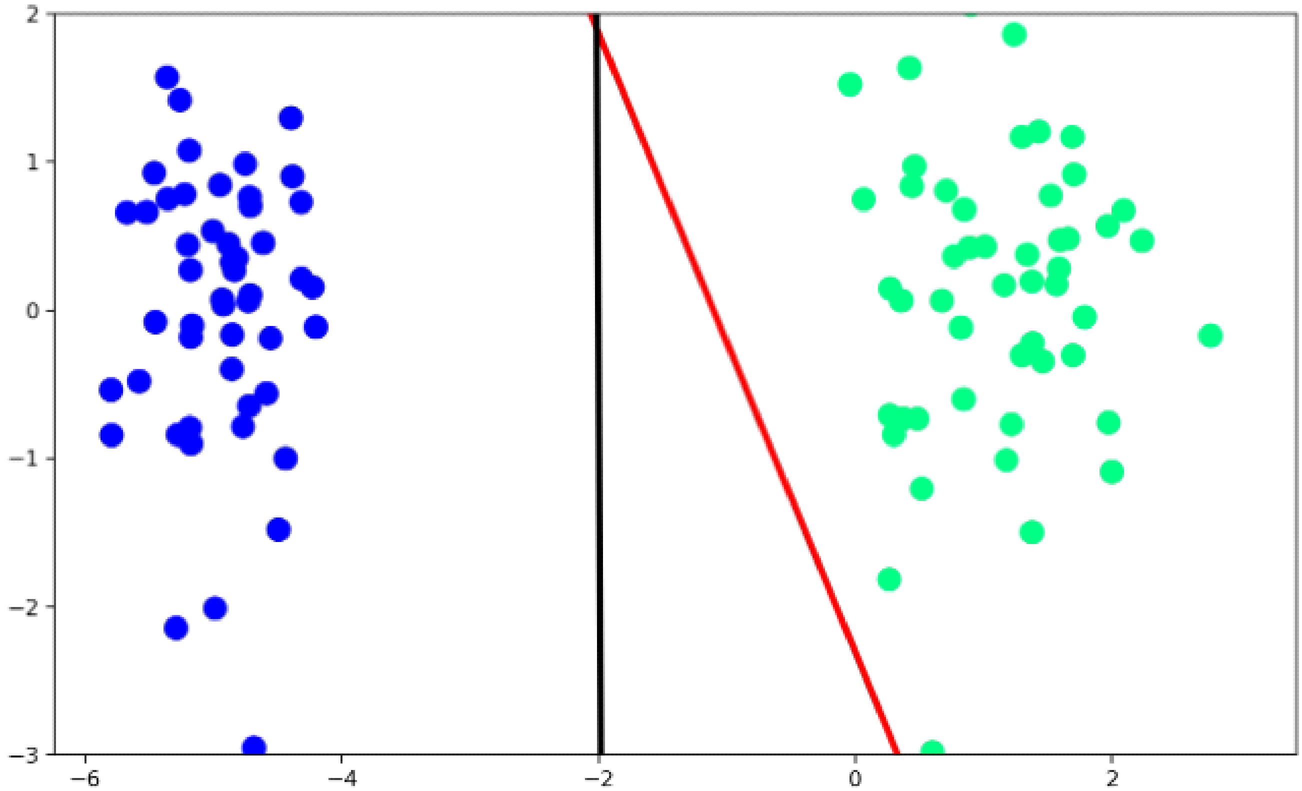
$\boxed{m} = -\frac{A}{B}$ coef [− __] [,]

$c = -\frac{C}{B}$


vs
training error
test

Problem with Perceptron

Thursday, June 17, 2021 12:18 PM



Possible Solution?

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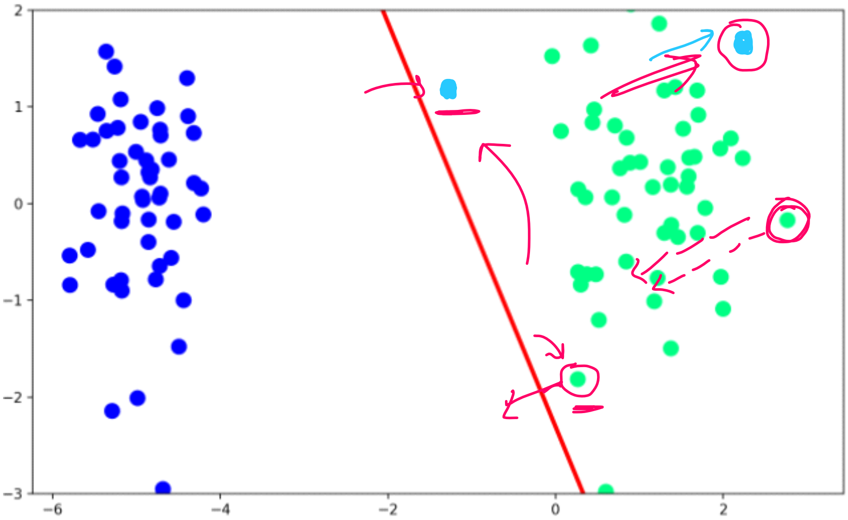


$$\boxed{w_n = w_0 + (y_i - \hat{y}_i) x_i}$$

→ misclassified

line - pull

→ correctly

└ line push

push pull

y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
1	1	0
0	0	0
1	0	1
0	1	−1

$$\underbrace{w_n = w_0}_{} + \eta \underbrace{(y_i - \hat{y}_i)}_{0} x_i$$

$(y_i - \hat{y}_i) \neq 0$ model predict

$\sum w_i x_i = \boxed{0, 1}$

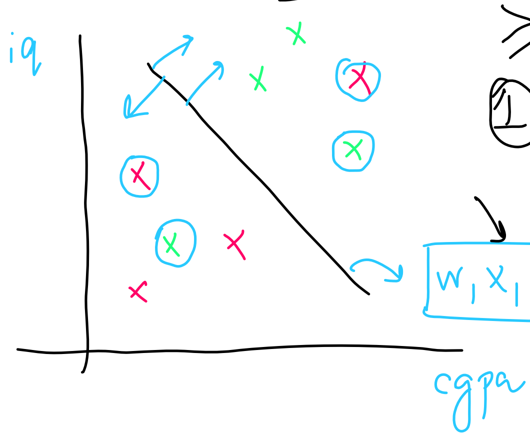
$(y_i - \hat{y}_i)$

$w_1 \times 8.1 + w_2 \times 81 + w_0 =$

>0 <0

1 0

$\{8.1, 81\}$



$\boxed{w_1 x_1 + w_2 x_2 + w_0 = 0}$

cgpa	iq	(x_i) placed
9	91	0
8.8	78	1
8.1	102	1
7.9	98	1

The Sigmoid Function

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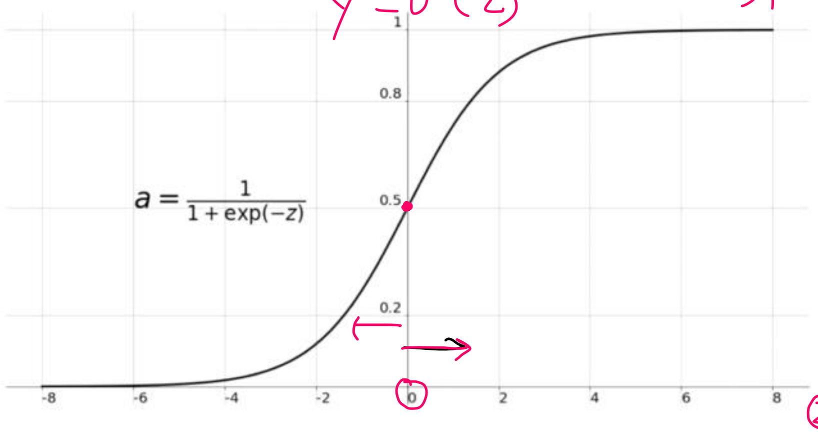
Sigmoid Function

$Y = \sigma(z)$

$\boxed{\sigma(z) = \dfrac{1}{1+e^{-z}}}$



$\rightarrow 1$

z

$\boxed{-\infty < z < \infty}$

$0 < Y < 1$

$\{7.5, 81\} \rightarrow w_1 \; w_2 \; w_0$

$\hat{y}_i = \underbrace{w_1 \times 7.5 + w_2 \times 81 + w_0}_{z} =$

$z \geq 0 \rightarrow 1$

$z < 0 \rightarrow 0$

$\sigma > 0.5 \leftarrow z = \sum w_i x_i = \sigma(z)$

$\sigma(z) < 0.5$

$\{7.5, 81\} \rightarrow z \rightarrow \sigma(z) = 0.5 \geq$

if 1 $P(N) = 1 - P(Y)$

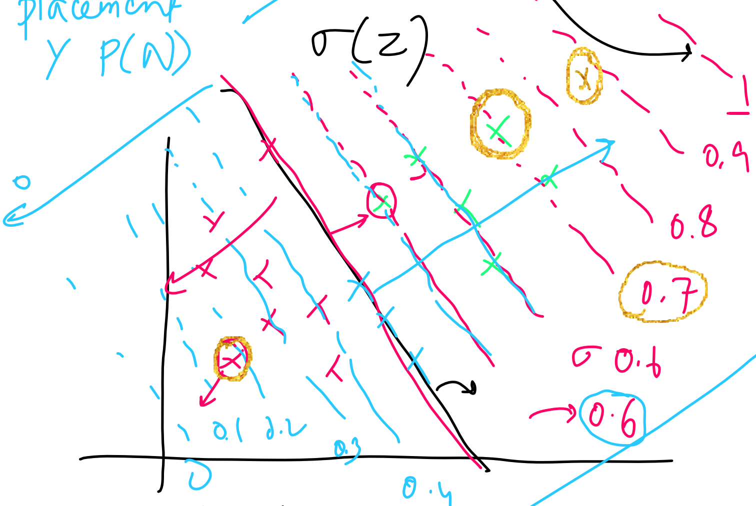
else 0

$0.5 <$ $P(Y) =$

placement

Y P(N)

$P(y) = 0.7$



$\sum w_i x_i = 0$ $\sigma = 0.5$ $P(y) = 0.6$

$\sum w_i x_i > 0$ $\sigma > 0.5$

$\sum w_i x_i$ $\sigma > 0.6$

$\sigma(\;) = P(\;)$ $P(y)$ $P(N)$

$P(y) = 0.1$

$\sigma = 0.5$

$\sigma = 0.5$ $\boxed{P(0.5)} \rightarrow P(y) = 0.5$

0

Impact of Sigmoid

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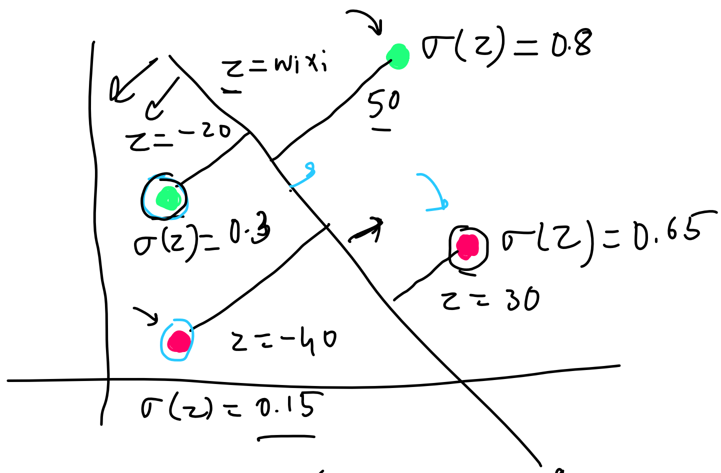
$$w_n = w_0 + \eta (y_i - \hat{y}_i) x_i$$

$$\hat{y}_i = \sigma(z)$$

where $z = \sum w_i x_i$



y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
1	0.8	0.2
0	0.65	-0.65
1	0.3	0.7
0	0.15	-0.15

$$w_n = w_0 + \eta \times 0.2 \times x_i$$

$$w_n = w_0 - \eta * 0.65 * x_i$$

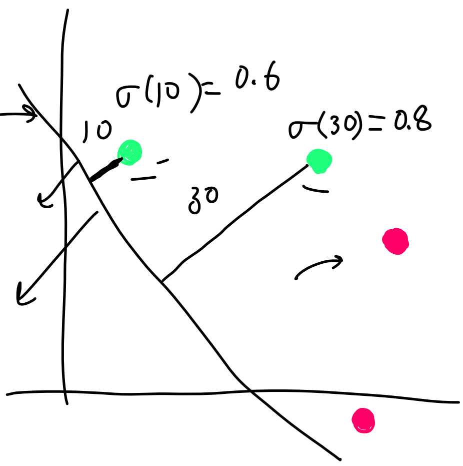
$$w_n = w_0 + \eta * 0.7 * x_i$$

$$w_n = w_0 - \eta * 0.15 * x_i$$

$$w_n = w_0 + \eta (y_i - \hat{y}_i) x_i$$

$$\hat{y}_i = \sigma(z)$$

where $z = \sum w_i x_i$



y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
1	0.6	0.4
1	0.8	0.2
0		
0		

1st

$$w_n = w_0 + \boxed{\eta * 0.4 * x_i} = x_1$$

$$w_n = w_0 + \boxed{\eta * 0.2 * x_i} = x_2$$

$x_1 > x_2$ → code implement

The Problem

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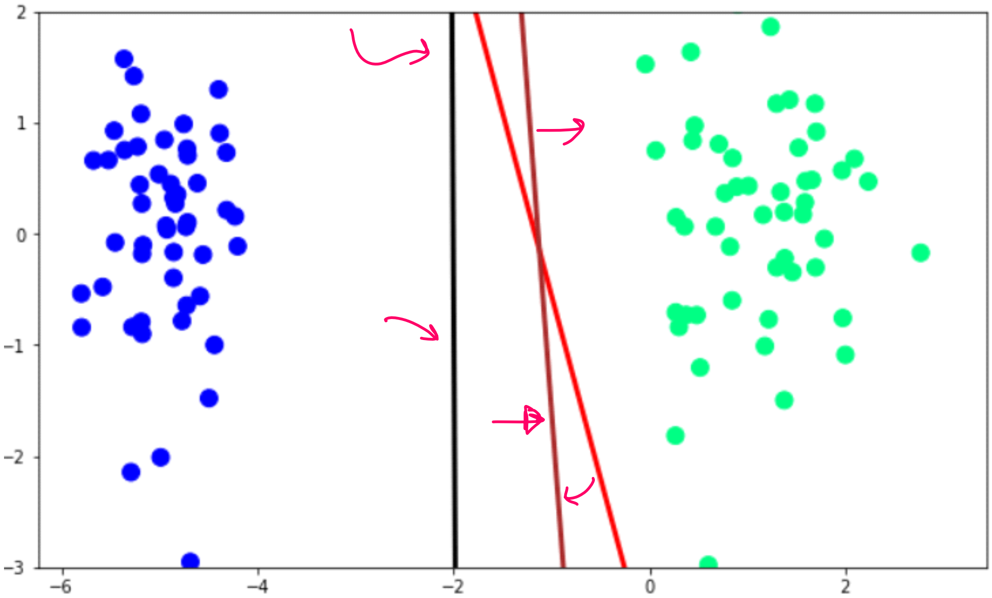

red → step

brown → sigmoid

black → sklearn

for i in 1000

random

Loss function

error function

Maximum Likelihood and Cross-Entropy

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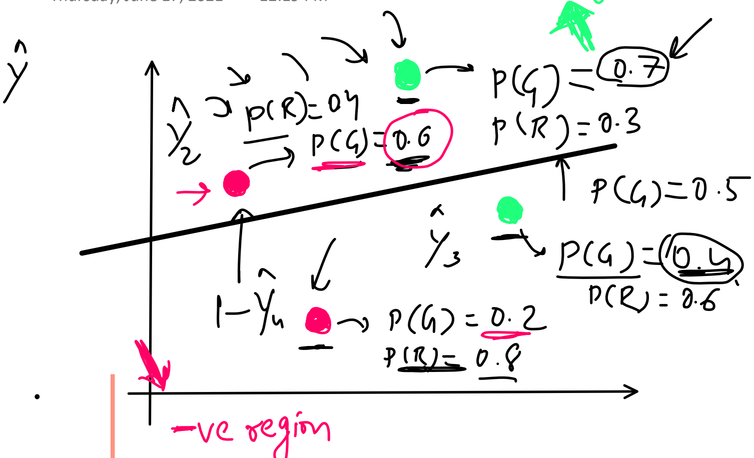
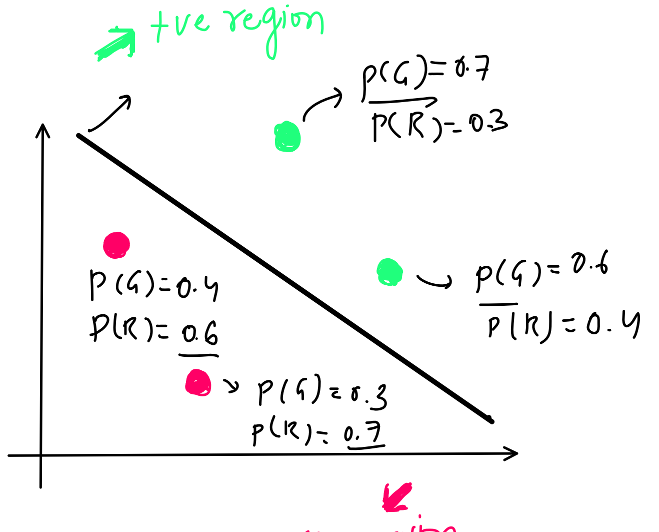


model 1



model 2

$\hat{y}_i$

$\hat{y} = \sigma(z)$

$\hookrightarrow z = \sum w_i x_i$

model 2 $\rightarrow$ $0.7 \times 0.6 \times 0.6 \times 0.7$

$= 0.176$

model 1 $\rightarrow$ $0.7 \times 0.4 \times 0.4 \times 0.8 = 0.089$

$\log(0.1) > \log(0.9)$

$-1 \qquad -0.04$

$\log(ab) = \log a + \log b$

$\log(\max) = \boxed{-\log(0.7)} - \log(0.4) - \log(0.4) - \log(0.8)$

0-1 = -ve

cross entropy minimize

No

$-\log(\hat{y}_1) - \log(\hat{y}_2) - \log(\hat{y}_3) - \log(\hat{y}_4)$

$(1-\hat{y})$

$y_i = 1$

$-y_i \log(\hat{y}_i) - (1-y_i)\log(1-\hat{y}_i)$

$-y_1 \log(\hat{y}_1) = -\log(\hat{y}_1)$

$= -\log(0.7)$

$y_2 = 0 \qquad y_3 = 1$

$Y_2 = 0 \qquad Y_3 = 1 \qquad = -\log(0.7)$

$-Y_2 \log(\hat{y}_2) - (1-y_2)\log(1-\hat{y}_2)$

$- \log(1-\hat{y}_2) = -\log 0.4$

$-Y_3 \log(\hat{y}_3)$

$-\log(\hat{y}_3) = -\log(0.4)$

$Y_4 = 0$

$-(1-Y_4)\log(1-\hat{y}_4)$

$-\log(1-\hat{y}_4)$

$-\log(0.8)$

$$L = \sum_{i=1}^{n} -y_i \log(\hat{y}_i) - (1-y_i)\log(1-\hat{y}_i)$$

MSE

$$\boxed{L = -\frac{1}{n}\sum_{i=1}^{n} y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)}$$

closed form

gradient descent

→ log-loss error

→ Binary cross entropy

$\min_{w_1, w_2, w_0}$

Derivative of Sigmoid

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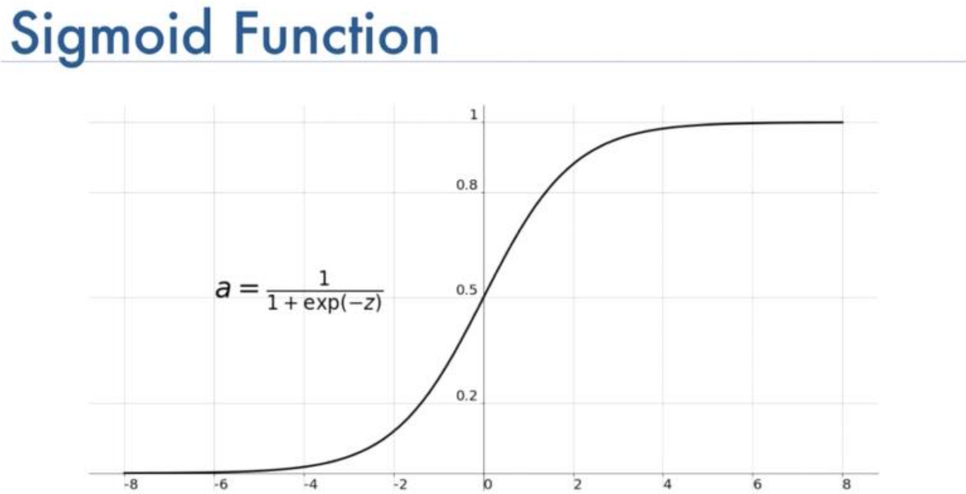


$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \frac{d}{dx}\left(\frac{1}{1+e^{-x}}\right)$$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x)^{-1}$$

$$= -x^{-2} = -\frac{1}{x^2}$$

$$\frac{d}{dx}\left[\frac{1}{1+e^{-x}}\right] = \frac{d}{dx}\left[(1+e^{-x})^{-1}\right] = -\frac{1}{(1+e^{-x})^2}\frac{d}{dx}(1+e^{-x})$$

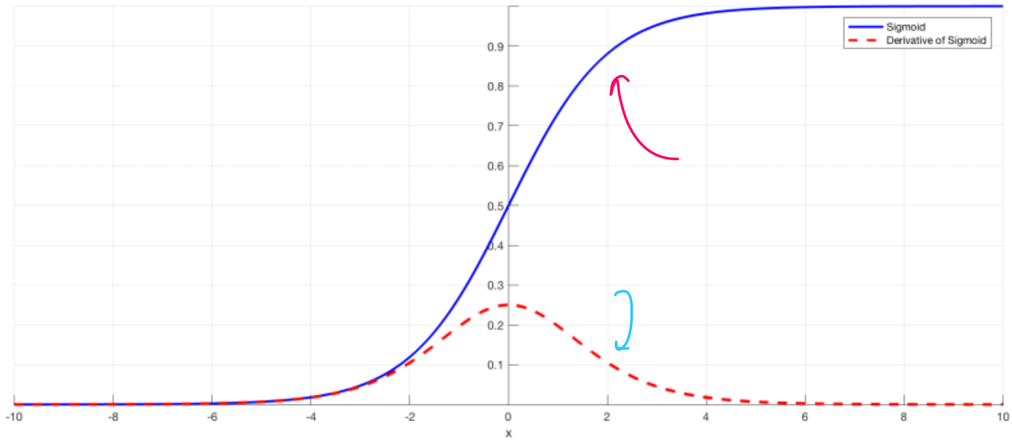
$$e^{-x} = e^{-x} \qquad -x = -1$$

$$= -\frac{1}{(1+e^{-x})^2}\frac{d}{dx}(e^{-x}) = -\frac{e^{-x}}{(1+e^{-x})^2}\frac{d}{dx}(-x) = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$\frac{1 \cdot e^{-x}}{(1+e^{-x})(1+e^{-x})} = \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}} = \sigma(x)\left[\frac{e^{-x}}{1+e^{-x}}\right]$$

$$= \sigma(x)\left[\frac{1+e^{-x}-1}{1+e^{-x}}\right] = \sigma(x)\left[\frac{1+\cancel{e^{-x}}}{1+\cancel{e^{-x}}} - \frac{1}{1+e^{-x}}\right]$$

$$\sigma(x)\left[1-\sigma(x)\right] \Rightarrow \sigma'(x) = \boxed{\sigma(x)\left[1-\sigma(x)\right]}$$


Sigmoid
Derivative of Sigmoid
x

Gradient Descent

Monday, June 21, 2021 11:50 AM

$\boxed{\text{wow }m_?}$ → classification $\{x_1, x_2\} \to y$

$w_1x_1 + w_2x_2 + w_0 = 0$

$x_0 \quad x_1 \quad x_2 \quad y = \{0, 1\}$

$\hat{y}_i = \sigma(z) =$

$\hat{y}_i = \sigma(z) \quad z = \sum_{j=0}^{n} w_j x_j$

$w_1x_1 + w_2x_2 + w_0x_0$

$\sum_{j=0}^{2} w_j x_j = z$

$$L = -\frac{1}{m}\sum_{i=1}^{m} y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)$$

$L(w_1, w_2, w_0) =$ minimum

gradient descent

argmin $(w_0\, w_1\, w_2)$

$w_1\, w_2\, w_0 = 1$

for i in epochs

0.01

$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$

$$\underbrace{w_0 = w_0 - \eta\frac{\partial L}{\partial w_0}, \quad w_1 = w_1 - \eta\frac{\partial L}{\partial w_1}, \quad w_2 = w_2 - \eta\frac{\partial L}{\partial w_2}}_{}$$

n cols → n+1 derivatives

$w_n = w_n - \eta\frac{\partial L}{\partial w_n}$ $\qquad j = 0, 1, 2 \ldots n$

$$L = -\frac{1}{m}\sum_{i=1}^{m} y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)$$

$$\frac{\partial L}{\partial w_j} = -\frac{1}{m}\sum_{i=1}^{m}\left[\frac{\partial L}{\partial w_j}\left(y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right)\right]$$

$\hat{y}_i = \sigma(z) \quad z = \sum_{j=0}^{n} w_j x_j$

$$\frac{\partial L}{\partial w_j} y_i \log(\hat{y}_i) = \frac{\partial L}{\partial w_j} y_i \log(\sigma(z))$$

$- y_i \quad \partial L \quad \sigma\left(\sum^{n} w_j x_j\right)$

$$= \frac{y_i}{\hat{y}_i} \frac{\partial L}{\partial w_j} \sigma\left(\sum_{j=0}^{n} w_j x_j\right)$$

$$\frac{\partial L}{\partial w_j} y_i \log\left(\sigma\left(\sum_{j=0}^{n} w_j x_j\right)\right)$$

$$= y_i \frac{\partial L}{\partial w_j} \log\left(\underbrace{\sigma\left(\sum_{j=0}^{n} w_j x_j\right)}_{\hat{y}_i}\right) = \frac{y_i}{\cancel{\hat{y}_i}} \cancel{\hat{y}_i}(1-\hat{y}_i) \frac{\partial L}{\partial w_j} \sum_{j=0}^{n} w_j x_j$$

$$y_i (1-\hat{y}_i) \sum_{j=0}^{n} \frac{\partial L}{\partial w_j} w_j x_j = \boxed{y_i (1-\hat{y}_i) \sum_{j=0}^{n} x_j}$$

$$\frac{\partial L}{\partial w_j} (1-y_i)\log(1-\hat{y}_i) \Rightarrow (1-y_i) \frac{\partial L}{\partial w_j} \log(1-\hat{y}_i) \qquad \hat{y}_i = \sigma(z) \quad \sigma\left(\sum_{j=0}^{n} w_j x_j\right)$$

$$\frac{(1-y_i)}{(1-\hat{y}_i)} \frac{\partial L}{\partial w_j} (1-\hat{y}_i) \Rightarrow -\frac{(1-y_i)}{(1-\hat{y}_i)} \frac{\partial L}{\partial w_j} \hat{y}_i \Rightarrow -\frac{(1-y_i)}{(1-\hat{y}_i)} \frac{\partial L}{\partial w_j} \sigma(z)$$

$$\Rightarrow -\frac{(1-y_i)}{(1-\hat{y}_i)} \underbrace{\sigma(z)(1-\sigma(z))}_{\hat{y}_i} \frac{\partial L}{\partial w_j} \sigma(z) = -\frac{(1-y_i)}{\cancel{(1-\hat{y}_i)}} \hat{y}_i \cancel{(1-\hat{y}_i)} \frac{\partial L}{\partial w_j} \sigma(z)$$

$$\Rightarrow -\hat{y}_i (1-y_i) \frac{\partial L}{\partial w_j} \sum_{j=0}^{n} w_j x_j \Rightarrow -\hat{y}_i (1-y_i) \sum_{j=0}^{n} \frac{\partial L}{\partial w_j} (w_j x_j)$$

$$\Rightarrow \boxed{-\hat{y}_i (1-y_i) \sum_{j=0}^{n} x_j}$$

$$\frac{\partial L}{\partial w_j} = -\frac{1}{m} \sum_{i=1}^{m} \left[y_i (1-\hat{y}_i) \sum_{j=0}^{n} x_j - \hat{y}_i (1-y_i) \sum_{j=0}^{n} x_j \right]$$

$$\frac{\partial L}{\partial w_j} \ldots \sum^{m} \left[\ldots - \hat{y}_i (1-y_i) \right] \sum^{n} x_j$$

$$\frac{\partial L}{\partial w_j} = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i(1-\hat{y}_i) - \hat{y}_i(1-y_i)\right]\sum_{j=0}^{n} x_j$$

$$= -\frac{1}{m}\sum_{j=1}^{m}\left[y_i - \cancel{y_i\hat{y}_i} - \hat{y}_i + \cancel{y_i\hat{y}_i}\right]\sum_{j=0}^{n} x_j$$

$$\boxed{\frac{\partial L}{\partial w_j} = -\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)\sum_{j=0}^{n} x_j}$$

$$\frac{\partial L}{\partial w_0} = -\frac{1}{2}\left[1 + 0\right]\left[1+1\right]$$

$$= -\frac{1}{\cancel{2}}\left[1\right]\left[\cancel{2}\right] = -1$$

$$= -\frac{1}{2}\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}1 \\ 1\end{bmatrix}$$

$$= -\frac{1}{2}\left[1\right] = -\frac{1}{2}$$

x_1	x_2	y_i	$\hat{y}_i$
1	2	1	0
3	4	0	0

no. of rows $= m = 2$

no. of cols $= n = 2$

Gradient Descent

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Rows = m Columns = n

n input

	1	2	3		n	y	$\hat{y}$
1	x_{11}	x_{12}	x_{13}	.....	x_{1n}	y_1	$\hat{y}_1$
2	x_{21}	x_{22}	x_{23}	....	x_{2n}	y_2	$\hat{y}_2$
⋮	⋮						
m	x_{m1}	x_{m2}	x_{m3}	......	x_{mn}	y_m	$\hat{y}_m$

$\{ w_1 \; w_2 \; w_3 \; ... \; w_n \}$

w_0

$(n+1)$

$$\sigma(w_1 x_{11} + w_2 x_{12} + w_3 x_{13} + \dots + w_n x_{1n} + w_0) = \hat{y}_1$$

$$\sigma(w_1 x_{21} + w_2 x_{22} + w_3 x_{23} + \dots + w_n x_{2n} + w_0) = \hat{y}_2$$

$\hat{y}$

$$\underline{\hat{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} = \begin{bmatrix} \sigma(w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n}) \\ \sigma(w_0 + w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2n}) \\ \vdots \\ \sigma(w_0 + w_1 x_{m1} + w_2 x_{m2} + \dots + w_n x_{mn}) \end{bmatrix}$$

$$\hat{y} = \sigma\left(\overset{C}{\begin{bmatrix} w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n} \\ w_0 + w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2n} \\ \vdots \\ w_0 + w_1 x_{m1} + w_2 x_{m2} + \dots + w_n x_{mn} \end{bmatrix}} \right)$$

w_0 A   X   B   W

$$\hat{y} = \sigma\left(\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & & & & \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \end{bmatrix} \right)$$

$$\hat{y} = \sigma\left(\begin{bmatrix} 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & & & & \\ 1 & x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \begin{bmatrix} \vdots \\ w_n \end{bmatrix}\right)$$

$$\boxed{\hat{y} = \sigma(XW)}$$

$$L = -\frac{1}{m}\sum_{i=1}^{m} y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)$$

$$L = -\frac{1}{m}\left[\sum_{i=1}^{m} y_i \log(\hat{y}_i) + \sum_{i=1}^{m} (1-y_i)\log(1-\hat{y}_i)\right]$$

$(1-y)\log$

$\sigma(1-XW)$

$$\sum_{i=1}^{m} y_i \log(\hat{y}_i) = y_1 \log \hat{y}_1 + y_2 \log \hat{y}_2 + y_3 \log \hat{y}_3 + \cdots + y_m \log \hat{y}_m$$

$$\begin{bmatrix} y_1 & y_2 & y_3 & \cdots & y_m \end{bmatrix} \begin{bmatrix} \log \hat{y}_1 \\ \log \hat{y}_2 \\ \vdots \\ \log \hat{y}_m \end{bmatrix}$$

$$\begin{bmatrix} y_1 & y_2 & y_3 & \cdots & y_m \end{bmatrix} \log\left(\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix}\right)$$

$$\boxed{Y \log \hat{Y} = y \log(\sigma(XW))}$$

$$L = -\frac{1}{m}\left[y \log \hat{y} + (1-y)\log(1-\hat{y})\right]$$ min

where $\hat{y} = \sigma(XW)$ GD $[W]$ find where $\hat{y} = \sigma(XW)$ L (GD) [W]...

Loss function in Matrix form

$$L = -\frac{1}{m}\left[y \log(\sigma(wx)) + (1-y)\log(1-\sigma(wx)) \right]$$

minimum

gradient descent

$W = [\quad]$

for i in epochs:

$\rightarrow W = W - \eta \frac{\Delta L}{\Delta W}$

$w_0 \longrightarrow w_n$

$(n+1)$

$\left\{ \frac{\Delta L}{\Delta W} \right\}$

$$\frac{\Delta L}{\Delta W} = \left[\frac{\partial L}{\partial w_0}, \frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \cdots \frac{\partial L}{\partial w_n} \right]$$

$\frac{\Delta L}{\Delta W}$

$$L = -\frac{1}{m}\left[\boxed{y \log \hat{y}} + (1-y)\log(1-\hat{y}) \right]$$

$\frac{dL}{dW} =$

$$\frac{d}{dW} y \log \hat{y} \Rightarrow y \frac{d}{dW} \log \hat{y} \Rightarrow \frac{y}{\hat{y}} \frac{d}{dL}(\hat{y})$$

$$\Rightarrow \frac{y}{\hat{y}} \frac{d}{dL} \sigma(wx) \Rightarrow \frac{y}{\hat{y}} \underbrace{\sigma(wx)}_{\Rightarrow \hat{y}}[1-\sigma(wx)] \frac{d}{dW}(wx)$$

$$= \frac{y}{\cancel{\hat{y}}} \cancel{\hat{y}}(1-\hat{y})\, x = \boxed{y(1-\hat{y})X}$$

$\hat{y} = \sigma(wx)$

$$\frac{d}{dw}(1-y)\log(1-y) \Rightarrow (1-y)\frac{d}{dw}\log(1-\hat{y}) \Rightarrow \frac{(1-y)}{(1-\hat{y})}\frac{d}{dw}[1-\hat{y}]$$

$$= -\frac{(1-y)}{(1-\hat{y})}\frac{d}{dw}\sigma(wx) \Rightarrow -\frac{(1-y)}{(1-\hat{y})}[\sigma(wx)[1-\sigma(wx)]] \quad \frac{d}{dL}(wx)$$

$$\Rightarrow -\frac{(1-y)}{\cancel{(1-\hat{y})}}\hat{y}\cancel{(1-\hat{y})}x = \boxed{-\hat{y}(1-y)x}$$

$$\frac{dL}{dw} = -\frac{1}{m}\left[y(1-\hat{y})x - \hat{y}(1-y)x\right]$$

$$= -\frac{1}{m}\left[y(1-\hat{y}) - \hat{y}(1-y)\right]x$$

$$= -\frac{1}{m}\left[y - \cancel{y\hat{y}} - \hat{y} + \cancel{y\hat{y}}\right]x$$

$$\boxed{\frac{\Delta L}{\Delta w} = -\frac{1}{m}(y-\hat{y})x}$$

gd

2

$$\boxed{w = w + \eta\frac{1}{m}(y-\hat{y})x}$$

$$\boxed{\underline{w} = \underline{w} + \eta \frac{1}{\underline{m}} \underline{(y - \hat{y})} X}$$

$$\overset{(n+1),1}{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \qquad X = \underset{m,\, n+1}{\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & & & & \\ 1 & x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}} \qquad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \rightarrow m,1$$

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} \leftarrow (m,1)$$

$$\underline{w} = \underline{w} + \boxed{\frac{\eta}{m}} \underline{(y - \hat{y})} \underline{X}$$

$$\underset{(n+1,1)}{\underline{\underline{}}} = \underline{(n+1,1)} \qquad (1, m) \quad m, (n+1) \rightarrow \underline{(1, n+1)}$$

$$\downarrow$$

$$(n+1), 1$$

Accuracy

Tuesday, June 22, 2021 1:12 PM

binary ✓

Actual Label	Logistic Regression Prediction	Decision Tree Prediction
1	✓ 1	✓ 1
0	✗ 1	✗ 1
0	✓ 0	✓ 0
0	✓ 0	✓ 0
1	✓ 1	✓ 1
1	✓ 1	✓ 1
0	✗ 1	✓ 0
0	✓ 0	✓ 0
0	✓ 0	✓ 0
1	✓ 1	✓ 1

cgpa | iq | placement 1000

train 800 test 200

LoR DT

$$\text{accuracy} = \frac{\text{no. of } ✓}{\text{total prediction}} \qquad \frac{8}{10} = 0.8 \rightarrow 80\%$$

$$\frac{9}{10} = 90\%$$

Accuracy of multi-classification problem

Wednesday, June 23, 2021 8:44 AM

Actual Label	Logistic Regression Prediction	Decision Tree Prediction
0	✓ 0	0
0	✓ 0	0
0	✓ 0	0
2	✓ 2	2
0	✓ 0	0
2	✓ 2	2
0	✓ 0	0
2	✓ 2	2
1	✓ 1	1
1	✓ 1	1

iris

setosa, virginica / versicolor
0 1 2

$$\text{accuracy} = \frac{\#\ \text{correct prediction}}{\#\ \text{total}}$$

$$= \frac{10}{10} = 1 = 100\%$$

How much accuracy is good?

Wednesday, June 23, 2021 11:14 AM

The Problem with Accuracy

Wednesday, June 23, 2021 11:14 AM

90% → $\boxed{10\%}$ →

binary

Y | N

→ Y ✓ → X

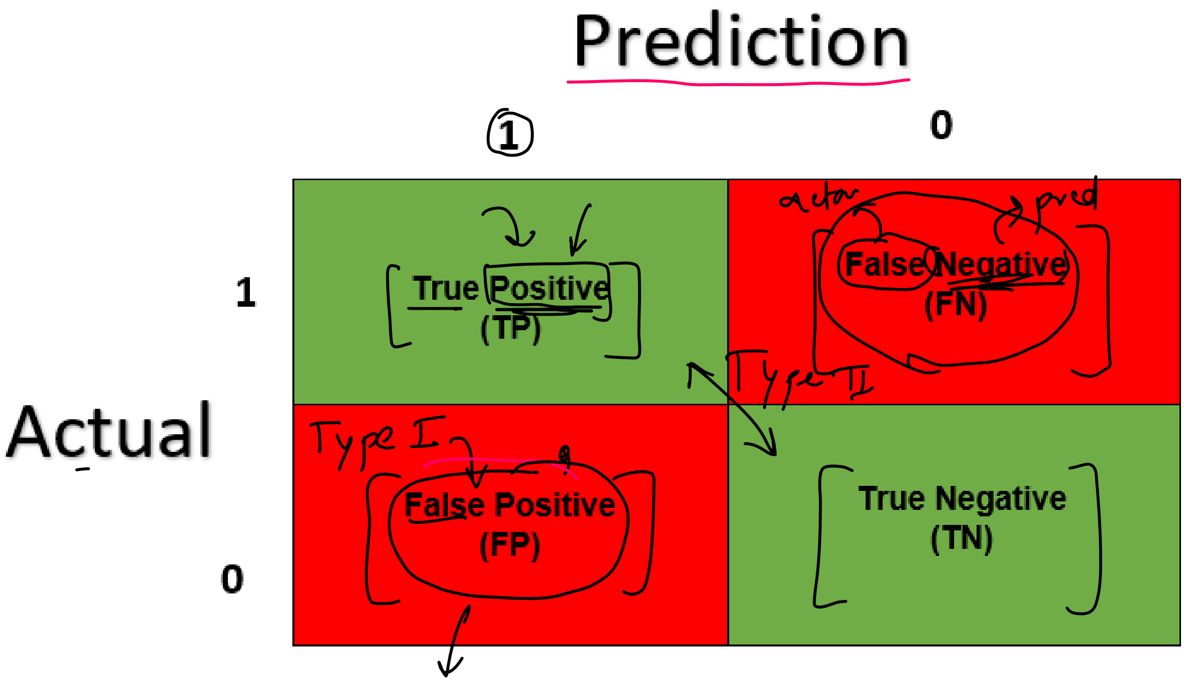
→ N X → ✓

Y ✓ → X

N X → ✓

Confusion Matrix

Tuesday, June 22, 2021 1:12 PM



$$acc = \frac{TP + TN}{TP + TN + FP + FN}$$

Extented

↳ { echo or not }

duplicate

↳ echo

$1+2+3=6$ → outcome

$$\boxed{\frac{xy}{2y}}$$

Type 1 Error

Wednesday, June 23, 2021 8:45 AM

Type 2 Error

Wednesday, June 23, 2021 8:45 AM

Confusion Matrix for Multi-classification Problem

Wednesday, June 23, 2021 8:45 AM

0, 1, 2 →

$\boxed{10 \times 10}$

Predicted

Actual	0	1	2
0	7	0	5
1	2	21	6
2	9	1	13

3 x 3

binary

2 x 2

$$\frac{\text{True predict}}{\text{total}} = \text{accuracy}$$

3 x 3

When accuracy is misleading?

Wednesday, June 23, 2021 8:45 AM

Imbalanced Dataset

1000 people

999 Not terrorist

1 terrorist

model → No one is terrorist

		Predicted 1	Predicted 0
Actual	1	0	1
	0	0	999

$$\text{Accuracy} = \frac{999}{999+1}$$

$$= \boxed{99.9\%}$$

Precision

Wednesday, June 23, 2021 8:46 AM

{spam:1, not spam:0}

A Predicted B

Actual

	Sent to Spam	Not sent to spam
Spam	100	170 FN
Not Spam	30 FP	700

	Sent to Spam	Not sent to spam
Spam	100	190
Not Spam	10	700

$$P_A = \frac{100}{100+30}$$

$$P_B = \frac{100}{100+10}$$

$$P_A < \boxed{P_B}$$

80%

FP → Not spam → Spam

FN → spam → Not spam

What proportion of predicted Positives is truly Positive?

1 0

Binary

+ve

	Sent to Spam	Not sent to spam
Spam	True Positive	False Negative
Not Spam	False Positive	True Negative

Precision= TP/(TP+FP)

Recall

Wednesday, June 23, 2021 8:46 AM

has cancer : 1 , no cancer : 0

Predicted

Actual

	Detected Cancer	Not Detected
Has Cancer	1000	200 FN
No Cancer	800 FP	8000

A 90%

	Detected Cancer	Not Detected
Has Cancer	1000	500 ← B
No Cancer	500	8000

B 90%

$$Recall_A = \frac{1000}{1200}$$

$$R_B = \frac{1000}{1500}$$

↑↓

FP

$\boxed{R_A} > R_B$

What proportion of actual Positives is correctly classified?

	Detected Cancer	Not detected Cancer
Has Cancer	True Positive	False Negative
No Cancer	False Positive	True Negative

Recall= TP/(TP+FN)

F1 Score

Wednesday, June 23, 2021 8:46 AM

$$F1score = \frac{2PR}{P+R}$$

$P = 0$ $R = 100$

$F1score = 50$

$F1 = 0$

$$F1score = \frac{P+R}{2}$$

A

$P = R = 80$

$$\frac{80+80}{2} = 80$$

$F1 = 80$

$$\frac{2 \times 80 \times \cancel{80}^{40}}{\cancel{160}\,2} = 80$$

B

75

$P = 60$ $R = 100$

$$\frac{100+60}{2} = 80$$

$$= \frac{2 \times 60 \times 100}{160}$$

$= 75$

Multi-class Precision and Recall

Wednesday, June 23, 2021 8:46 AM

Positive 1 Yes NO

Binary 2 class

Multi 3

Dog Cat Rabbit

$$\boxed{\frac{2 \times 0.86 + 0.66}{0.86 + 0.66}}$$

R_D, R_C, R_R

$$F1_D = \frac{2 P_D R_D}{P_D + R_D}$$

$$F1_C = \frac{2 P_C R_P}{P_C + R_D}$$

$F1_R$

Predicted

Actual

	Dog	Cat	Rabbit	Total	Recall
Dog	25	5	10	40	0.62
Cat	0	30	4	34	0.88
Rabbit	4	10	20	34	0.58
Total	29	45	34		
Precision	0.86	0.66	0.58		

$$R_D = \frac{25}{40} = 0.62, \quad R_C = \frac{30}{34} = 0.88, \quad R_R = \frac{20}{34} = 0.58$$

- macro recall
- weighet recall

$F1_D, F1_C, F1_R$

- macro f1
- weighted f1

Multi-class F1 Score

Thursday, June 24, 2021 11:14 AM

Softmax Regression

Monday, June 28, 2021 3:35 PM

cgpa	iq	place?
7.1	71	0
8.5	85	1
9.5	95	2

$\{0, 1, 2\} \rightarrow$ classifier

$\left\{ \begin{array}{l} \text{Yes} \\ \text{No} \end{array} \right\} \begin{array}{l} 1 \\ 0 \end{array}$

Optout 2

$\{6.5, 65\}$

$\rightarrow$ Yes / No / optout

Binary classification

$\left\{ \begin{array}{l} \text{Softmax} \\ \text{Multinomial} \\ \text{LR} \end{array} \right\} \rightarrow$ Deep Learning

softmax $\rightarrow$ general

for 2class

$\boxed{0 < \sigma < 1}$

softmax function

K = no. of class $\{3\}$

yes → 1
no → 2
opt → 3

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

$$\sigma(z)_3 = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}} \rightarrow 3$$

(Yes)

$$\sigma(z)_1 = \left\{ \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} \right\}$$

No

$$\sigma(z)_2 = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

Training Intuition

Monday, June 28, 2021 3:35 PM

$\{\overset{0}{Y}, \overset{1}{N}, \overset{2}{D}\}$

cgpa	iq	place?
—	—	0
—	—	1
—	—	2

transform

OHE

5

cgpa	iq	= 0	= 1	= 2
✓	✓	1	0	0
		0	1	0
		0	0	1

3 coeff

$w_1\ w_2\ w_0$

D

D_1 D_2 D_3

cgpa | iq | = 0 → m_1 → $w_1^{(0)}, w_2^{(0)}, w_0^{(0)}$

cgpa | iq | = 1 → m_2 → $w_1^{(1)}, w_2^{(1)}, w_0^{(1)}$

cgpa | iq | = 2 → m_3 → $w_1^{(2)}, w_2^{(2)}, w_0^{(2)}$

{ Loss function
gradient descent }

Loss Function

Monday, June 28, 2021 3:47 PM

$$L = -\frac{1}{m}\sum_{i=1}^{m} y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)$$

$k = \{1, 2, 3\}$ $i=1$

$$L = -\frac{1}{m}\boxed{\sum_{i=1}^{m}}\boxed{\sum_{k=1}^{K}} y_k^{(i)} \log(\hat{y}_k^{(i)})$$

$w_1^{(1)}\ w_2^{(1)}\ w_0^{(1)}$

x_1	x_2	y	$y_{k=1}$	$y_{k=2}$	$y_{k=3}$
1 x_{11}	x_{12}	1	1	0	0
2 x_{21}	x_{22}	2	0	1	0
3 x_{31}	x_{32}	3	0	0	1

$$y_1^{(1)}\log(\hat{y}_1^{(1)}) + \boxed{y_2^{(1)}\log(\hat{y}_2^{(1)})} + \boxed{y_3^{(1)}\log(\hat{y}_3^{(1)})} +$$

$$+ \boxed{y_1^{(2)}\log(\hat{y}_1^{(2)})} + y_2^{(2)}\log(\hat{y}_2^{(2)}) + \boxed{y_3^{(2)}\log(\hat{y}_3^{(2)})} +$$

$$+ \boxed{y_1^{(3)}\log(\hat{y}_1^{(3)})} + \boxed{y_2^{(3)}\log(\hat{y}_2^{(3)})} + y_3^{(3)}\log(\hat{y}_3^{(3)})$$

$$L = y_1^{(1)}\log(\hat{y}_1^{(1)}) + y_2^{(2)}\log(\hat{y}_2^{(2)}) + y_3^{(3)}\log(\hat{y}_3^{(3)})$$

$\hat{y}_1^{(1)},\ \hat{y}_2^{(2)},\ \hat{y}_3^{(3)}$

softmax

$$\hat{y}_1^{(1)} = \sigma(w_1^{(1)}x_{11} + w_2^{(1)}x_{12} + w_0^{(1)})$$

$$y_2^{(2)} = \sigma(w_1^{(2)}x_{21} + w_2^{(2)}x_{22} + w_0^{(2)})$$

$$y_3^{(3)} = \sigma(w_1^{(3)}x_{31} + w_2^{(3)}x_{32} + w_0^{(3)})$$

$$\begin{bmatrix} w_1^{(1)} & w_2^{(1)} & w_0^{(1)} \\ w_1^{(2)} & w_2^{(2)} & w_0^{(2)} \\ w_1^{(3)} & w_2^{(3)} & w_0^{(3)} \end{bmatrix}$$

L

$$\frac{\partial L}{\partial w_1^{(1)}}, \frac{\partial L}{\partial w_2^{(1)}}, \frac{\partial L}{\partial w_0^{(1)}} \cdots \quad 9 \text{ du}$$

gradient

9-values init = 1

$$\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} =$$

loop $\Rightarrow$ 1000 epochs

$$w_1^{(1)} = w_1^{(1)} - \eta \frac{\partial L}{\partial w_1^{(1)}}$$

$$w_2^{(1)} = w_2^{(1)} - \eta \frac{\partial L}{\partial w_2^{(1)}}$$

$\vdots$

Prediction

Monday, June 28, 2021 3:35 PM

$S_x = \{7, 70\} \Rightarrow Y, N, Opt$

$\boxed{m_1}$ Yes $\qquad \boxed{m_2}$ No $\qquad \boxed{m_3}$ Opt out

$w_1^{(1)} \; w_2^{(1)} \; w_0^{(1)} \qquad w_1^{(2)} \; w_2^{(2)} \; w_0^{(2)} \qquad w_1^{(3)} \; w_2^{(3)} \; w_0^{(3)}$

$z_1 = 7 \times w_1^{(1)} + 70 \times w_2^{(2)} + w_0^{(1)}$

$z_2 = 7 \times w_1^{(2)} + 70 \times w_2^{(2)} + w_0^{(3)}$

$z_3 = 7 \times w_1^{(3)} + 70 \times w_2^{(3)} + w_0^{(3)}$

$\sigma(y) = \dfrac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$

$= 0.40$

$\sigma(N) = \dfrac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$

0.35

$e(0) = \dfrac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$

0.25

Sigmoid Vs Softmax

Monday, June 28, 2021 3:47 PM

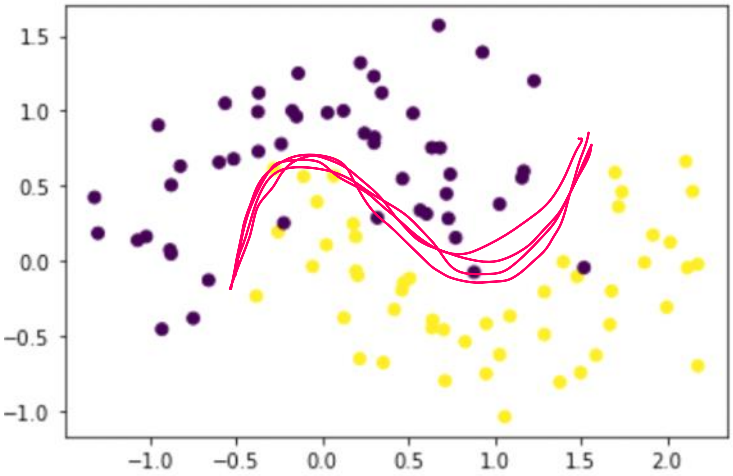
Code Sample

Monday, June 28, 2021 3:36 PM

Polynomial Logistic Regression

Monday, June 28, 2021 6:43 PM

$X_1 \quad x_2 \quad y \quad \{0,1\}$



Linear

degree

polynomial

$x^0 \quad x^1 \quad x^2 \quad x^3$

degree $= \boxed{2}$

$\boxed{X_1 \quad X_2} \rightarrow$ 2 cols

$\rightarrow$ 6 cols

$x_1^0 \quad x_1^1 \quad x_1^2 \quad x_2^0 \quad x_2^1 \quad x_2^2 \quad y$

W_0

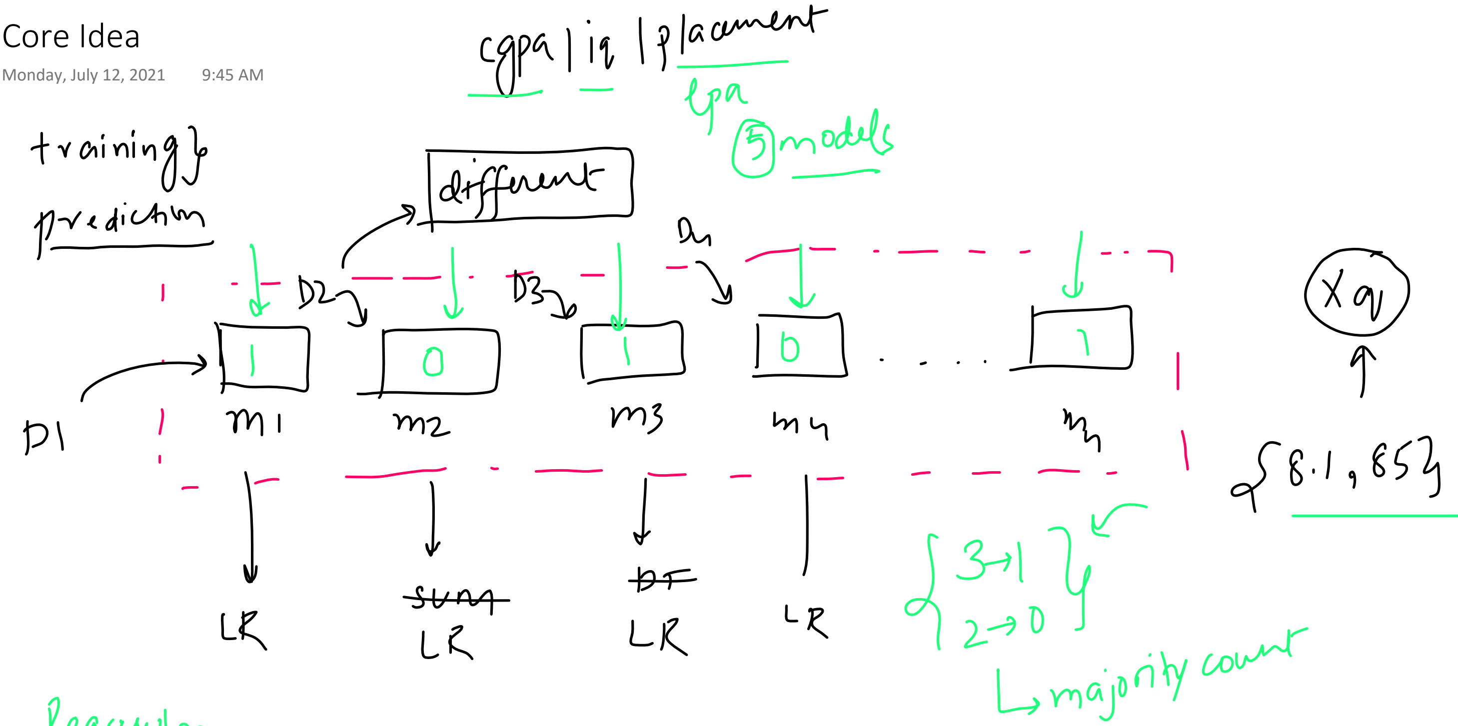
$w_1 \quad w_2 \quad w_3 \quad w_6$

Wisdom of the Crowd

Monday, July 12, 2021 9:56 AM

Core Idea

Monday, July 12, 2021 9:45 AM



Regression

$$\frac{8.1 + 3.2 + 4.1 + 7.5}{4} = 6.6 \text{ Lpa}$$

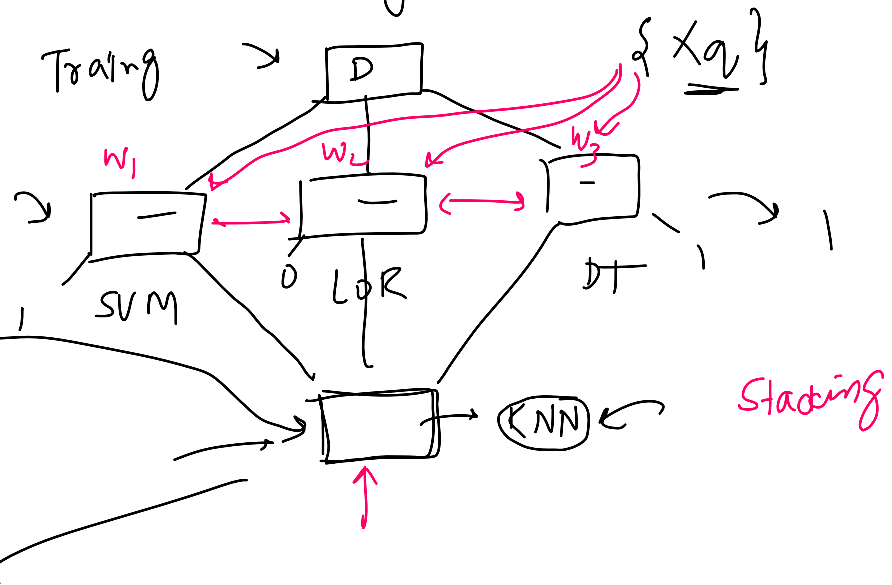
Type of Ensemble Learning

Monday, July 12, 2021 9:45 AM

Ensemble

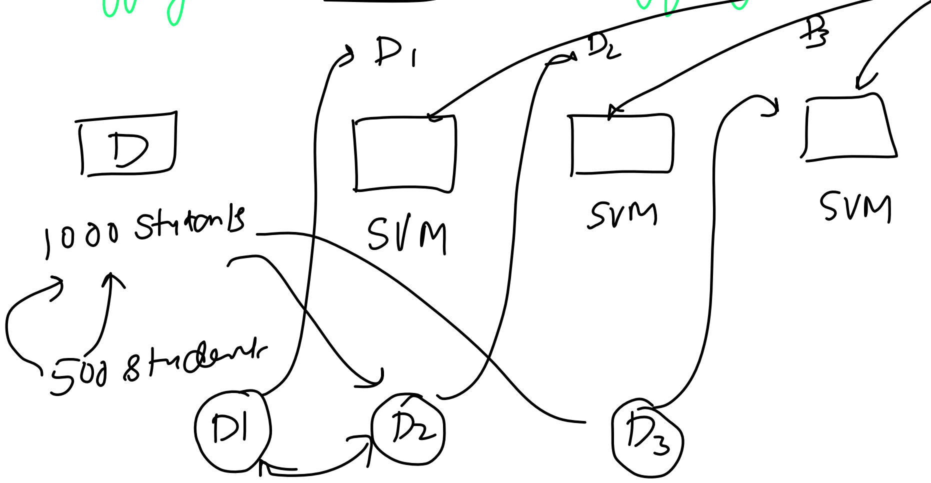
- { Voting ensemble }
- { Bagging } → Random Forest
- { Boosting }
  - Adaboosting
  - Gradient Boosting
  - XgBoost
- { stacking }

→ Base models diff. algo



SVM	LoR	DT	R
1	0	1	1
0	0	1	1

Bagging → Bootstrapped Aggregation

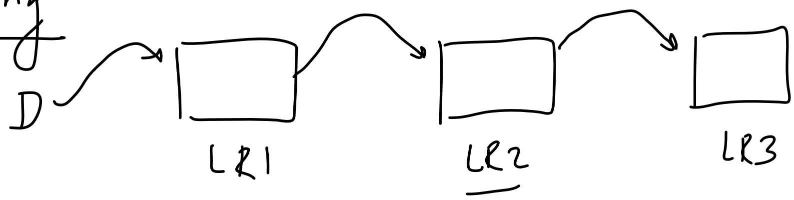


→ Random forest { Decision Tree }

boosting your data

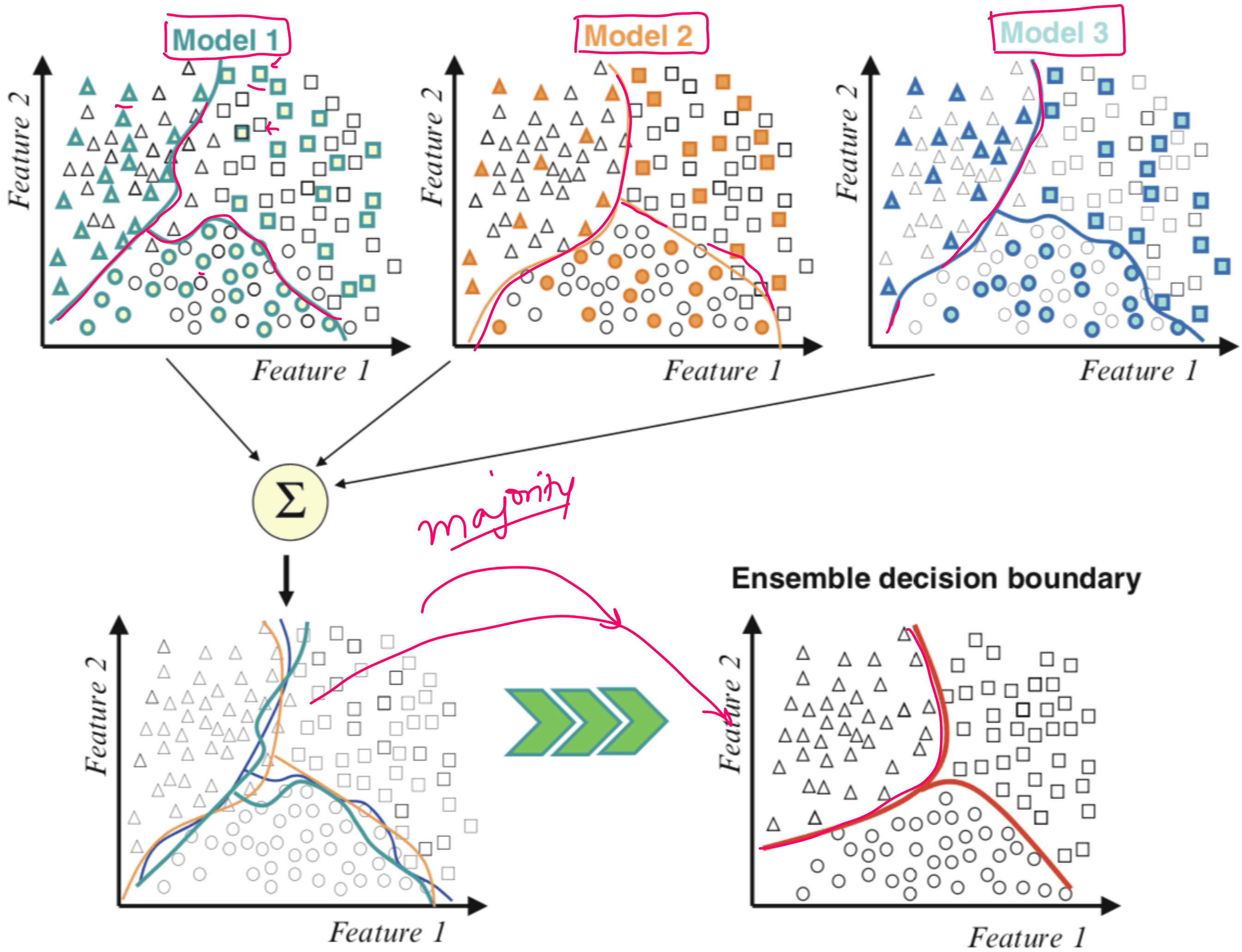
boosting your data

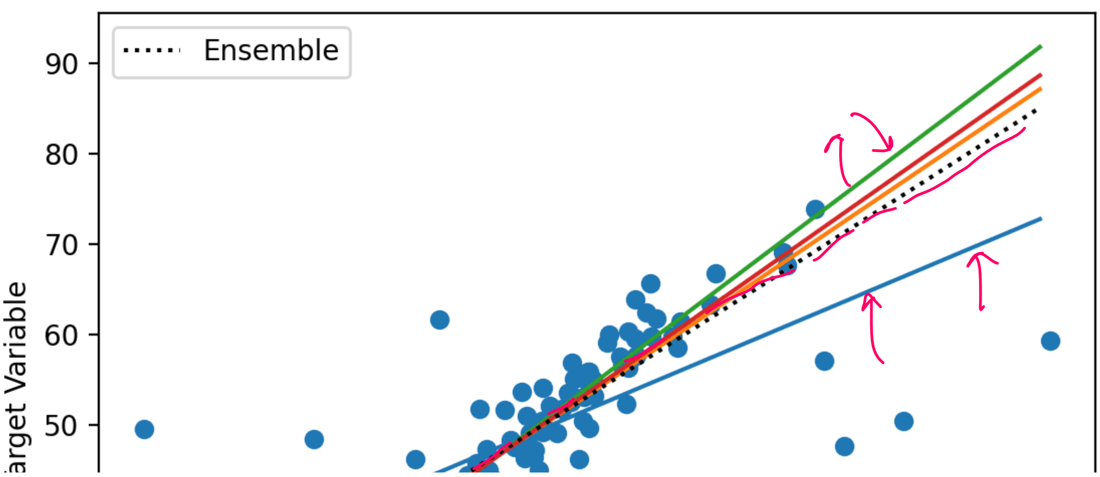
Boosting

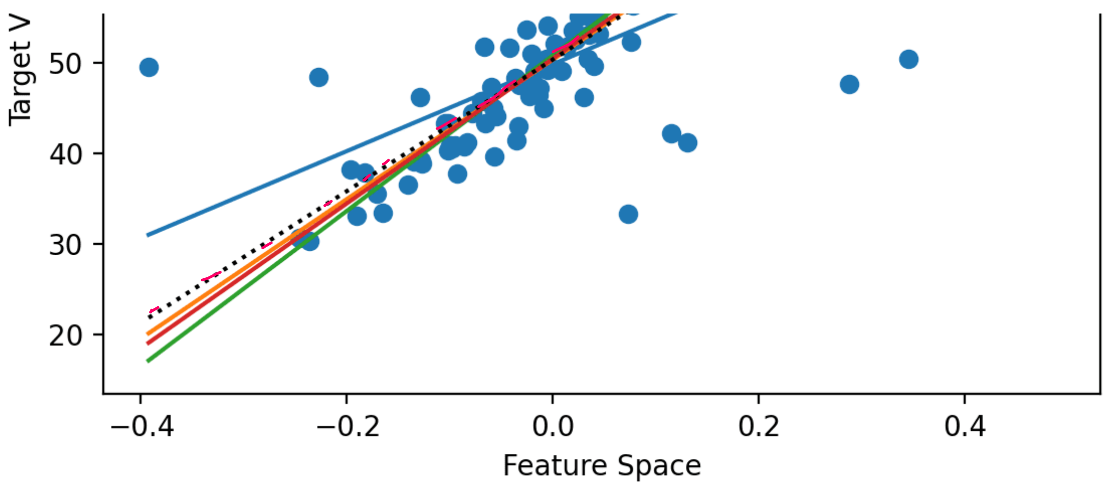


Why it works?

Monday, July 12, 2021 9:45 AM






Target V
50
40
30
20
−0.4
−0.2
0.0
0.2
0.4
Feature Space

Benefits

Monday, July 12, 2021 9:45 AM

1) Improvement in performance

2) Bias Variance

Low Bias + Low Variance

High bias Low variance

Low bias LV

Low Bias High variance

DT → Ensem

Low Bias → LV

3) Robustness

When to use?

Monday, July 12, 2021 9:46 AM

Always

Intuition

Monday, July 19, 2021 4:53 PM

Random forest → group of trees

Bagging → Bootstrapped Aggregation

Bootstrapped → Sampling

Decision Tree

Bootstrapping Agg

D_1 = 1, 18, 32, 43, 103 998

D= 1000 rows → 10 cols

D_1 D_2 D_3 D_{100}

- row sampling
- col sampling
- combination

5 cols

DT1 DT2 DT3 . . . DT100

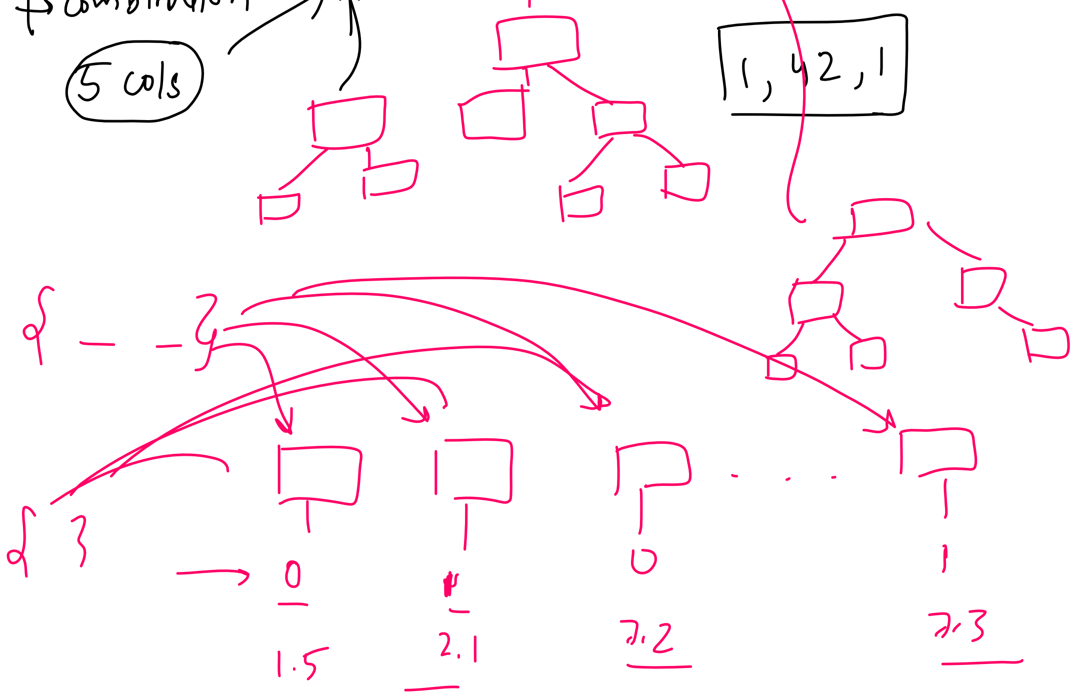
D

1
2
3
4
5
.
.
.
1000

1, 42, 1

with replacement

without replace



0 – 43

1 – 57

Demo

Monday, July 19, 2021 5:05 PM

Bias Variance and Random Forest

Tuesday, July 20, 2021 10:52 AM

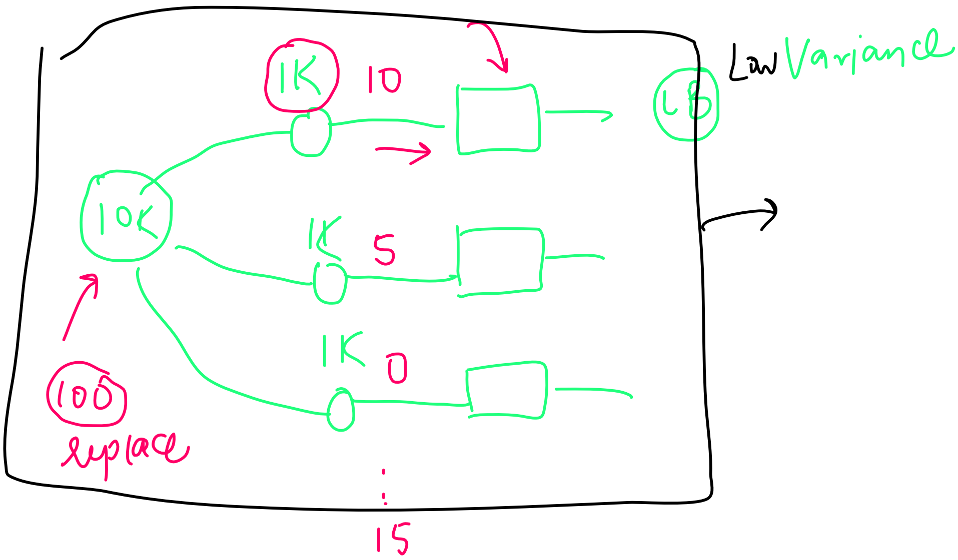
$\boxed{LB \ LV}$

$B \propto \frac{1}{V}$

LB HV	HB LV
→ Fully grown DT	→ LR
→ SVM	
→ KNN	

Random Forest

$\boxed{LBHV} \rightarrow \boxed{LBLV}$



Bagging Vs Random Forest

Monday, July 19, 2021 4:53 PM

RF → bagging ①

DT KNN SVM

RF → DT

Bagging → DT → Random forest ?

Sampling

row + feature

Bagging

1 3

4,5

D

5 cols

1 4

Col samp → 2 cols

DT

feature sampling

col

→ tree level col sampling

2 col

Randomness

node level

RF

1 2 3 4 5

RF > bagging

DT

D

col → 2

max-features

2 3

4 5

2 1

Hyperparameters

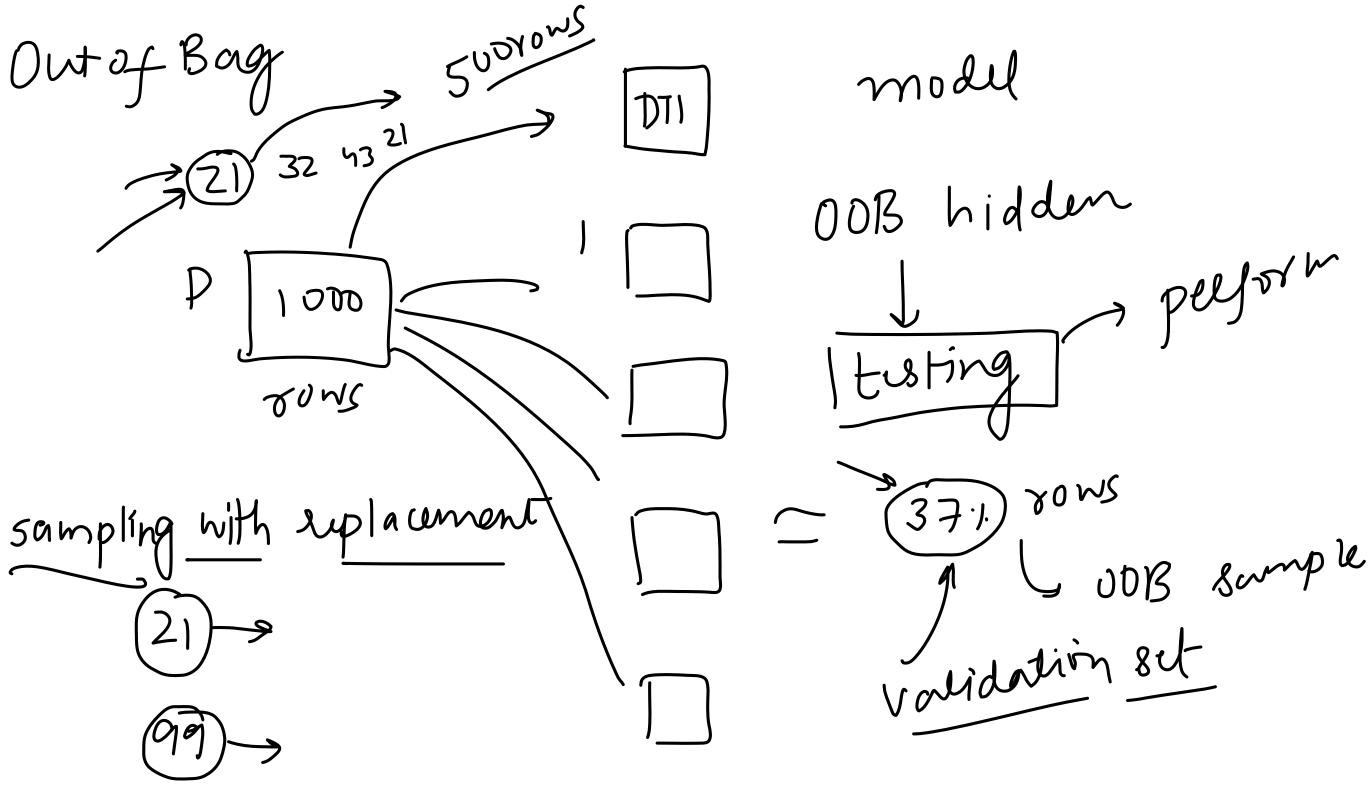
Monday, July 19, 2021 4:56 PM

Code Demo

Monday, July 19, 2021 5:05 PM

OOB Evaluation

Monday, July 19, 2021 5:14 PM



Out Of Bag samples

Feature Importance

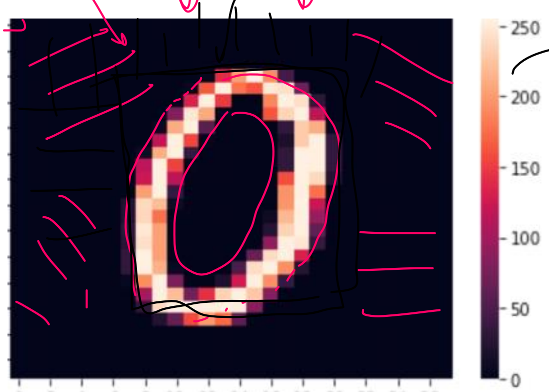
Monday, July 19, 2021 4:56 PM

MNIST

	label	pixel0	pixel1	pixel2	pixel3	pixel4	pixel5	pixel6	pixel7	pixel8	...	pixel774	pixel775	pixel776	pixel777	pixel778
0	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
2	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
3	4	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0

42000 imgs

28x28 → 784



→ feature import

Formula

Ntr → 4 5   NtL-1 5

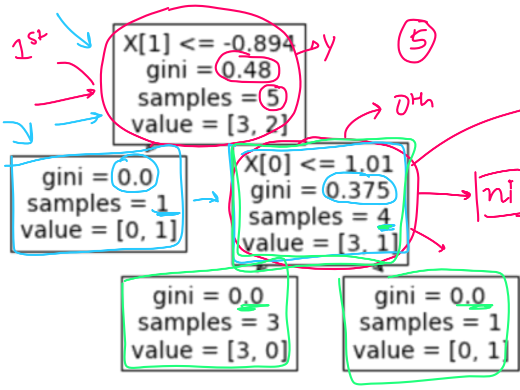
$$ni = \frac{N\_t}{N}\left[impurity - \left(\frac{N\_t\_r}{N\_t} \times right\_impurity\right) - \left(\frac{N\_t\_L}{N\_t} \times left\_impurity\right)\right]$$

fi

0, 1

$$fi_K = \frac{\sum_{j \in \text{node split on feature } K} ni}{\sum_{j \in \text{all nodes}} ni}$$

1st



y

5

0th

ni

$0^{th} = \frac{x}{x+y}$

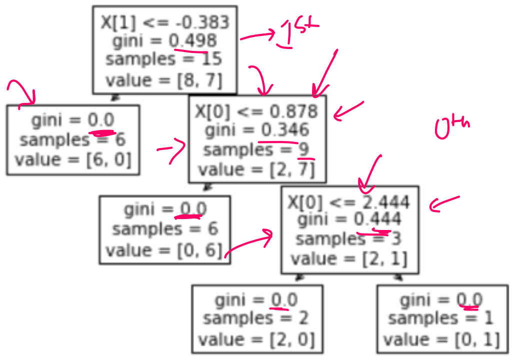
$1^{st} = \frac{y}{x+y}$ → 0.18

$$= \frac{5}{5}\left[0.48 - \frac{4}{5} \times 0.375 - \frac{1}{5} \times 0.\right] = y \Rightarrow \left[0.48 - (0.8 \times 0.375)\right]$$

$$\Rightarrow 0.48 - 0.30 = \boxed{0.18}$$

$$\frac{4}{5}\left[0.375\right] = x = 0.8 \times 0.375 = 0.30 = x$$

$$0^{th} = \frac{0.3}{0.3 + 0.18} = 0.625$$



$$0^{th} = \frac{0.3}{0.3 + 0.18} = 0.625$$

$$1^{st} = \frac{0.18}{0.3 + 0.18} = 0.375$$

0th Node

$$\frac{15}{15}\left[0.49 - \frac{9}{15} \times 0.34\right]$$

$$= \boxed{0.290}$$

$$\frac{9}{15}\left[0.34 - \frac{3}{15} \times 0.44\right]$$

$$= 0.118$$

$$\frac{3}{15}\left[0.44\right]$$

$$= 0.088$$

$$fi[0] = \frac{0.11 + 0.08}{0.29 + 0.11 + 0.08} = \frac{\quad}{0.48}$$

$$\boxed{= 0.39}$$

$$fi[1] = \frac{0.29}{0.48} = 0.60$$

Extra Trees

Monday, July 19, 2021 4:57 PM

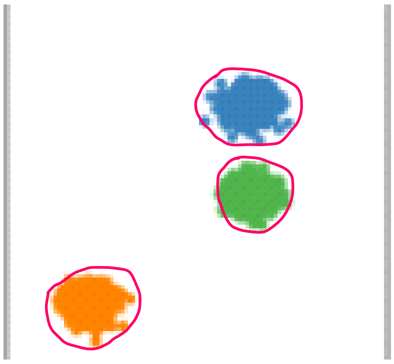
Before Starting

Monday, August 9, 2021 10:42 AM

1) Weak classifiers (> 50%)

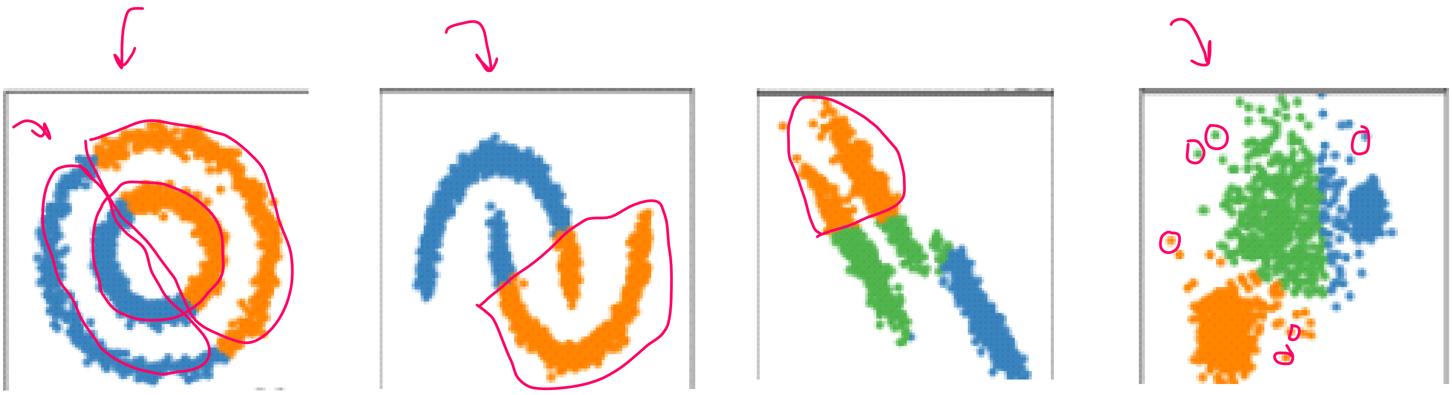
Need of Other Clustering Methods

Saturday, November 6, 2021 7:38 AM



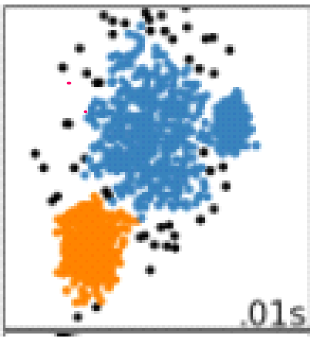
distance to centroid

noise



2 More clustering methods:

1. Hierarchical Clustering
2. Density Based Clustering DBSCAN

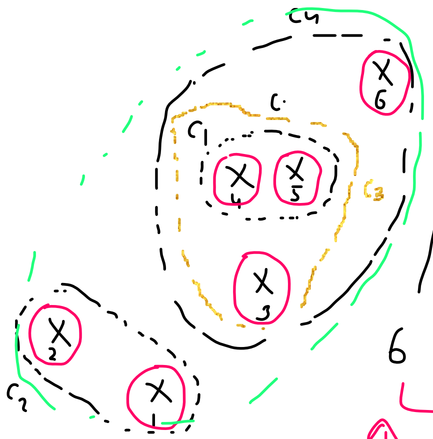

.01s

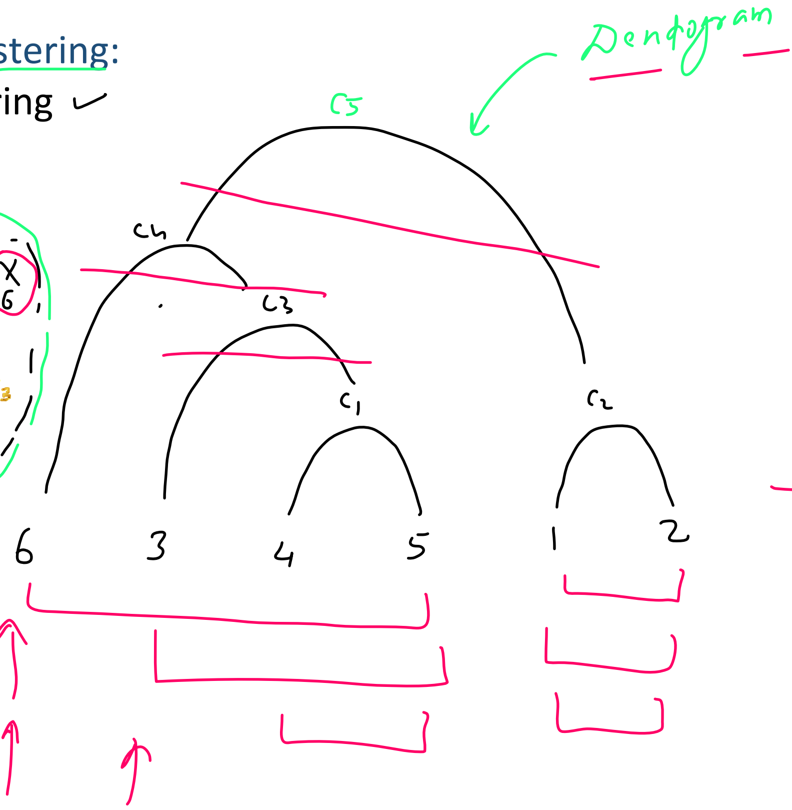
Introduction

Saturday, November 6, 2021 7:39 AM

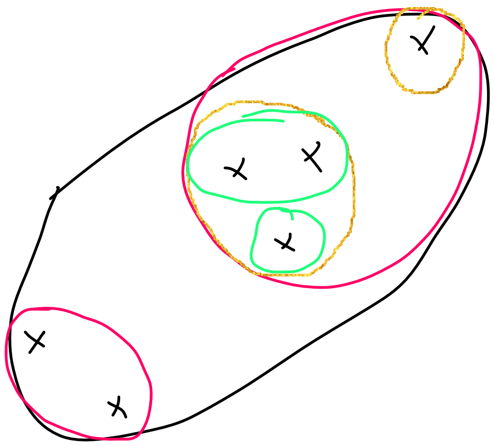
Types of Hierarchical Clustering:

1. Agglomerative Clustering
2. Divisive Clustering





6 clusters
↓
5 cluster
↓
4 cluster
↓
3
↓
2
↓
1

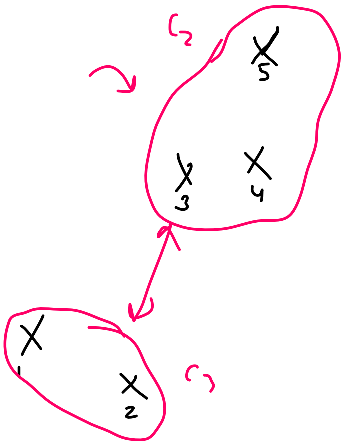


1 cluster
↓
2 cluster
↓
3 cluster
↓
4 cluster
↓
6 clusters

Algorithm

Saturday, November 6, 2021 7:39 AM

1. Initialize the Proximity Matrix
2. Make each point a cluster
3. Inside a loop
    a. Merge the 2 closest clusters
    b. Update the Proximity Matrix
4. Until only one cluster is left



n points nxn

	P_1	P_2	P_3	P_4	P_5
P_1	0	–	–	–	–
P_2	–	0	–	–	–
P_3					
P_4					
P_5					

C_2 C_3

C_2

C_3

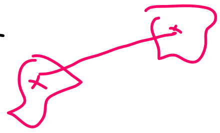
P_1 P_2 C_2

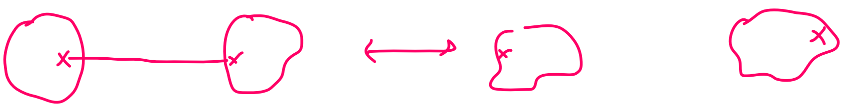
P_1

P_2

C_2

	P_1	P_2	C_1	P_5
P_1	0	–	?	–
P_2	–	0	?	–
C_1	?	?	0	?
P_5	–	–	?	0





Types

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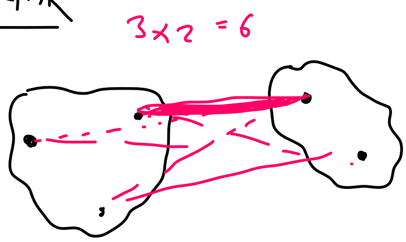
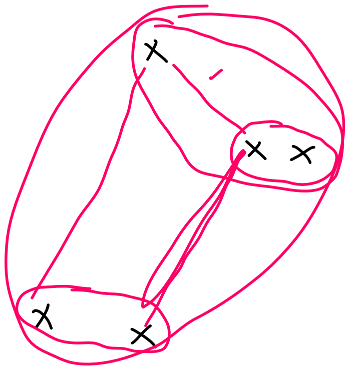
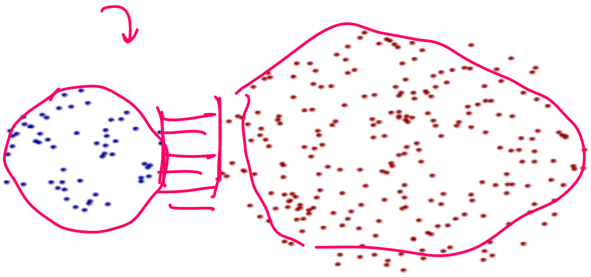
Types of Agglomerative Clustering

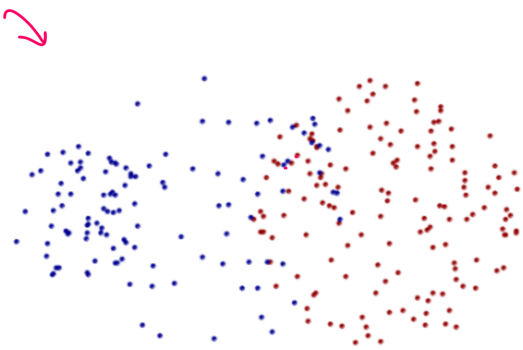
1. Min (Single-link)
2. Max (Complete Link)
3. Average
4. Ward

1. Single Link

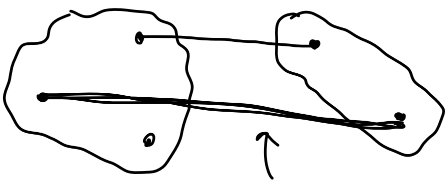


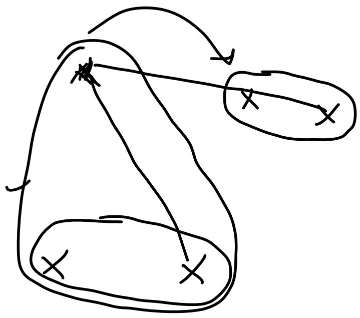


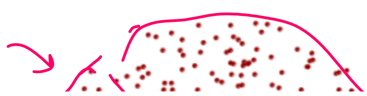


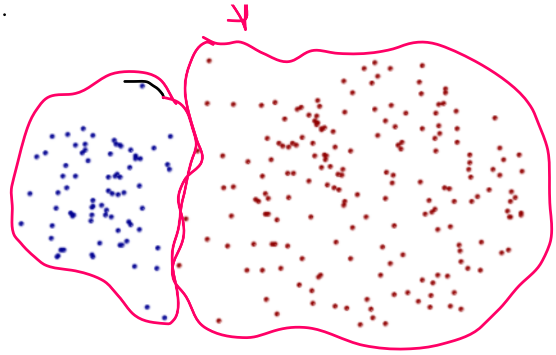


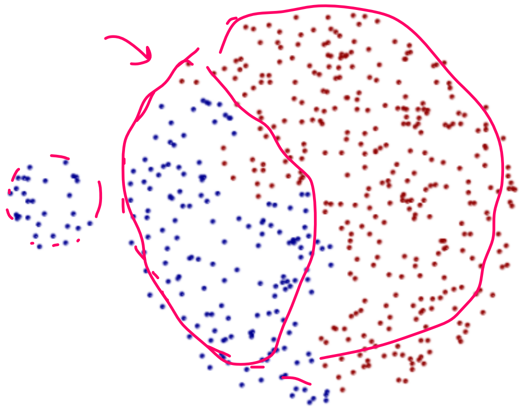
2. Complete Link (Max)



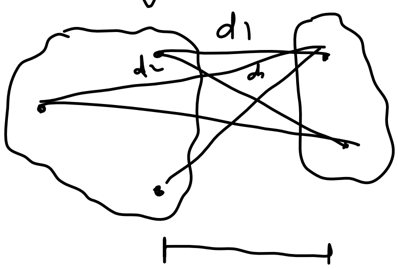








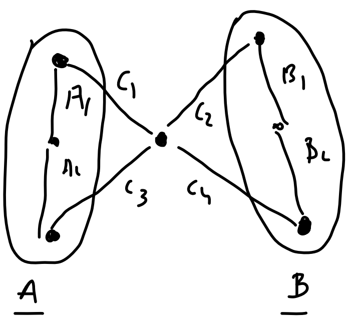
3. Group Average —



2×3

$$\frac{d_1+d_2+d_3+\dots+d_6}{\boxed{2\times 3}} =$$

4. Ward → sklearn



$$\text{dist}_{A\to B} = C_1^2 + C_2^2 + C_3^2 + C_4^2 - A_1^2 - A_2^2 - B_1^2 - B_2^2$$

Variance minimize

How to find the ideal number of clusters

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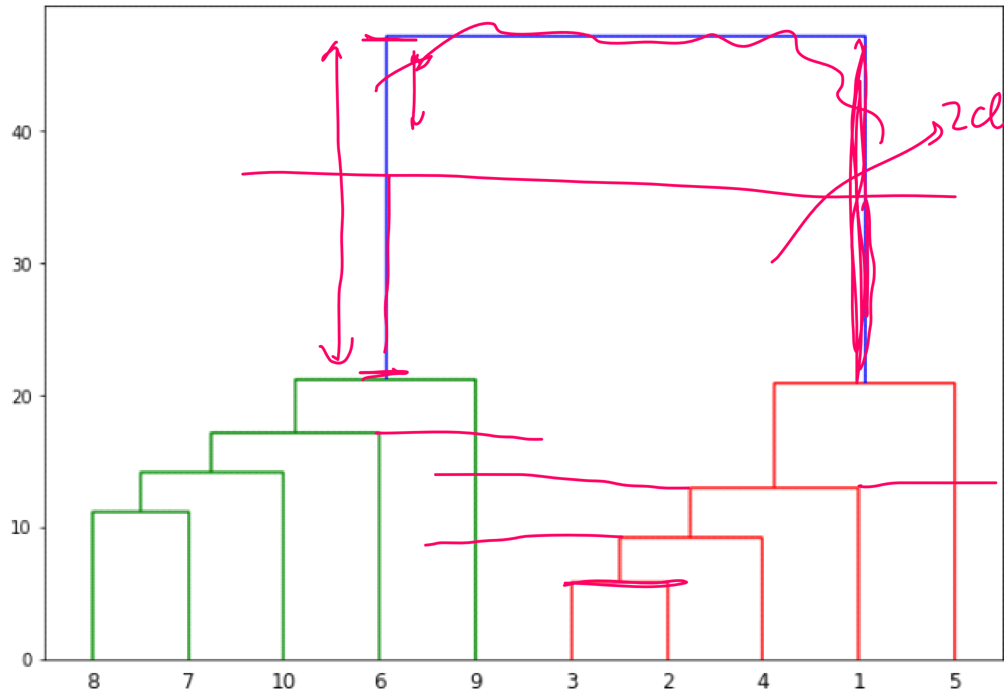
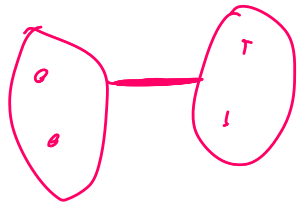


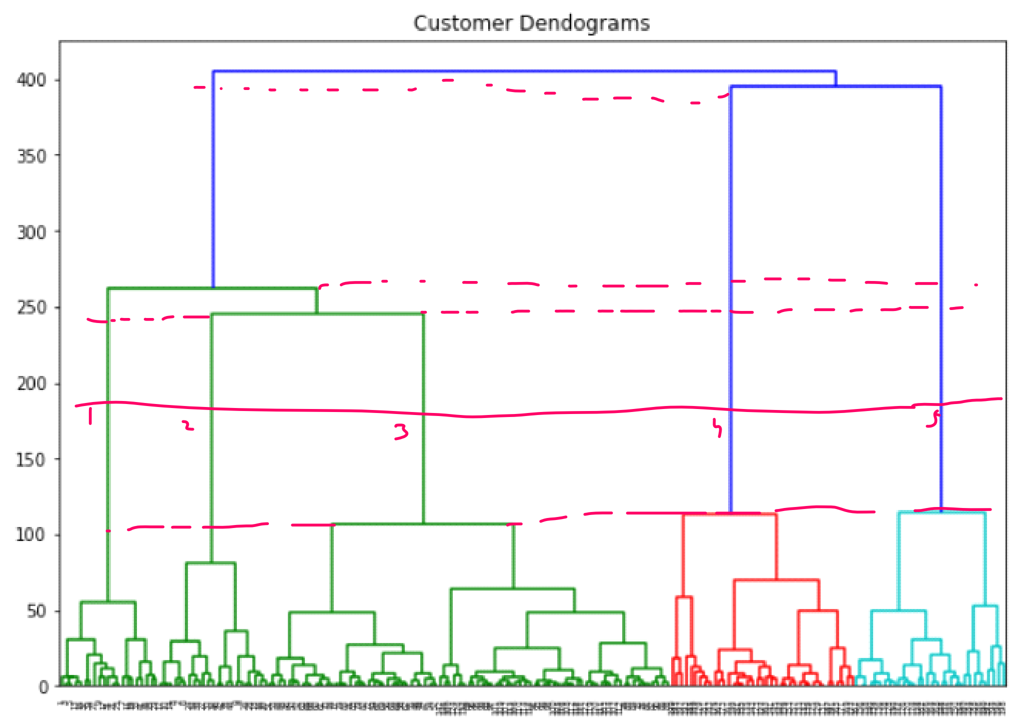
2clusters

Kmeans →

elbow method

inter cluster similarity





5 clusters

Hyperparameter

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Code Example

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Benefits/Limitations

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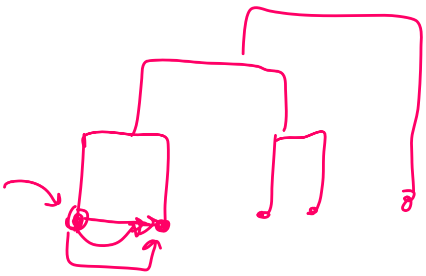
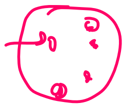
Benefits

1) Widely applicable

2) Dendrogram



Limitation



n $\boxed{n \times n}$

$\boxed{10^6}$ → $\boxed{10^{12} \text{ bytes}}$

10 GB

RAM